Composition nodes: reference and evidence

For senior review, 2026-11-05. Code: `feat/tile-nodes` at `b1d1328`.

Section	What it is	Read it for
1. Reference	Part A: every native node, reference binding and derived node, as the code stands	**The review. Start here, at A.0 and A.9**
2. Figures	All 31 image sheets from `docs/eval/syntax-audit/`, each marked current or older	Seeing the variants and their breaks. Each page zooms to full resolution
3. Syntax audit	The free audit of 2026-11-04, unedited, with its decisions and measurements	The evidence behind the changes
4. Design record	Part B: the spec as approved from 2026-10-27, with notes where it is out of date	Why the convention is what it is

The position in five lines

- **Six derived nodes are built,** with 28 variants: `chart`, `timeline`, `kpi_row`, `card_grid`, `table`, `bullet_list`. Nine plan kinds are still written by hand as native POM.
- **They are not on by default.** With no settings, only KPI text is bound by reference.
- **Free checks are strong:** all 28 variants compile, no words lost, 0 attributes silently ignored. These use sample plans, not LLM-written tags.
- **Proof with a real LLM is incomplete.** The first paid run missed its gate, its fixes are in, and the rerun's result is not in the repo. The full-deck run has not been run.
- **Decisions wanted:** A.9 lists 16 open points. The first three concern the evidence itself.

In these documents, "the user" means the project owner, who made the decisions and gave the blind marks. A "blind mark" is a side-by-side vote between two slides without knowing which route drew which.

1. Reference

Status: for senior review, 2026-11-05. This document has two parts:

- **Part A (reference)** describes the code on `feat/tile-nodes` at `b1d1328`: every native node, every reference binding and every derived node. Variant tables are generated from `block-variants.yaml`, and each behaviour stated was run through the real expander and compiler.
- **Part B (design record)** is the spec as approved on 2026-10-27 and extended on 2026-11-03, kept so its decisions and reasons stay readable. Other documents cite its section numbers, so they are unchanged. **Where the two differ, Part A is right.**

A.0 Summary

A slide reaches the `.pptx` like this:

```
plan JSON -> generator LLM writes POM XML -> composed.expand (derived nodes) -> element_refs / tiles fill (references)
          -> normalize_xml -> fit-grow (sizes type to the boxes) -> POM 10.3.0 compile -> pptx-post
```

Topic	Position today
Families	Three: **native** (the LLM writes POM XML), **reference binding** (the LLM lays out, code writes the words), **derived** (the LLM places one tag, code draws the block)
Derived nodes built	Six: `chart`, `timeline`, `kpi_row`, `card_grid`, `table`, `bullet_list`, with 28 variants in all
Plan kinds still written by hand	Nine: `title`, `narrative`, `caption`, `layer`, `process_arrow`, `flow`, `pyramid`, `tree`, `matrix`
On by default?	**No.** With no settings, only KPI text is bound by reference. All six nodes need `POM_NODES` (A.1)
Proof with a real LLM	One paid same-plan run; its rerun's result is not in the repo, and the full-deck run has not been run (A.8)

Please review first: the open points in A.9, especially the missing rerun result, the three overlapping variant vocabularies, and the 280 px box.

A.1 The three families and the switches

Family	What the LLM writes	What code does	Example
Native	The POM node in full, with its text	Normalizes and fits it; never changes POM's syntax	`<Pyramid h="240"><PyramidLevel label="Top"/> </Pyramid>`
Reference binding	Its own boxes, with a reference where each piece of plan text goes	Writes the text, sizes and ids	`<Text ref="kpis.0.value"/>`

Family	What the LLM writes	What code does	Example
Derived (also "composition node")	One tag naming a plan component	Expands it into native nodes from the plan	`<CardGrid ref="levers" variant="banded" grow="1"/>`

A naming trap: `variant="native"` on a derived node is still derived. The LLM writes `<Table ref="gaps" variant="native"/>`, and code writes the real `<Table><Tr><Td>`. `blocks.*.family: native` in the YAML means the same thing, and `spec.DERIVED_TAGS` lists only the two tags that are not POM names.

Two settings choose the route (`.env`, read per process):

Route	`POM_BLOCKS`	`POM_NODES`	Effect
Old	`off`	empty	Everything by hand
Default	`kpi` (also when unset)	empty	KPI text by field reference; everything else by hand
A	`tiles`	empty	Older tile tags `<KpiGrid>` / `<CardGrid>`; the rest by hand
B	`tiles`	`chart,timeline,kpi_row,card_grid,table,bullet_list`	All six derived. `python -m scripts.eval_run ... --nodes` sets this

Each kind in `POM_NODES` switches on alone. `kpi_row` also needs `POM_BLOCKS` to be `kpi` or `tiles`, because `<KpiRow>` expands into field references that `fill_refs` then fills.

A.2 Every plan kind, by route

These are the 15 kinds of `state.ComponentKindLiteral`. "Hand" means the LLM writes native POM XML from the plan's text.

Plan kind	Plan content (`planner_schema_typed.py`)	Default	Route A	Route B
`kpi_row`	`kpi_labels`, `kpi_values`, optional `kpi_deltas`, `kpi_directions`, `kpi_tones`, `kpi_icons`	`<Text ref>`	`<KpiGrid ref>`	`<KpiRow ref>`
`card_grid`	`card_layout`, `cards[]` (`title`, `tag`, `body`, `bullets`, `tone`, `key_phrase`, `icon`), `focus_card`	Hand	`<CardGrid ref>`, tile variants	`<CardGrid ref>`, compositions
`card_grid`, `card_layout: matrix`	also `columns`, `rows`	Hand	Hand	Hand (deferred)
`chart`	`chart_title`, `chart_labels`, `chart_values` or `chart_series`	Hand	Hand	`<Chart ref>`
`table`	`table_columns`, `table_rows`	Hand	Hand	`<Table ref>`
`timeline`	`direction`, `timeline_items[]` (`date`, `label`, `description`)	Hand	Hand	`<Timeline ref>`
`bullet_list`	`bullets`	Hand	Hand	`<Ul ref>`
`process_arrow`	`direction`, `process_steps`	Hand	Hand	Hand (node dropped)

Plan kind	Plan content (`planner_schema_typed.py`)	Default	Route A	Route B
`flow`	`direction`, `flow_steps`	Hand	Hand	Hand (node dropped)
`pyramid`	`direction`, `pyramid_levels`	Hand	Hand	Hand
`tree`	`layout`, `tree_nodes`	Hand	Hand	Hand
`matrix`	`x_axis`, `y_axis`, `quadrants`, `matrix_items`	Hand	Hand	Hand
`title`	`title`, `subtitle`	Hand	Hand	Hand
`narrative`, `caption`	`text`	Hand	Hand	Hand
`layer`	`description`, `layers`, `center`, `nodes`, `connections`	Hand	Hand	Hand

When each was introduced

Date	What	Code	Evidence
2026-10-11	KPI field references, on by default	`element_refs.py`	`docs/kpi-node-pilot-2026-10-11.md`
2026-10-12	KPI tile tags `<KpiGrid>` / `<Kpi>`, route A	`tiles.py`	`docs/tile-nodes-spec-2026-10-12.md`
2026-10-13	Card tile tags `<CardGrid>` / `<Card>`, route A	`cards.py`	same, §11
2026-10-28	Chart node; timeline, process arrow and flow nodes	`charts.py`, `sequences.py`, `composed.py`	`docs/eval/chart-node/`, `docs/eval/sequence-nodes/`
2026-11-01	KPI and card compositions; table node	`tile_groups.py`, `tables.py`	`docs/eval/composition-convention/`, `docs/eval/table-node/`
2026-11-02	**Process arrow and flow nodes dropped**: POM's fixed boxes hold about two words, and the code-drawn steps were card grids under another name	`sequences.py`	`docs/eval/sequence-nodes/`
2026-11-04	Bullet node; **five table variants and card `rail` deleted**	`bullets.py`	`docs/eval/bullet-node/`, `docs/syntax-audit-2026-11-04.md`

A.3 Native nodes

These are the POM 10.3.0 tags the LLM may write, from `knowledge/core/nodes.yaml`. Every tag below compiled in a test slide at this commit. The attribute lists come from `nodes.yaml`; `tests/unit/test_upstream_reference.py` compiles POM's own reference examples. This project never changes POM's syntax.

Structure and layout

Tag	Own attributes	Children
`Slide`	none	any layout or content node
`Theme`	one per colour token, 6-digit hex only	none
`VStack`, `HStack`	`gap`, `alignItems`, `justifyContent`, `flexWrap`	any layout or content node
`Layer`	`w`, `h`, `padding`, `margin`, `backgroundColor`, `border`, `borderRadius`, `opacity`, `shadow`	any node, each with `x` and `y`

Content

Tag	Own attributes	Children
`Text`	`fontSize`, `color`, `textAlign`, `bold`, `italic`, `underline`, `strike`, `subscript`, `superscript`, `fontFamily`, `lineHeight`, `letterSpacing`, `textGradient`, `glow.*`, `outline.*`, `rotate`	inline tags
`Shape`	`shapeType` (required), `text`, `fill.*`, `line.*`, `outline.*`, `glow.*`, `rotate`, and the text attributes	none
`Icon`	`name`, `size`, `color`, `variant`, `bgColor`, `rotate`	none
`Svg`	`w`, `h`, `color`	one `<svg>`
`Ul`	`fontSize`, `color`, `textAlign`, `bold`, `italic`, `strike`, `subscript`, `superscript`, `underline`, `fontFamily`, `lineHeight`	`Li`
`Ol`	as `Ul`, plus `numberType`, `numberStartAt`	`Li`
`Li`	`bold`, `italic`, `strike`, `subscript`, `superscript`, `underline`, `color`, `fontSize`, `fontFamily`	inline tags

Inline tags: `B`, `I`, `U`, `S`, `Sub`, `Sup`, `Span` (`color`, `fontFamily`, `fontSize`, `letterSpacing`), `A` (`href`).

Common attributes on most nodes (`core/attributes.yaml`): `id`, `w`, `h`, `grow`, `minW`, `maxW`, `minH`, `maxH`, `padding`, `margin`, `backgroundColor`, `backgroundGradient`, `border`, `borderTop` / `Right` / `Bottom` / `Left`, `borderRadius`, `opacity`, `zIndex`, `position`, `top`, `right`, `bottom`, `left`, `alignSelf`, `shadow`.

Data and diagrams

Tag	Own attributes	Children
`Table`	`defaultRowHeight`, `cellBorder.color` / `width` / `dashType`	`Col`, `Tr`
`Col`	`width` (not `w`)	none
`Tr`	`height`	`Td`
`Td`	`fontSize`, `color`, `textAlign`, `bold`, `italic`, `underline`, `strike`, `subscript`, `superscript`, `fontFamily`, `backgroundColor`, `colspan`, `rowspan`	inline tags

Tag	Own attributes	Children
`Chart`	`chartType` (`bar`, `line`, `pie`, `area`, `doughnut`, `radar`), `h` (required), `showLegend`, `showTitle`, `title`, `chartColors`, `radarStyle`, `sparkline`	`ChartSeries`
`ChartSeries`	`name`	`ChartDataPoint`
`ChartDataPoint`	`label`, `value` (both required)	none
`Timeline`	`direction`, `dateColor`, `titleColor`, `descriptionColor`, `connectorColor`, `connectorGradient`, `useColorForDate`, `fontFamily`, `w`, `h`	`TimelineItem`
`TimelineItem`	`date`, `title` (both required), `description`, `color`, `dateColor`	none
`ProcessArrow`	`direction`, `itemWidth`, `itemHeight`, `gap`, `fontSize`, `bold`, `italic`, `strike`, `underline`, `highlight`, `fontFamily`, `w`, `h`	`ProcessArrowStep`
`ProcessArrowStep`	`label` (required), `color`, `textColor`	none
`Flow`	`direction`, `nodeWidth`, `nodeHeight`, `nodeGap`, `connectorStyle.*`, `w`, `h`	`FlowNode`, `FlowConnection`
`FlowNode`	`id`, `shape`, `text` (all required), `color`, `textColor`, `width`, `height`	none
`FlowConnection`	`from`, `to` (both required), `label`, `color`, `labelColor`	none
`Pyramid`	`direction`, `fontSize`, `bold`, `fontFamily`, `w`, `h`	`PyramidLevel`
`PyramidLevel`	`label` (required), `color`, `textColor`	none
`Tree`	`layout`, `nodeShape`, `textColor`, `nodeWidth`, `nodeHeight`, `levelGap`, `siblingGap`, `connectorStyle.*`, `w`, `h`	one `TreeItem`
`TreeItem`	`label` (required), `color`, `textColor`	`TreeItem`
`Matrix`	`w`, `h`, `axisLabelColor`, `quadrantLabelColor`, `itemLabelColor`	`MatrixAxes`, `MatrixQuadrants`, `MatrixItem`
`MatrixAxes`	`x`, `y` (label strings, both required)	none
`MatrixQuadrants`	`topLeft`, `topRight`, `bottomLeft`, `bottomRight` (all required)	none
`MatrixItem`	`label`, `x`, `y` (0 to 1; all required), `color`, `textColor`	none
`Line`	`x1`, `y1`, `x2`, `y2` (all required), `color`, `lineWidth`, `dashType`, `beginArrow`, `endArrow`	none
`Arrow`	`from`, `to` (both required), `color`, `lineWidth`, `dashType`, `beginArrow`, `endArrow`	none

`Image` is deferred and not offered to the LLM.

Limits to review against

- **POM's flexbox is Yoga.** There is no `alignContent` and no min-content width, `grow="0"` is rejected on `Text`, and `w="max"` grows along the parent's main axis (`docs/syntax-audit-2026-11-04.md` §6).
- **Item tags reject borders:** `Td`, `Li`, `Col`, `FlowNode`, `TimelineItem`, `ProcessArrowStep`, `PyramidLevel`, `MatrixItem`. The normalizer removes them (`ITEM_BORDER_REMOVED`).
- **`Td` holds no `Icon` or stack.** Enrichment is composed around core nodes, never inside them.
- **Native diagrams have small fixed boxes** that only shrink: chevrons 150 × 80, flow nodes 120 × 60, timeline titles 120 px. This is why process arrow and flow left the derived set.
- **`Theme` is hex only,** with no fonts or alpha.
- **The normalizer repairs some native XML without a retry,** for example `SHAPE_TYPE_ALIAS`, `ALPHA_COLOR_BLENDED`, `TEXT_BOX_UNSQUEEZED`, `GROW_BAND_ADDED`, `GROW_ROWS_ADDED`, `TEXT_CONTAINER_FLATTENED`, `UNKNOWN_ICON_REMOVED`.

A.4 Reference bindings

A reference ties something on the slide to the plan by id, so the LLM never types the content. There are five forms.

Form	Written as	Binds to	Available on
Group	`<CardGrid ref="levers"/>`	A whole plan component	The six derived tags (and `<KpiGrid>` in route A)
Item	`<Card ref="levers.1"/>`	One item, by index from 0	`<Kpi>`, `<Card>` only
Field	`<Text ref="kpis.0.value"/>`	One field of one item	`kpi_row` only: `label`, `value`, `note`
Marks by name	`<Highlight column="Fix" color="good"/>` inside the tag	Rows, columns, cells, points or phrases, by their text	`<Table ref>`, `<Ul ref>` only
POM ids	`<Arrow from="s1" to="s2"/>`	Another node's `id`	POM's own, unrelated to `ref`

Group and item references

```
<CardGrid ref="levers" grow="1">
  <Card ref="levers.1" title.color="$negativeText" item.backgroundColor="$warningSoft"/>
</CardGrid>

<HStack grow="1" gap="18">
  <Card ref="levers.0" w="40%"/>
  <CardGrid ref="levers" cols="2" w="1" grow="1"/>
</HStack>
```

An item tag **inside** its group restyles that item. **Outside**, it places the item anywhere, and the group skips it; this is how the LLM gets flexbox freedom with no new syntax.

Field references

The LLM draws its own tiles and code fills the text:

```
<HStack gap="12" grow="1">
  <VStack padding="16" gap="6" backgroundColor="$surfaceAlt">
    <Text ref="kpis.0.label"/>
    <Text ref="kpis.0.value"/>
    <Text ref="kpis.0.note"/>
  </VStack>
</HStack>
```

Field	Plan key	Code also sets
`label`	`kpi_labels`	Size 16; a role or tone colour if the LLM gave no `$token`
`value`	`kpi_values`	One size per component, the unit as a span, peer ids so values grow together (28 to 96 px; a hero starts at 72 and reaches 120)
`note`	`kpi_deltas`	Size 16; a role or tone colour if the LLM gave no `$token`

Everything else on the tile (boxes, fills, bars, icons) is the LLM's. Text on a dark fill gets a readable colour. If the LLM leaves out a label, code draws it (`NODE_LABEL_PLACED`, a warning). **Cards have no field references**: `<Text ref="levers.0.title"/>` is an error. They were prototyped for the audit and rejected as a default (`docs/syntax-audit-2026-11-04.md` §5: 46 × the tokens, and swapped fields pass unseen).

Marks by name

On	Target	Effect	Rules
`<Table ref>`	`row`, `column`, `cell="row / column"`, `phrase`	`fill` (background), `color` (text)	`above` / `below` / `else`, `max` / `min`; `show="arrows"` (`invert`), `show="pills"` and `show="bars"` (those two on `rows` and `ledger` only)
`<Ul ref>`	`item`, `phrase`	`color`, `fill` (on rows only), `bold`	

Tones are `good`, `bad`, `watch`, `focus`. A name or phrase not in the plan is `NODE_HIGHLIGHT_UNKNOWN` (error). Other tags drop `<Highlight>` with a report.

Binding rules

- **One nesting level.** A reference is never placed inside another. `<Table ref>` inside `<CardGrid ref>` is removed and becomes `NODE_ITEM_MISSING` (error), because the table is then missing from the slide.
- **Each component once.** A second copy, or the same item twice, is `NODE_REF_DUPLICATE` (error).
- **Each drawn component must appear.** Leaving it out, or typing it by hand, is `NODE_ITEM_MISSING` (error).
- **Wrong kind.** A derived tag on a component with no node (a narrative written as `<CardGrid ref>`) is `NODE_REF_UNKNOWN` (error); the message hands over the component's words so the repair can write it by hand.
- **Kinds that can never be referenced:** `title` and `layer` (`spec.NO_REF_KINDS`).

Ids code writes

fit-grow reads these to size peers together. The LLM does not see or write them.

Id prefix	Marks
`kpiv<rank>-`, `kpih<rank>-`	KPI values, and a hero value; rank comes from the plan's `weight`
`kpil-`	KPI labels
`kpig-`, `cardg-`	A KPI or card group whose columns code chose, so fit-grow may re-grid it
`card-<cid>-<i>`, `card-<cid>x-<n>`	Items, and their inner boxes
`cardt-`, `cardb-`, `cardn-`, `cardf-`	Card title, body, numeral, and the `bento` focus block
`tbl-`	Table
`lst-`, `lsth-`	Bullet list, and a row head

A.5 Derived nodes

How a tag resolves

1. **Alias:** an old variant name maps to its replacement (`NODE_ATTR_ALIAS`, a report). Saved slides keep working.
2. **No variant:** on KPI and cards, the design hint can name a word (hero or spotlight gives `bento`, ruled gives `rows`, strip gives `strip`; `NODE_LOOK_FROM_HINT`). Otherwise the first variant whose `requires` hold, along the fallback chain (`NODE_DEFAULT_FROM_CONTENT`).
3. **Unknown variant:** the default (`NODE_VARIANT_UNKNOWN`, a warning).
4. **`requires` fails:** the `fallback` chain (`NODE_VARIANT_UNMET`, a warning). The prompt line offers only the variants that fit.
5. **Emphasis** always comes from the plan (`focus_card`, the hint, or the last step), never from the LLM.

The tables below come from `nodes.<kind>` in `block-variants.yaml`. "In the `.pptx`" was read from compiled files. **Only `table native`, `chart native` / `native labelled` and `bullet_list native` / `native numbered` stay real PowerPoint objects** that a person can edit as one; everything else is loose shapes.

`requires` **checks** (`spec.check`), all on the plan, none on the box: `count_le_N` / `count_ge_3` count items; `layout_grid` / `layout_steps` read `card_layout`; `has_emphasis` means the plan singles out one item; `no_own_numbers` means no item already carries a number or date; `bullets_every_card`; `rows_fit` estimates bullet height; `columns_le_6`; `labels_short` means every label is at most 24 characters with no detail; `one_series`; `chart_is_bar`.

`chart: <Chart ref>`

The chart type, colours and focus bar are code's. `direction="horizontal"` turns native bars. Chart text takes no attribute, so style power lives in `rows`.

Variant	What it draws	Needs → else	Ignores (reported if written)	In the `.pptx`
`native`	the chart, one hue, the focus bar in the accent	nothing	`alignItems`, `label`, `value`; only `bar.fill.color`	real chart
`native labelled`	each bar's value written on it, no value axis	`one_series`, `count_le_8`, `chart_is_bar` → `native`	`alignItems`, `value`; only `label.color/bold`, `bar.fill.color`	real chart
`rows`	one row per point: label, a bar, the value as written	`one_series`, `count_le_8` → `native`	`showLegend`	loose shapes

- **Tag attributes:** `ref`, `variant`, `direction`
- **Parts:** `title.` (Text), `label.` (Text), `value.` (Text), `bar.` (Shape)
- **Native attributes passed on:** `showLegend`. **Code-owned, dropped if written:** `chartColors`, `chartType`, `title`, `showTitle`
- **Recipes:** none
- **Aliases:** `plain` → `native`, `labelled` → `native labelled`, `focus_label` → `native labelled`, `horizontal` → `native`

`timeline: <Timeline ref>`

The default is chosen from the content: `native` needs short labels with no detail, so a timeline with detail starts at `rail`. A vertical timeline that is not native draws `rows`. With this node on, the planner fills `description` as its own field.

Variant	What it draws	Needs → else	Ignores (reported if written)	In the `.pptx`
`native`	POM's timeline: dots on a line, short labels only	`labels_short` → `rail`	`alignItems`, `gap`, `justifyContent`, `band`, `item`; only `date.color`, `title.color`, `description.color`, `marker.fill.color`, `connector.fill.color`	loose shapes
`rail`	dots on a line, date and label alternating above and below	`count_le_7` → `columns`	`band`, `item`	loose shapes

Variant	What it draws	Needs → else	Ignores (reported if written)	In the `.pptx`
`columns`	a line with a dot per point, one box per point under it	nothing	`band`	loose shapes
`rows`	one point per ruled row: date, head, detail; what a vertical timeline with details draws	nothing	`marker`, `connector`, `band`	loose shapes
`banded`	equal boxes, the date in a solid band: done / now / ahead	nothing	`marker`, `connector`	loose shapes

- **Tag attributes:** `ref`, `variant`, `direction`
- **Parts:** `date.` (Text), `title.` (Text), `description.` (Text), `marker.` (Shape), `connector.` (Shape), `band.` (Box), `item.` (Box)
- **Native attributes passed on:** none. **Code-owned, dropped if written:** `dateColor`, `titleColor`, `descriptionColor`, `connectorColor`, `useColorForDate`
- **Recipes:** none
- **Aliases:** `vertical` → `native`, `cards` → `columns`, `alternating` → `rail`

`kpi_row`: `<KpiRow ref>` and `<Kpi ref>`

Expands into field references (A.4). `cols` is a measured knob that code may overrule. `size="hero"` on an item needs the plan's emphasis item.

Variant	What it draws	Needs → else	Ignores (reported if written)	In the `.pptx`
`grid`	separate tiles, icon and label over the value and delta (default)	nothing	`band`, `divider`, `value.bold`, `item.grow`, `item.borderRadius`	loose shapes
`strip`	one shared box, items side by side split by hairlines, one row (at most 5)	`count_le_5` → `grid`	`borderRadius`, `band`, `value.bold`, `item.borderRadius`	loose shapes
`rows`	one item per ruled row: icon badge, label over delta, the value at the right	nothing	`borderRadius`, `band`, `value.bold`, `item.borderRadius`	loose shapes
`banded`	equal boxes, each with a solid coloured band holding icon + label (at most 5)	`count_le_5` → `grid`	`marker`, `divider`, `value.bold`, `item.borderRadius`	loose shapes
`bento`	the emphasis item as one large dark block, the others as rows beside it	`count_ge_3`, `has_emphasis` → `grid`	`band`, `divider`, `value.bold`, `item.borderRadius`	loose shapes

- **Tag attributes:** `ref`, `variant`, `cols`, `color`, `icon`. On the item tag: `ref`, `variant`, `color`, `icon`, `size`
- **Parts:** `label.` (Text), `value.` (Text), `delta.` (Text), `icon.` (Icon), `marker.` (Shape), `band.` (Box), `divider.` (Rule), `item.` (Box)
- **Native attributes passed on:** none. **Code-owned, dropped if written:** none

- **Recipes:** `filled`, `toned`, `bar`, `dot`, `bare`
- **Aliases:** `plain` → `grid`, `filled` → `grid`, `toned` → `grid`, `bar` → `grid`, `dot` → `grid`, `bare` → `grid`, `inverted` → `grid`, `divided` → `strip`

`card_grid:` `<CardGrid ref>` and `<Card ref>`

`rows_fit` (`cards.rows_fit`) estimates bullet lines against a full slide, because bullets set a row's height while body text can shrink. A `card_layout: matrix` plan has no node and stays hand-written.

Variant	What it draws	Needs → else	Ignores (reported if written)	In the `.pptx`
`grid`	separate cards: icon, tag, title, body, bullets (default for grid plans)	`layout_grid` → `steps`	`number`, `band`, `bullets`, `divider`	loose shapes
`grid numbered`	cards with a large 01 02 03 above each title	`layout_grid`, `no_own_numbers` → `grid`	`band`, `bullets`, `divider`	loose shapes
`rows`	one card per ruled full-width row: icon, tag over title, body, bullets	`count_le_6`, `rows_fit` → `grid`	`number`, `band`, `bullets`, `item.grow`	loose shapes
`rows numbered`	rows led by a large 01 02 03	`count_le_6`, `no_own_numbers`, `rows_fit` → `rows`	`icon`, `band`, `bullets`, `item.grow`	loose shapes
`columns`	one column per card: the tag, a coloured pill head, the body, each bullet in its own box	`bullets_every_card`, `count_le_5` → `grid`	`number`, `divider`, `item.grow`	loose shapes
`banded`	boxes with a solid coloured band (icon, title, tag) over body and bullets	`count_le_6` → `grid`	`number`, `bullets`, `divider`, `item.grow`	loose shapes
`bento`	the focus card as one large dark block, the others beside it; at least 500 px wide	`layout_grid`, `count_ge_3`, `count_le_5`, `has_emphasis` → `grid`	`number`, `band`, `bullets`, `divider`	loose shapes
`steps`	cards in a row joined by arrows, the destination dark (default for steps plans)	`layout_steps` → `grid`	`gap`, `number`, `band`, `bullets`, `divider`	loose shapes

- **Tag attributes:** `ref`, `variant`, `cols`, `color`, `icon`. On the item tag: `ref`, `variant`, `color`, `icon`
- **Parts:** `tag.` (Text), `title.` (Text), `body.` (Text), `icon.` (Icon), `number.` (Text), `band.` (Box), `bullets.` (Box), `divider.` (Rule), `item.` (Box)
- **Native attributes passed on:** none. **Code-owned, dropped if written:** none
- **Recipes:** `column`, `filled`, `outline`, `toned`, `accent_top`
- **Aliases:** `lifted` → `grid`, `rail` → `steps`, `column` → `grid`, `filled` → `grid`, `outline` → `grid`, `toned` → `grid`, `accent_top` → `grid`, `numerals` → `grid numbered`, `inverted` → `grid`

`bullet_list: <Ul ref>`

`native` is the default and the end of every fallback. `spacing="tight"` keeps points at the top, and fit-grow splits 4+ short points into two real lists. `icons` works on `rows`, up to 6 points.

Variant	What it draws	Needs → else	Ignores (reported if written)	In the `.pptx`
`native`	a real editable list: bold lead-ins, accent markers, points spaced to fill its box; 4+ short points split into two columns by themselves	nothing	none	real list (one text box)
`native numbered`	the same list, auto-numbered	`no_own_numbers` → `native`	none	real list (one text box)
`rows`	one ruled row per point: a dot (or an icon), the head over the detail	nothing	none	loose shapes
`rows numbered`	rows with 01 02 03 leading each	`no_own_numbers` → `rows`	none	loose shapes

- **Tag attributes:** `ref`, `variant`, `icons`, `spacing`
- **Parts:** `head.` (Run), `detail.` (Run), `marker.` (Marker), `item.` (Box)
- **Native attributes passed on:** `color`, `textAlign`, `italic`. **Code-owned, dropped if written:** none
- **Recipes:** none
- **Aliases:** `plain` → `native`, `icon` → `rows`, `columns` → `native`

`table: <Table ref>`

`native` is the only real PowerPoint table, so a person can drag its columns. Pills and bars work only on `rows` and `ledger`; `icons` only on `ledger`.

Variant	What it draws	Needs → else	Ignores (reported if written)	In the `.pptx`
`native`	POM's table: solid header band, striped rows, no borders, bold first column, the hinted row / column tinted	nothing	`alignItems`, `gap`, `justifyContent`	real table
`rows`	one ruled row per table row: the first cell as its title, the other cells to the right	nothing	`cellBorder.color`, `cellBorder.dashType`, `cellBorder.width`, `stripe`	loose shapes
`ledger`	one solid header band (an icon over each column name) over joined, tinted rows	`columns_le_6` → `native`	`cellBorder.color`, `cellBorder.dashType`, `cellBorder.width`, `stripe.bold`, `stripe.color`, `stripe.italic`, `stripe.textAlign`, `stripe.underline`	loose shapes

- **Tag attributes:** `ref`, `variant`, `icons`
- **Parts:** `header.` (Cell), `label.` (Cell), `cell.` (Cell), `stripe.` (Cell), `item.` (Box)

- **Native attributes passed on:** `cellBorder.color`, `cellBorder.width`, `cellBorder.dashType`. **Code-owned, dropped if written:** `defaultRowHeight`, `labels`
- **Recipes:** `borderless`, `bordered`, `dark_header`, `accent_edge`, `lifted`
- **Aliases:** `plain` → `native`, `rows numbered` → `rows`, `columns` → `native`, `banded` → `native`, `grid` → `native`, `grid numbered` → `native`, `highlight_row` → `native`, `zebra` → `native`, `dark_header` → `native`

Names that no longer exist

Table and card names still work as aliases, so saved slides keep drawing.

Removed	Now draws	Why
Table `rows numbered`	`rows`	Hand-editability: only `native` is a real table (user, 2026-11-04)
Table `columns`, `banded`, `grid`, `grid numbered`	`native`	Same
Card `rail`	`steps` (`grid` on a grid plan)	Spilled from 4 cards (user, 2026-11-04)
Bullet `grid`, `banded`	Unknown, so the default	Removed before merging, as duplicates of the card grid (user, 2026-11-04)
`process_arrow`, `flow` nodes	Hand-written POM	See A.2

Older names still read: KPI `plain`, `filled`, `toned`, `bar`, `dot`, `bare`, `inverted`, `divided`, and card `lifted`, `column`, `filled`, `outline`, `toned`, `accent_top`, `numerals`, `inverted`. `arrange` and `layout` also map to compositions, and `<KpiGrid>` to `<KpiRow>` (`NODE_TAG_ALIAS`).

Never built

`Callout` and `SlideHeader` appear in `spec.DERIVED_TAGS` and `blocks`, but no expander exists. Narratives and titles are written by hand. Matrix `grid` was rendered only.

A.6 Who owns what, and how style works

Owner	Controls
Code	Every word and number, verbatim and in plan order; which parts appear and their order; type sizes; word fit; the emphasis item; ids
LLM	Layout around the tag; `w`, `h`, `grow`, `min*`, `max*`, `alignSelf`; the variant; every style attribute

Later layers win: **composition `look` → `tone` → `emphasis` → the hint's or alias's recipe → group attributes → item tag.**

Written as	Reaches	Example
A box attribute on the group	The group's own box only, no cascade	`backgroundColor="$surfaceAlt"`
`item.<box attribute>` on the group	Every item's box	`item.borderLeft.color="$tone"`

Written as	Reaches	Example
`<part>.<attribute>` on the group	That part in every item	`title.color="$toneText"`
Either, on an item tag	That one item	`<Card ref="levers.1" title.color="$negativeText"/>`

Part attribute types (`compose.part_attrs`):

Type	Attributes
Text	`color`, `bold`, `italic`, `underline`, `letterSpacing`, `textAlign`, `textGradient`, `glow.size`, `glow.opacity`, `glow.color`, `outline.size`, `outline.color`
Icon	`name`, `color`, `variant`, `bgColor`, `glow.size`, `glow.opacity`, `glow.color`, `outline.size`, `outline.color`
Shape	`shapeType`, `w`, `h`, `fill.color`, `fill.transparency`, `line.color`, `line.width`, `line.dashType`, `glow.size`, `glow.opacity`, `glow.color`
Rule	`color`, `width`
Cell	`backgroundColor`, `color`, `bold`, `italic`, `underline`, `textAlign`
Run	`color`, `bold`, `italic`, `underline`
Marker	`color`
Box	the box attributes below

Box attributes: `backgroundColor`, `backgroundGradient`, `border`, `border.color`, `border.width`, `border.dashType`, `borderTop.color`, `borderTop.width`, `borderRight.color`, `borderRight.width`, `borderBottom.color`, `borderBottom.width`, `borderLeft.color`, `borderLeft.width`, `padding`, `padding.top`, `padding.right`, `padding.bottom`, `padding.left`, `gap`, `alignItems`, `justifyContent`, `shadow.type`, `shadow.blur`, `shadow.offset`, `shadow.color`, `shadow.opacity`, `opacity`, `w`, `h`, `grow`, `minW`, `maxW`, `minH`, `maxH`, `alignSelf`, `margin`.

Colours are `$tokens` only. A typed hex is removed (`LITERAL_COLOR`, a warning), so a theme switch restyles the slide. `$tone`, `$toneSoft`, `$toneText` and `$toneDark` follow each item's plan tone: good is `$positive*`, bad is `$negative*`, watch is `$warning*`, and none is `$accent*`. Charts and timelines have no item tones, so there `$tone*` becomes the accent, with a report.

Dropped with a report, never followed: `fontSize`, `fontFamily`, `lineHeight`, `text`, `size`, `highlight` on parts; `position`, `top`, `right`, `bottom`, `left`, `zIndex`, `id` on boxes; and each node's code-owned attributes.

Style recipes (`styles:`) are prompt examples, not values the tag accepts as a recipe name. The LLM can write any allowed combination.

A.7 Validation codes

An **error** goes to the repairer, which costs a call. A **warning** or **report** is logged and the slide proceeds.

Code	Level	Trigger
`NODE_REF_UNKNOWN`	Error	Bad id or index; a tag on a kind with no node; `<Text ref>` on a card or with field `delta`; a tag from another route
`NODE_REF_DUPLICATE`	Error	The same group or item placed twice
`NODE_ITEM_MISSING`	Error	A drawn component left out, nested in another reference, or missing items
`NODE_ATTR_UNKNOWN`	Error	An invented attribute or part (`foo="1"`, `nosuch.color`)
`NODE_HIGHLIGHT_UNKNOWN`	Error	A `<Highlight>` target or attribute not in the plan or grammar
`NODE_LABEL_MISSING`	Error	A label typed by hand in a tile, where it could sit beside the wrong value
`NODE_XML_INVALID`	Error	A slide with KPI references that cannot be parsed
`NODE_VARIANT_UNKNOWN`, `NODE_VARIANT_UNMET`	Warning	Unknown variant; `requires` failed, fallback drawn
`LITERAL_COLOR`	Warning	A typed colour
`NODE_HIGHLIGHT_LIMIT`, `NODE_HIGHLIGHT_RULE`	Warning	More than 6 marks (all drawn); a value rule on cells without one unit
`NODE_LABEL_PLACED`, `NODE_FIELD_MISSING`	Warning	Code placed a label; a field reference left out
`NODE_ATTR_DROPPED`, `NODE_ATTR_NO_EFFECT`	Report	Code-owned attribute removed; the variant does not draw it
`NODE_ATTR_ALIAS`, `NODE_TAG_ALIAS`	Report	An old name was read
`NODE_DEFAULT_FROM_CONTENT`, `NODE_LOOK_FROM_HINT`, `NODE_TEXT_SPLIT`, `NODE_VERTICAL_ROWS`	Report	Code chose a default; a hint named the look; a label split into head and detail; a vertical timeline drew `rows`

A.8 Evidence

Measured at this commit (free):

Check	Result
Every variant compiled on a sample plan	28 of 28
Attribute × variant pairs that silently do nothing	0 of 3,904 on five nodes; `bullet_list` 0 flagged of 344 on its own probe, after discarding on/off values equal to the default
Unit tests	2120 pass, with the 4 known failures. Left out: one new test in `b1d1328` that leaks `POM_NODES` and fails 6 later tests; the next local commit, `b2ed3c5`, fixes it

Free blind marks. The user's marks on bare tags against the LLM's own drawing, with no LLM writing the tags. The gate is at least 0.5, and by the user's rule a tie counts as satisfied. The other two columns show how much rests on that rule.

Node	Node wins	LLM wins	Same	Ties satisfied	Ties split	Ties left out	Source (`docs/eval/`)
`kpi_row`, `card_grid`	11	2	26	0.949	0.615	0.846	`composition-convention/replay.md`
`chart`	3	**7**	43	0.868	**0.462**	**0.300**	`chart-node/replay.md`
`timeline`	14	3	6	0.870	0.739	0.824	`sequence-nodes/replay.md`
`table`	10	2	54	0.970	0.561	0.833	`table-node/replay.md`
`bullet_list`	17	5	6	0.821	0.714	0.773	`bullet-node/replay.md`

Card stress review (8 variants × 8 cases, `docs/syntax-audit-2026-11-04.md` §7): 33 pass, 23 warn, 8 break, and all 8 breaks are the 280 px box. The other nodes had the wider audit run before this round's changes, not a re-run.

Paid same-plan check (22 saved plans, each drawn by routes A and B, first run 2026-11-04, $1.40, `docs/eval/nodes-paid/README.md`):

	Route A	Route B
Compiled first try	21 of 22	16 of 22
Repair passes	1	10
Placeholder slides	0	1
Plan words lost	0	12, on 2 slides
Cost	$0.675	$0.726
Blind marks	4	8 (10 the same)

So the first run did **not** meet the gate (no words lost, no placeholder, B wins the marks). Five causes were fixed, in `05a69db` and `98e800d`, and the 6 failed slides were set for a rerun. See A.9 item 1 for that rerun's status.

Not yet run: the full-deck run on fresh plans (`docs/eval/nodes-deck/README.md`, about $1.05, stops at $2). It is the first test of the planner's timeline `description` field and of the real repair cost.

A.9 Open points for review

Evidence

1. **The rerun's result is not in the repo.** `docs/eval/nodes-deck/README.md` says the same-plan check passed the gate; `docs/eval/nodes-paid/README.md` still says "gate not met yet", and `0aaeab5` ("gate confirmed") only records the gate's wording. No `nodes-b6` results exist in `docs/` or on the build PC. Until they are added, treat the pass as unverified.
2. `chart` **passes only because ties count as satisfied.** The LLM won 7 of the 10 decisive pairs (0.462 with ties split, 0.300 with ties left out). Every other node stays above 0.5 under all three rules.

3. **The silent-attribute probe proves an attribute reaches the XML,** not that it changes the picture.

Design

4. **Three variant vocabularies share one YAML.** `nodes` is what route B reads, `tile_nodes` is route A's, and `blocks` holds the 1a route's plus the prompt phrases for four nodes. Route A's code (`tiles.py`, `cards.py`) sits beside `tile_groups.py`, which reuses parts of it. Should route A go once B is the default?
5. **Code that production does not call.** `validate.check_skeleton`, `prompt_lines.py` and most `native.py` fillers are reached only from the 1a harnesses (`scripts/node_test.py`, `scripts/phase0b/expand.py`) and tests. In `native.py`, `x`, `number`, `chart`, `_direction` and `LAYOUT` are live.
6. **Words that break spec D2** ("a word means the same on every block"): timeline `rail` and `columns` draw differently from what the words meant on cards. Field `note` and part `delta.` also name the same thing.
7. **Should `NODE_ATTR_UNKNOWN` stay an error?** Code draws the block correctly without the attribute, so an invented name pays for a repair. The first paid run had 5 such first-try errors, since aliased.
8. **Only 5 of 28 variants stay real PowerPoint objects.** The user weighs hand-editing heavily; `rows` and `ledger` on tables were kept knowingly.

Known limits

9. **Narrow boxes.** All variants break in a 280 px box, and `requires` checks only content, never the box. `reflowNarrow` in fit-grow helps columns of code-drawn items, but a composition cannot switch at that point, because fit-grow runs after expansion. How often real slides get such a box is not measured.
10. **`rows_fit` assumes a full slide body.** Card `rows` beside another component can still overflow.
11. **`grid` spills on 5 to 6 cards with three bullets each.** Left unfixed on purpose: none of the 63 saved `grid` plans with 5+ cards has bullets. Re-count if the planner changes.
12. **Layers over derived blocks.** The LLM cannot know where a block ends after fit-grow. Anchors (Part B §13.2) are not built, and a missed `<Arrow>` id is only a compile warning.
13. **No result is recorded for `<Highlight>` written by a real generator on lists,** which the bullet node still owes.
14. **Route A lacks the attribute filter.** `POM_BLOCKS=tiles` without `POM_NODES` does not report attributes that do nothing.
15. **Visual polish:** a long tag on one `columns` card drops its pill below its peers; a `bento` box needs at least 500 px, stated in the prompt but not enforced.
16. **Still open from the audit:** `docs/syntax-audit-2026-11-04.md` §7 decisions 3, 6, 7, 8, 9 and 10, and whether callout and header stay hand-written.

A.10 Where the code is

Path	Holds
`src/knowledge/core/block-variants.yaml`	`nodes` (route B), `tile_nodes` (route A), `blocks`
`src/knowledge/core/nodes.yaml`, `attributes.yaml`	The native allowlist and attributes
`src/compiler/nodes/composed.py`	One interface for all six nodes; missing-node and wrong-kind checks

Path	Holds
`src/compiler/nodes/compose.py`	Shared attribute reading, `$tone`, the no-effect filter
`src/compiler/nodes/spec.py`	YAML loading, `requires` checks, `POM_NODES`
`charts.py`, `sequences.py`, `tile_groups.py`, `tables.py`, `bullets.py`	The six expanders
`element_refs.py`	Field references and `fill_refs`
`tiles.py`, `cards.py`	Route A's tile tags, parts reused by `tile_groups.py`
`src/agents/validator.py`	Where expansion runs, before `normalize_xml`
`src/node/fit-grow.js`	Type sizing, peer groups, regrid, reflow

The node tests, free and with no API key:

```
pytest tests/unit/test_block_variant_names.py tests/unit/test_tile_groups.py tests/unit/test_table_nodes.py tests/unit/test_bullet_nodes.py
tests/unit/test_sequence_nodes.py tests/unit/test_cards.py tests/unit/test_attribute_layer.py tests/unit/test_node_prompts.py -q
```

2. Figures

All 31 sheets in `docs/eval/syntax-audit/`. Each page is sized to its image, so zoom in to read a sheet at full resolution. In the PDF's bookmarks, each figure has its own entry.

Figure	File	Status	Group
1	`merged-all-variants.png`	Current	Current state
2	`card-variants.png`	Current	Current state
3	`rows-columns-after.png`	Current	Current state
4	`bento-after.png`	Current	Current state
5	`table-native-looks.png`	Current	Current state
6	`fixed-blank-kpi.png`	Current	Current state
7	`narrow-after.png`	Current	Current state
8	`bento-vs-flex.png`	Before the 2026-11-04 changes	Evidence for the decisions
9	`table-variants.png`	Before the 2026-11-04 changes	Evidence for the decisions
10	`atomic.png`	Older, still valid	Evidence for the decisions
11	`flex-probes.png`	Older, still valid	Evidence for the decisions
12	`flex-before-after.png`	Older, still valid	Evidence for the decisions
13	`duplicates.png`	Before the 2026-11-04 changes	Evidence for the decisions
14	`catalogue-kpi_row.png`	Before the 2026-11-04 changes	Audit catalogue: each variant as tag, styled, every attribute, atomic
15	`catalogue-card_grid.png`	Before the 2026-11-04 changes	Audit catalogue: each variant as tag, styled, every attribute, atomic
16	`catalogue-chart.png`	Before the 2026-11-04 changes	Audit catalogue: each variant as tag, styled, every attribute, atomic
17	`catalogue-timeline.png`	Before the 2026-11-04 changes	Audit catalogue: each variant as tag, styled, every attribute, atomic
18	`catalogue-table.png`	Before the 2026-11-04 changes	Audit catalogue: each variant as tag, styled, every attribute, atomic
19	`catalogue-bullets.png`	Before the 2026-11-04 changes	Audit catalogue: each variant as tag, styled, every attribute, atomic
20	`catalogue-callout.png`	Before the 2026-11-04 changes	Audit catalogue: each variant as tag, styled, every attribute, atomic
21	`catalogue-header.png`	Before the 2026-11-04 changes	Audit catalogue: each variant as tag, styled, every attribute, atomic
22	`stress-kpi_row.png`	Before the 2026-11-04 changes	Audit stress sheets
23	`stress-kpi_row-node.png`	Before the 2026-11-04 changes	Audit stress sheets

Figure	File	Status	Group
24	`stress-card_grid.png`	Before the 2026-11-04 changes	Audit stress sheets
25	`stress-card_grid-node.png`	Before the 2026-11-04 changes	Audit stress sheets
26	`stress-chart.png`	Before the 2026-11-04 changes	Audit stress sheets
27	`stress-chart-node.png`	Before the 2026-11-04 changes	Audit stress sheets
28	`stress-timeline.png`	Before the 2026-11-04 changes	Audit stress sheets
29	`stress-timeline-node.png`	Before the 2026-11-04 changes	Audit stress sheets
30	`stress-table.png`	Before the 2026-11-04 changes	Audit stress sheets
31	`stress-table-node.png`	Before the 2026-11-04 changes	Audit stress sheets

Figure 1 · Current

Every variant of the six derived nodes on one sample plan each: 28 compile, none loses a word, 25 pass and 3 warn. (merged-all-variants.png)

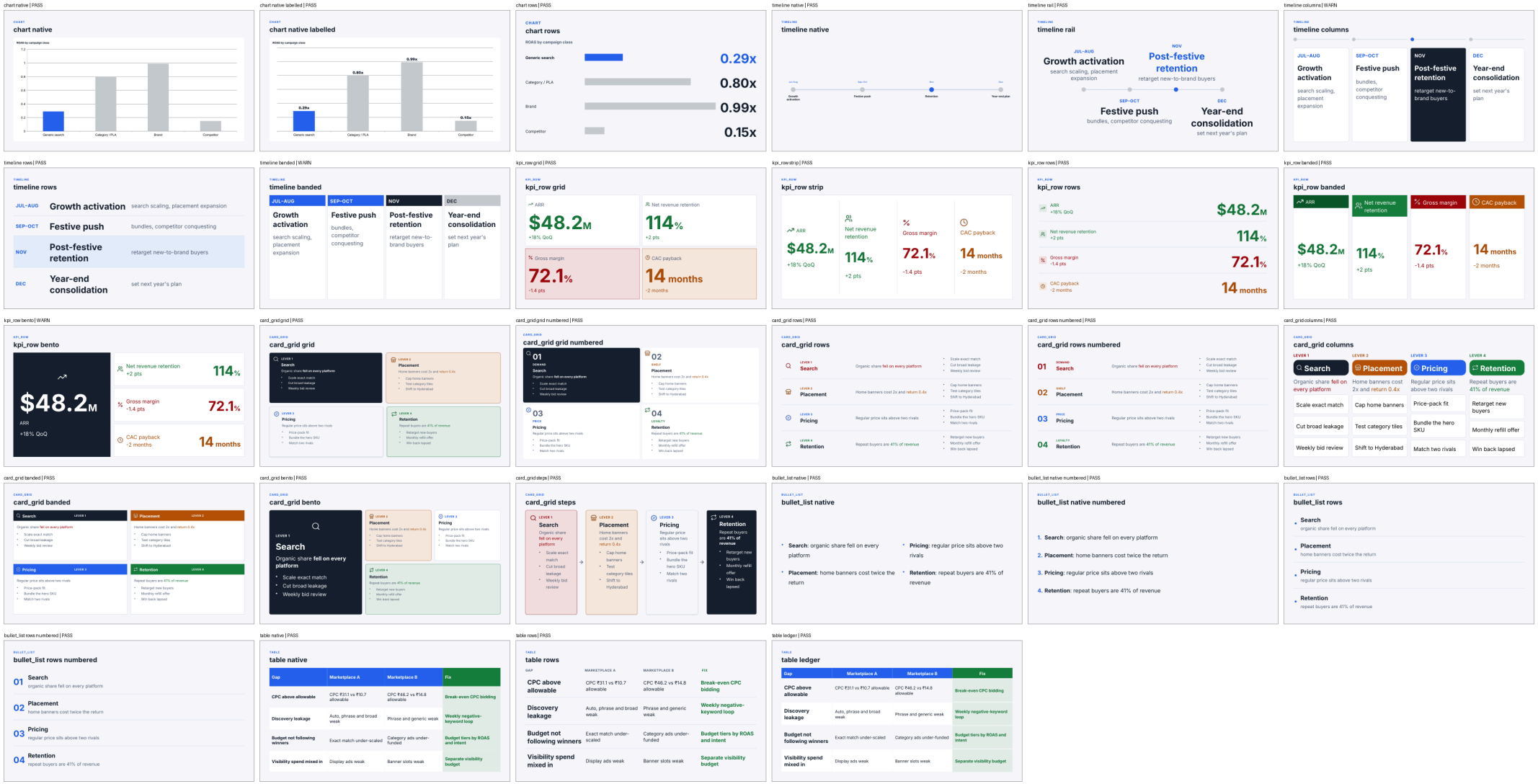


Figure 2 · Current

Card node: 8 variants x 8 stress cases, after the fixes. 33 pass, 23 warn, 8 break; all 8 breaks are the 280 px box. A warning is mostly a safe fallback, named in the frame's label. (card-variants.png)

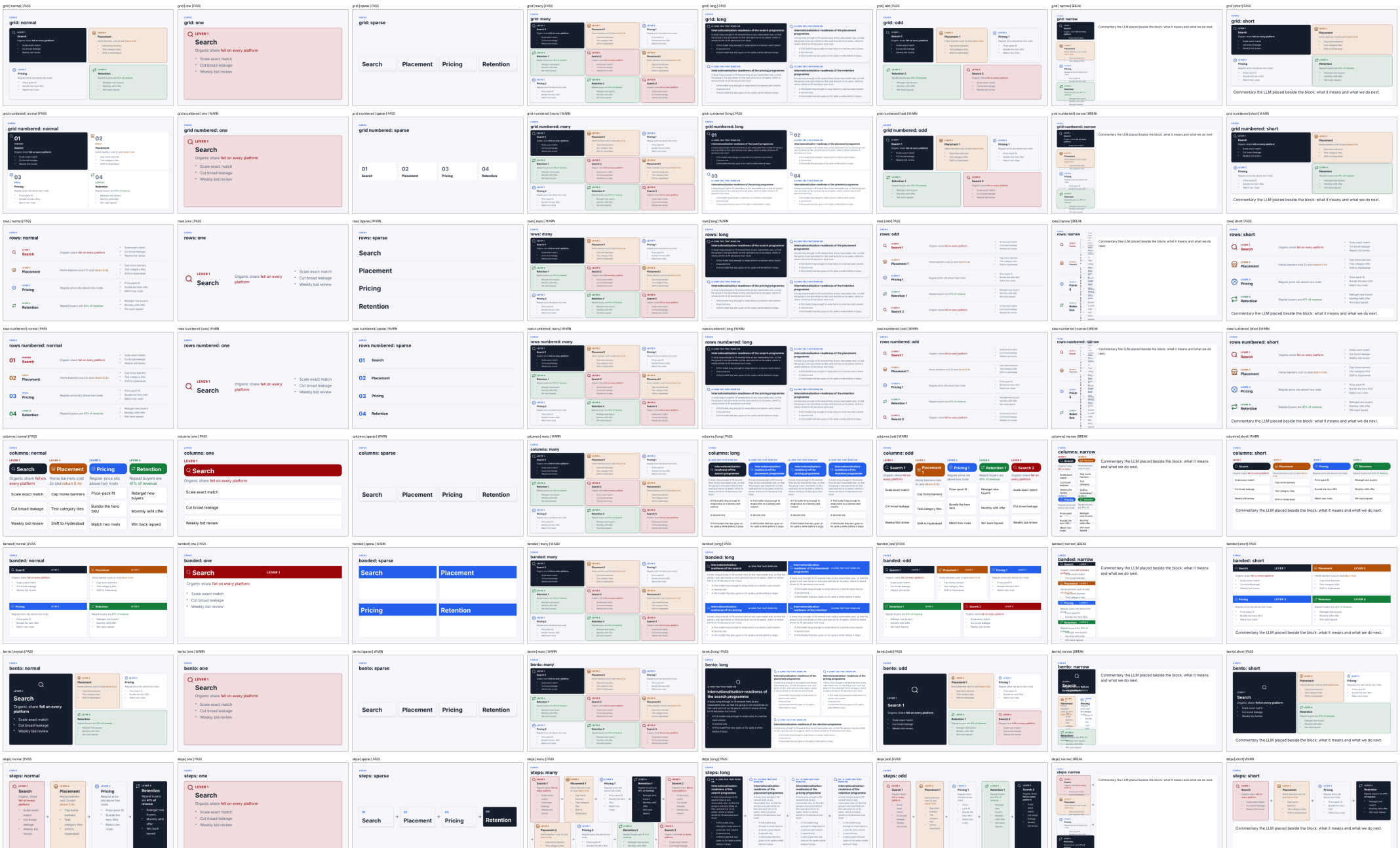
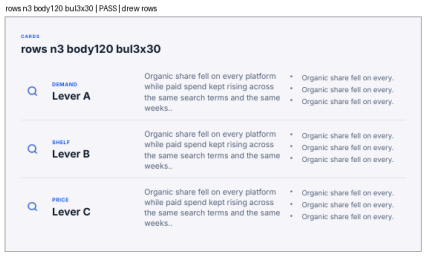


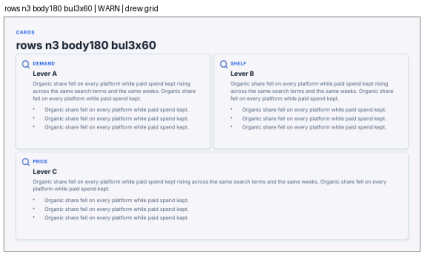
Figure 3 · Current

Card rows and columns near their limits. Each label says what was drawn. The BREAK slides fell back to grid, which also spills on 5 to 6 cards with three bullets each (A.9 item 11). (rows-columns-after.png)

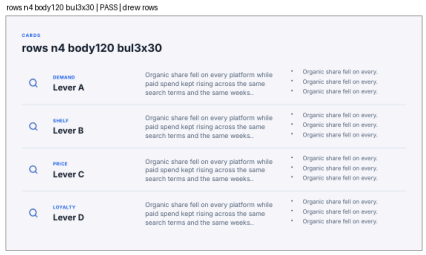
rows n3 body120 bul3x30 | PASS | drew rows



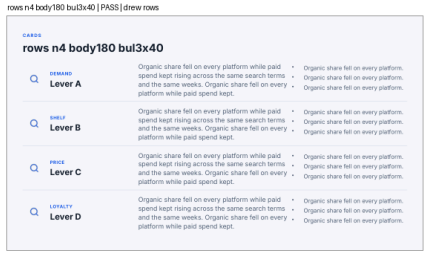
rows n3 body180 bul3x60 | WARN | drew grid



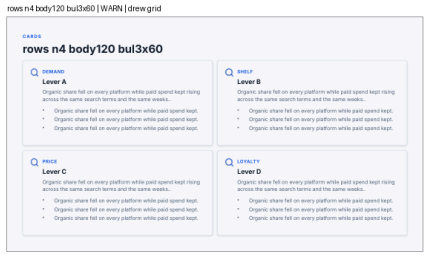
rows n4 body120 bul3x30 | PASS | drew rows



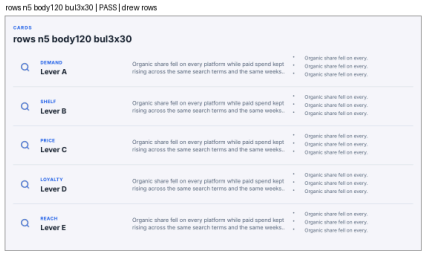
rows n4 body180 bul3x40 | PASS | drew rows



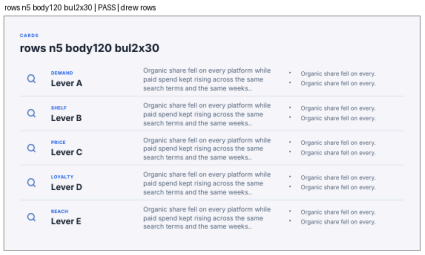
rows n4 body120 bul3x60 | WARN | drew grid



rows n5 body120 bul3x30 | PASS | drew rows



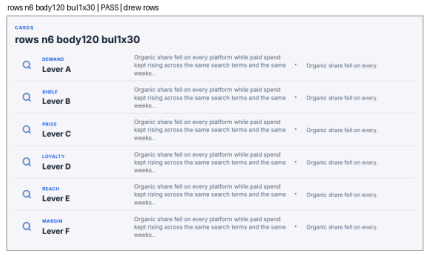
rows n5 body120 bul2x30 | PASS | drew rows



rows n5 body180 bul3x60 | BREAK | drew grid



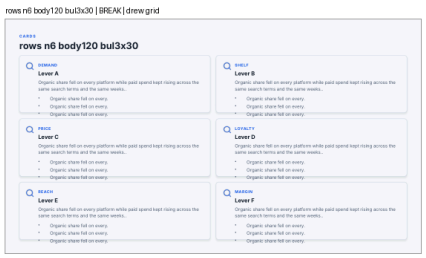
rows n6 body120 bul1x30 | PASS | drew rows



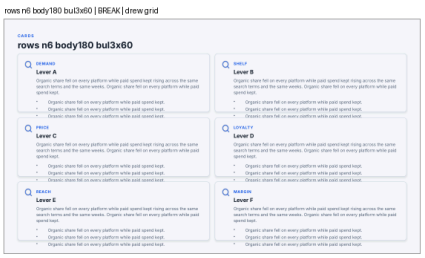
rows n6 body120 bul2x30 | PASS | drew rows



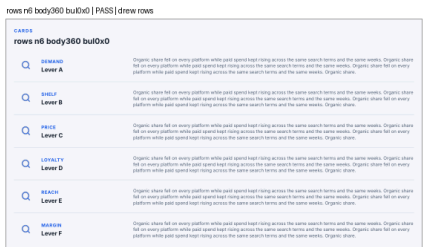
rows n6 body120 bul3x30 | BREAK | drew grid



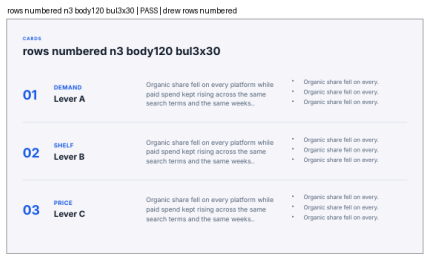
rows n6 body180 bul3x60 | BREAK | drew grid



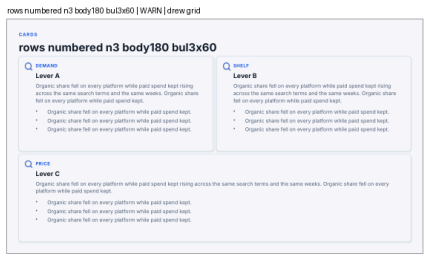
rows n6 body360 bul0x0 | PASS | drew rows



rows numbered n3 body120 bul3x30 | PASS | drew rows numbered



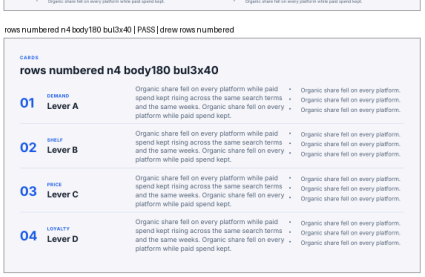
rows numbered n3 body180 bul3x60 | WARN | drew grid



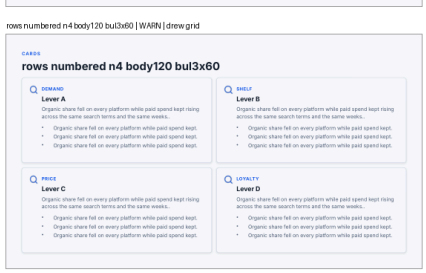
rows numbered n4 body120 bul3x30 | PASS | drew rows numbered



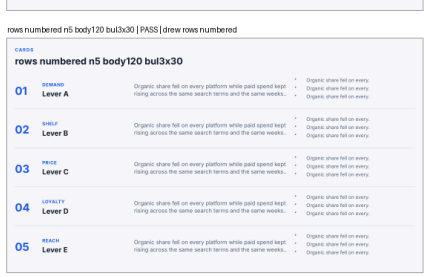
rows numbered n4 body180 bul3x40 | PASS | drew rows numbered



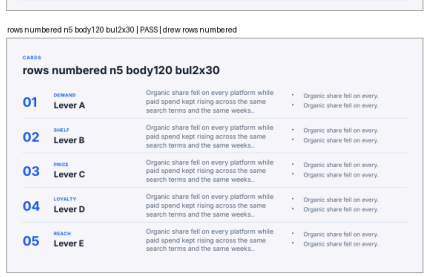
rows numbered n4 body120 bul3x60 | WARN | drew grid



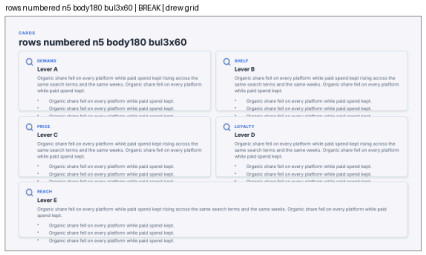
rows numbered n5 body120 bul3x30 | PASS | drew rows numbered



rows numbered n5 body120 bul2x30 | PASS | drew rows numbered



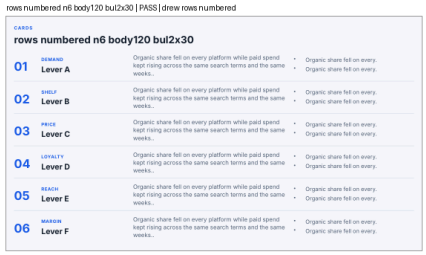
rows numbered n5 body180 bul3x60 | BREAK | drew grid



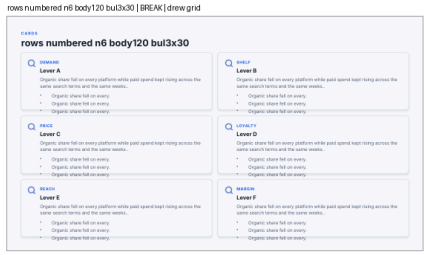
rows numbered n6 body120 bul1x30 | PASS | drew rows numbered



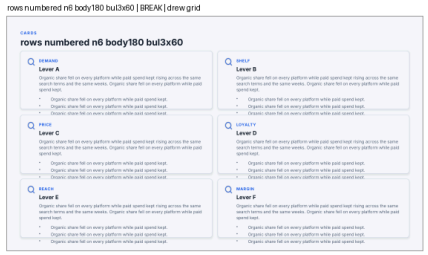
rows numbered n6 body120 bul2x30 | PASS | drew rows numbered



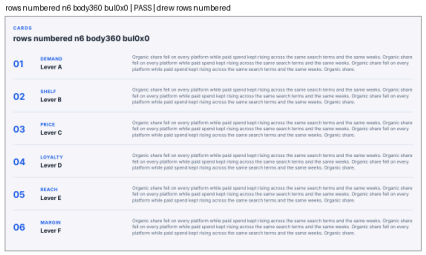
rows numbered n6 body120 bul3x30 | BREAK | drew grid



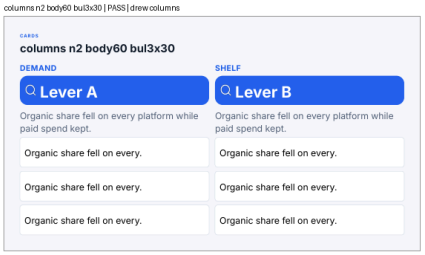
rows numbered n8 body180 bul3x60 | BREAK | drew grid



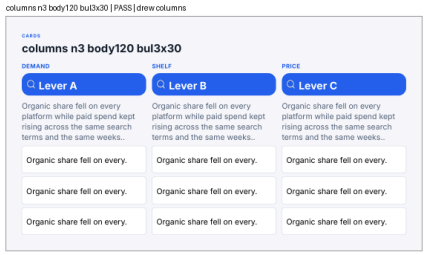
rows numbered n6 body360 bul0x0 | PASS | drew rows numbered



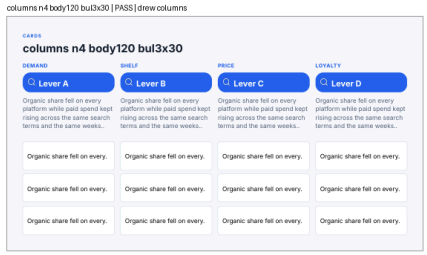
columns n2 body60 bul3x30 | PASS | drew columns



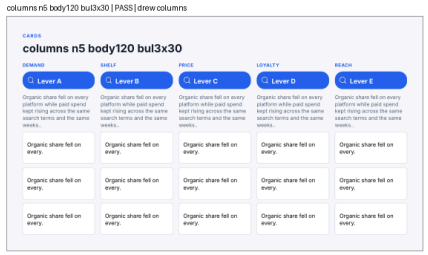
columns n3 body120 bul3x30 | PASS | drew columns



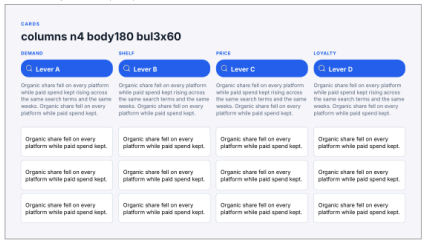
columns n4 body120 bul3x30 | PASS | drew columns



columns n5 body120 bul3x30 | PASS | drew columns



columns n4 body180 bul3x60 | PASS | drew columns



columns n5 body180 bul3x60 | PASS | drew columns



columns n4 one long tag | PASS | drew columns

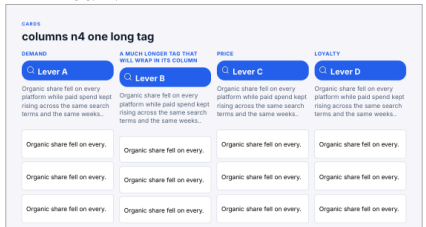


Figure 4 · Current

Card bento after small cards began drawing their tag. The last three frames show where it still overlaps: boxes 280 and 400 px wide, and long text on 5 cards. (bento-after.png)

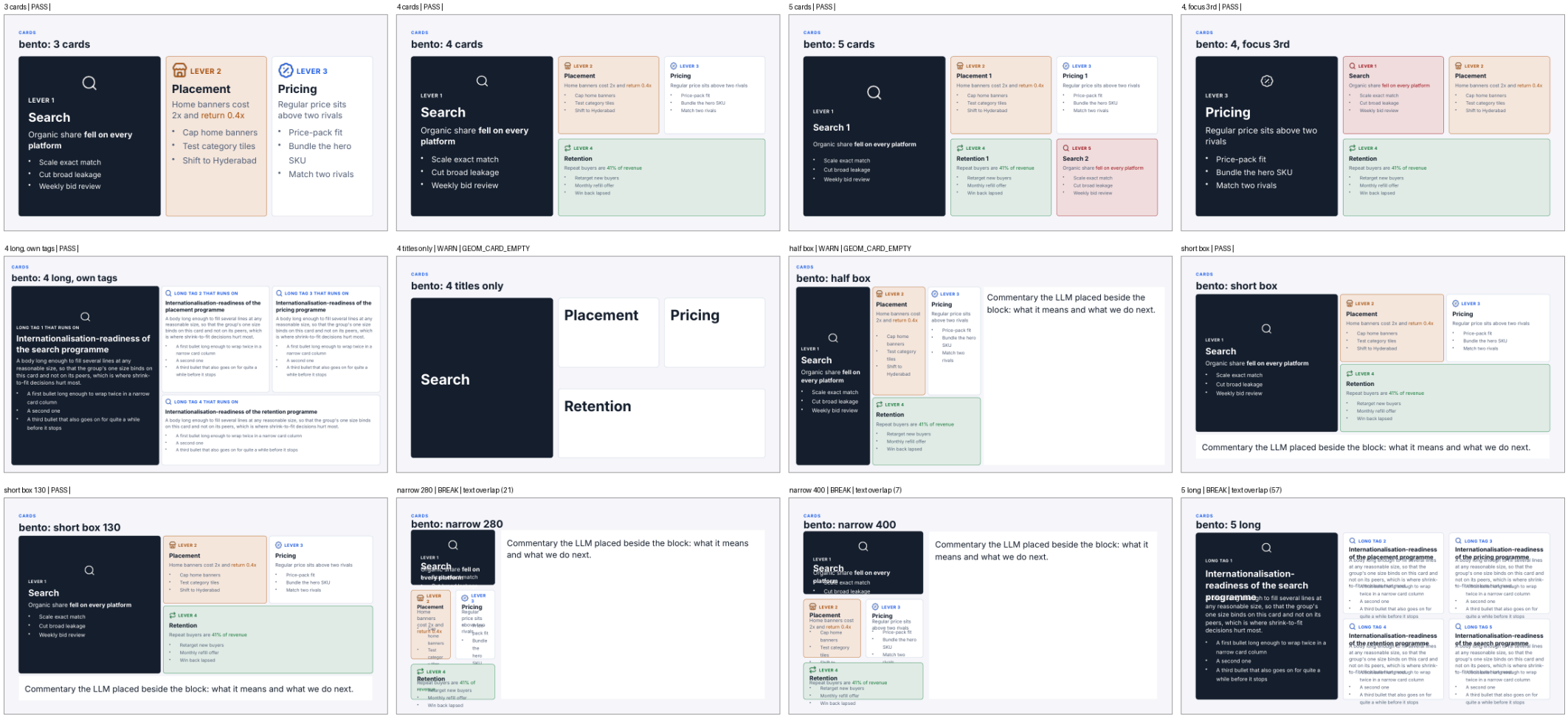


Figure 5 · Current

The native table, the only real PowerPoint table, in six looks reached by attributes alone. (table-native-looks.png)

native table: default | PASS

TABLE

native table: default

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

native table: zebra | PASS

TABLE

native table: zebra

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

native table: dark header | PASS

TABLE

native table: dark header

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

native table: borderless | PASS

TABLE

native table: borderless

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

native table: bordered | PASS

TABLE

native table: bordered

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

native table: bold first column, tinted last | PASS

TABLE

native table: bold first column, tinted last

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

Figure 6 · Current

Slides with a duplicate attribute: their KPI tiles used to come out blank, and now they fill. (fixed-blank-kpi.png)

KpiRow strip alignItems="center" (was blank)

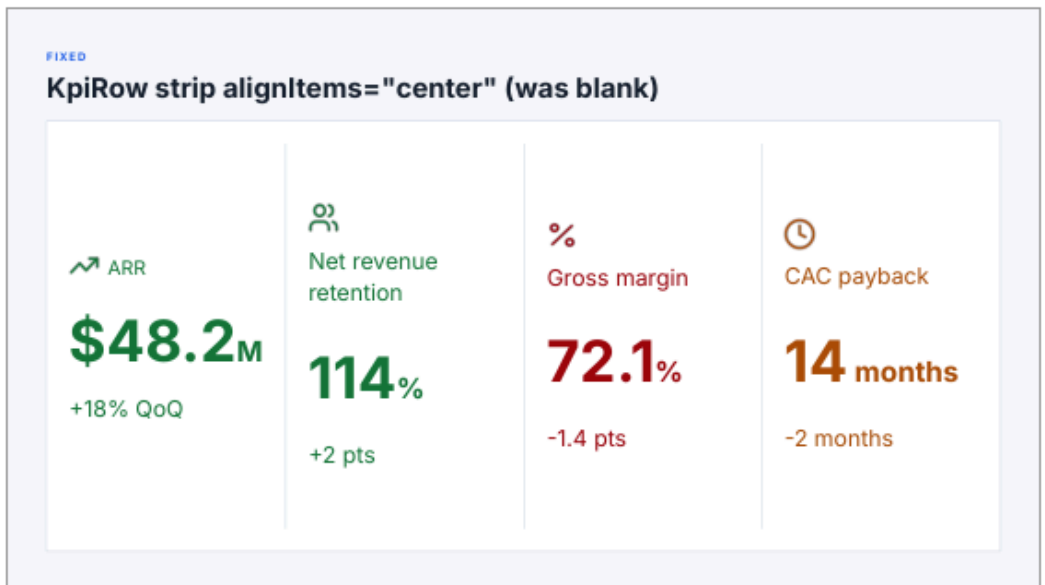


Table rows gap="20" beside a KpiRow (was blank)



Figure 7 · Current

Boxes 280 px wide, after fit-grow's reflow: all 30 cases now compile (25 before), but 24 still break and 6 warn. (narrow-after.png)

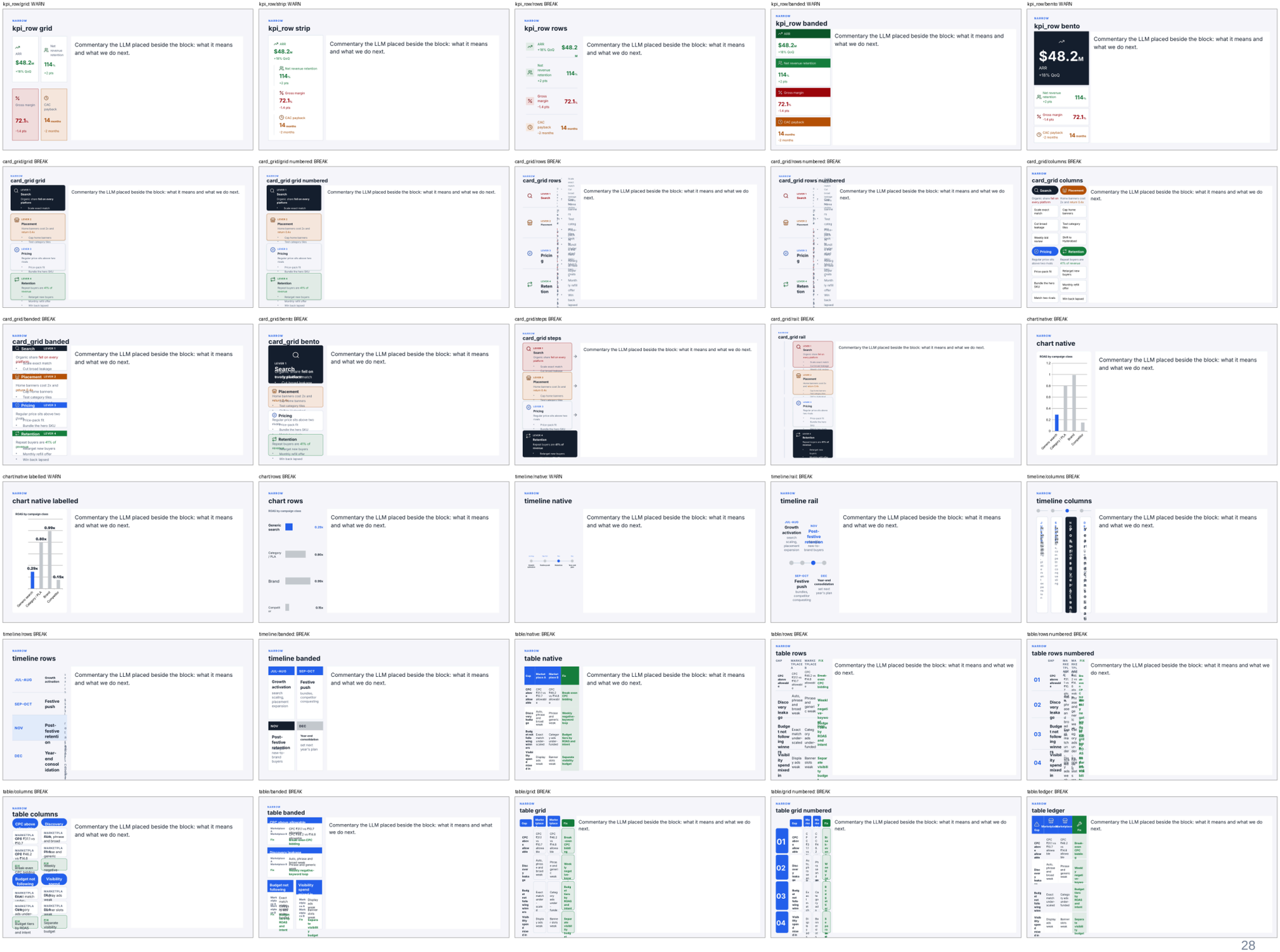


Figure 8 · Before the 2026-11-04 changes

Bento against flexbox placement around plain cards, on the same plans. Flexbox needs no new syntax, but its focus card gets no larger type. The bento column predates its tag fix, which is why it shows BREAK. (bento-vs-flex.png)



Figure 9 · Before the 2026-11-04 changes

All table variants under stress, and what each is inside the .pptx. Five were deleted after this sheet (A.5): only native is a real table. (table-variants.png)

native | normal | PASS | REAL TABLE

TABLE

native: normal

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

native | long | PASS | REAL TABLE

TABLE

native: long

Gap and its consequences for the plan	Internationalisation-readiness	Fix
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding

native | many | PASS | REAL TABLE

TABLE

native: many

Platform	Spend	ROAS	Target	Share	Note	Owner	Status
Platform 1	₹1.2L	1.8x	2.0x	3%	Scale	Team	On track
Platform 2	₹2.2L	2.8x	2.0x	6%	Scale	Team	On track
Platform 3	₹3.2L	3.8x	2.0x	9%	Scale	Team	On track
Platform 4	₹4.2L	4.8x	2.0x	12%	Scale	Team	On track
Platform 5	₹5.2L	5.8x	2.0x	15%	Scale	Team	On track
Platform 6	₹6.2L	6.8x	2.0x	18%	Scale	Team	On track
Platform 7	₹7.2L	7.8x	2.0x	21%	Scale	Team	On track
Platform 8	₹8.2L	8.8x	2.0x	24%	Scale	Team	On track
Platform 9	₹9.2L	9.8x	2.0x	27%	Scale	Team	On track
Platform 10	₹10.2L	10.8x	2.0x	30%	Scale	Team	On track
Platform 11	₹11.2L	11.8x	2.0x	33%	Scale	Team	On track
Platform 12	₹12.2L	12.8x	2.0x	36%	Scale	Team	On track

native | narrow | BREAK | REAL TABLE

TABLE

native: narrow

Commentary the LLM placed beside the block: what it means and what we do next.

rows | normal | PASS | 25 loose shapes

TABLE

rows: normal

GAP	MARKETPLACE A	MARKETPLACE B	FIX
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

rows | long | PASS | 25 loose shapes

TABLE

rows: long

GAP AND ITS CONSEQUENCES FOR THE PLAN	INTERNATIONALISATION-READINESS	FIX
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding

rows | many | PASS | 25 loose shapes

TABLE

rows: many

PLATFORM	SPEND	ROAS	TARGET	SHARE	NOTE	OWNER	STATUS
Platform 1	₹1.2L	1.8x	2.0x	3%	Scale	Team	On track
Platform 2	₹2.2L	2.8x	2.0x	6%	Scale	Team	On track
Platform 3	₹3.2L	3.8x	2.0x	9%	Scale	Team	On track
Platform 4	₹4.2L	4.8x	2.0x	12%	Scale	Team	On track
Platform 5	₹5.2L	5.8x	2.0x	15%	Scale	Team	On track
Platform 6	₹6.2L	6.8x	2.0x	18%	Scale	Team	On track
Platform 7	₹7.2L	7.8x	2.0x	21%	Scale	Team	On track
Platform 8	₹8.2L	8.8x	2.0x	24%	Scale	Team	On track
Platform 9	₹9.2L	9.8x	2.0x	27%	Scale	Team	On track
Platform 10	₹10.2L	10.8x	2.0x	30%	Scale	Team	On track
Platform 11	₹11.2L	11.8x	2.0x	33%	Scale	Team	On track
Platform 12	₹12.2L	12.8x	2.0x	36%	Scale	Team	On track

rows | narrow | BREAK | 25 loose shapes

TABLE

rows: narrow

Commentary the LLM placed beside the block: what it means and what we do next.

rows numbered | normal | WARN | 29 loose shapes

TABLE

rows numbered: normal

	GAP	MARKETPLACE A	MARKETPLACE B	FIX
01	CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
02	Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
03	Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
04	Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

rows numbered | long | PASS | 29 loose shapes

TABLE

rows numbered: long

	GAP AND ITS CONSEQUENCES FOR THE PLAN	INTERNATIONALISATION-READINESS	FIX
01	Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
02	Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
03	Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding

rows numbered | many | PASS | 29 loose shapes

TABLE

rows numbered: many

	PLATFORM	SPEND	ROAS	TARGET	SHARE	NOTE	OWNER	STATUS
01	Platform 1	₹1.2L	1.8x	2.0x	3%	Scale	Team	On track
02	Platform 2	₹2.2L	2.8x	2.0x	6%	Scale	Team	On track
03	Platform 3	₹3.2L	3.8x	2.0x	9%	Scale	Team	On track
04	Platform 4	₹4.2L	4.8x	2.0x	12%	Scale	Team	On track
05	Platform 5	₹5.2L	5.8x	2.0x	15%	Scale	Team	On track
06	Platform 6	₹6.2L	6.8x	2.0x	18%	Scale	Team	On track
07	Platform 7	₹7.2L	7.8x	2.0x	21%	Scale	Team	On track
08	Platform 8	₹8.2L	8.8x	2.0x	24%	Scale	Team	On track
09	Platform 9	₹9.2L	9.8x	2.0x	27%	Scale	Team	On track
10	Platform 10	₹10.2L	10.8x	2.0x	30%	Scale	Team	On track
11	Platform 11	₹11.2L	11.8x	2.0x	33%	Scale	Team	On track
12	Platform 12	₹12.2L	12.8x	2.0x	36%	Scale	Team	On track

rows numbered | narrow | BREAK | 29 loose shapes

TABLE

rows numbered: narrow

Commentary the LLM placed beside the block: what it means and what we do next.

columns | normal | BREAK | 46 loose shapes

TABLE

columns: normal

CPC above allowable	Discovery leakage	Budget not following winners	Visibility spend mixed in
MARKETPLACE A CPC ₹31.1 vs ₹10.7 allowable	MARKETPLACE A Auto, phrase and broad weak	MARKETPLACE A Exact match under-scaled	MARKETPLACE A Display ads weak
MARKETPLACE B CPC ₹46.2 vs ₹14.8 allowable	MARKETPLACE B Phrase and generic weak	MARKETPLACE B Category ads under-funded	MARKETPLACE B Banner slots weak
FIX Break-even CPC bidding	FIX Weekly negative-keyword loop	FIX Budget tiers by ROAS and intent	FIX Separate visibility budget

columns | long | BREAK | 46 loose shapes

TABLE

columns: long

Net revenue retention across enterprise subscription customers	Net revenue retention across enterprise subscription customers	Net revenue retention across enterprise subscription customers
INTERNATIONALISATION-READINESS A cell long enough to wrap several lines at any reasonable size in this column	INTERNATIONALISATION-READINESS A cell long enough to wrap several lines at any reasonable size in this column	INTERNATIONALISATION-READINESS A cell long enough to wrap several lines at any reasonable size in this column
FIX Break-even CPC bidding	FIX Break-even CPC bidding	FIX Break-even CPC bidding

columns | many | WARN (drew native) | 46 loose shapes

TABLE

columns: many

Platform	Spend	ROAS	Target	Share	Note	Owner	Status
Platform 1	₹1.2L	1.8x	2.0x	3%	Scale	Team	On track
Platform 2	₹2.2L	2.8x	2.0x	6%	Scale	Team	On track
Platform 3	₹3.2L	3.8x	2.0x	9%	Scale	Team	On track
Platform 4	₹4.2L	4.8x	2.0x	12%	Scale	Team	On track
Platform 5	₹5.2L	5.8x	2.0x	15%	Scale	Team	On track
Platform 6	₹6.2L	6.8x	2.0x	18%	Scale	Team	On track
Platform 7	₹7.2L	7.8x	2.0x	21%	Scale	Team	On track
Platform 8	₹8.2L	8.8x	2.0x	24%	Scale	Team	On track
Platform 9	₹9.2L	9.8x	2.0x	27%	Scale	Team	On track
Platform 10	₹10.2L	10.8x	2.0x	30%	Scale	Team	On track
Platform 11	₹11.2L	11.8x	2.0x	33%	Scale	Team	On track
Platform 12	₹12.2L	12.8x	2.0x	36%	Scale	Team	On track

columns | narrow | BREAK | 46 loose shapes

TABLE

columns: narrow

Commentary the LLM placed beside the block: what it means and what we do next.

banded | normal | BREAK | 42 loose shapes

TABLE

banded: normal

CPC above allowable		Discovery leakage	
Marketplace A	CPC ₹31.1 vs ₹10.7 allowable	Marketplace A	Auto, phrase and broad weak
Marketplace B	CPC ₹46.2 vs ₹14.8 allowable	Marketplace B	Phrase and generic weak
Fix	Break-even CPC bidding	Fix	Weekly negative-keyword loop
Budget not following winners		**Visibility spend mixed in**	
Marketplace A	Exact match under-scaled	Marketplace A	Display ads weak
Marketplace B	Category ads under-funded	Marketplace B	Banner slots weak
Fix	Budget tiers by ROAS and intent	Fix	Separate visibility budget

banded | long | BREAK | 42 loose shapes

TABLE

banded: long

Net revenue retention across enterprise subscription customers		Net revenue retention across enterprise subscription customers	
Internationalisation-readiness	A cell long enough to wrap several lines at any reasonable size in this column	Internationalisation-readiness	A cell long enough to wrap several lines at any reasonable size in this column
Fix	Break-even CPC bidding	Fix	Break-even CPC bidding

Net revenue retention across enterprise subscription customers	
Internationalisation-readiness	A cell long enough to wrap several lines at any reasonable size in this column
Fix	Break-even CPC bidding

banded | many | WARN (drew native) | 42 loose shapes

TABLE

banded: many

Platform	Spend	ROAS	Target	Share	Note	Owner	Status
Platform 1	₹1.2L	1.8x	2.0x	3%	Scale	Team	On track
Platform 2	₹2.2L	2.8x	2.0x	6%	Scale	Team	On track
Platform 3	₹3.2L	3.8x	2.0x	9%	Scale	Team	On track
Platform 4	₹4.2L	4.8x	2.0x	12%	Scale	Team	On track
Platform 5	₹5.2L	5.8x	2.0x	15%	Scale	Team	On track
Platform 6	₹6.2L	6.8x	2.0x	18%	Scale	Team	On track
Platform 7	₹7.2L	7.8x	2.0x	21%	Scale	Team	On track
Platform 8	₹8.2L	8.8x	2.0x	24%	Scale	Team	On track
Platform 9	₹9.2L	9.8x	2.0x	27%	Scale	Team	On track
Platform 10	₹10.2L	10.8x	2.0x	30%	Scale	Team	On track
Platform 11	₹11.2L	11.8x	2.0x	33%	Scale	Team	On track
Platform 12	₹12.2L	12.8x	2.0x	36%	Scale	Team	On track

banded | narrow | BREAK | 42 loose shapes

TABLE

banded: narrow

Commentary the LLM placed beside the block: what it means and what we do next.

grid | normal | PASS | 38 loose shapes

TABLE

grid: normal

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

grid | long | WARN | 38 loose shapes

TABLE

grid: long

Gap and its consequences for the plan	Internationalisation-readiness	Fix
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding

grid | many | WARN (drew native) | 38 loose shapes

TABLE

grid: many

Platform	Spend	ROAS	Target	Share	Note	Owner	Status
Platform 1	₹1.2L	1.8x	2.0x	3%	Scale	Team	On track
Platform 2	₹2.2L	2.8x	2.0x	6%	Scale	Team	On track
Platform 3	₹3.2L	3.8x	2.0x	9%	Scale	Team	On track
Platform 4	₹4.2L	4.8x	2.0x	12%	Scale	Team	On track
Platform 5	₹5.2L	5.8x	2.0x	15%	Scale	Team	On track
Platform 6	₹6.2L	6.8x	2.0x	18%	Scale	Team	On track
Platform 7	₹7.2L	7.8x	2.0x	21%	Scale	Team	On track
Platform 8	₹8.2L	8.8x	2.0x	24%	Scale	Team	On track
Platform 9	₹9.2L	9.8x	2.0x	27%	Scale	Team	On track
Platform 10	₹10.2L	10.8x	2.0x	30%	Scale	Team	On track
Platform 11	₹11.2L	11.8x	2.0x	33%	Scale	Team	On track
Platform 12	₹12.2L	12.8x	2.0x	36%	Scale	Team	On track

grid | narrow | BREAK | 38 loose shapes

TABLE

grid: narrow

Commentary the LLM placed beside the block: what it means and what we do next.

grid numbered | normal | PASS | 50 loose shapes

TABLE

grid numbered: normal

	Gap	Marketplace A	Marketplace B	Fix
01	CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
02	Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
03	Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
04	Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

grid numbered | long | WARN | 50 loose shapes

TABLE

grid numbered: long

	Gap and its consequences for the plan	Internationalisation-readiness	Fix
01	Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
02	Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
03	Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding

grid numbered | many | WARN (drew native) | 50 loose shapes

TABLE

grid numbered: many

Platform	Spend	ROAS	Target	Share	Note	Owner	Status
Platform 1	₹1.2L	1.8x	2.0x	3%	Scale	Team	On track
Platform 2	₹2.2L	2.8x	2.0x	6%	Scale	Team	On track
Platform 3	₹3.2L	3.8x	2.0x	9%	Scale	Team	On track
Platform 4	₹4.2L	4.8x	2.0x	12%	Scale	Team	On track
Platform 5	₹5.2L	5.8x	2.0x	15%	Scale	Team	On track
Platform 6	₹6.2L	6.8x	2.0x	18%	Scale	Team	On track
Platform 7	₹7.2L	7.8x	2.0x	21%	Scale	Team	On track
Platform 8	₹8.2L	8.8x	2.0x	24%	Scale	Team	On track
Platform 9	₹9.2L	9.8x	2.0x	27%	Scale	Team	On track
Platform 10	₹10.2L	10.8x	2.0x	30%	Scale	Team	On track
Platform 11	₹11.2L	11.8x	2.0x	33%	Scale	Team	On track
Platform 12	₹12.2L	12.8x	2.0x	36%	Scale	Team	On track

grid numbered | narrow | BREAK | 50 loose shapes

TABLE

grid numbered: narrow

Commentary the LLM placed beside the block: what it means and what we do next.

ledger | normal | PASS | 42 loose shapes

TABLE

ledger: normal

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

ledger | long | WARN | 42 loose shapes

TABLE

ledger: long

Gap and its consequences for the plan	Internationalisation-readiness	Fix
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding
Net revenue retention across enterprise subscription customers	A cell long enough to wrap several lines at any reasonable size in this column	Break-even CPC bidding

ledger | many | WARN (drew native) | 42 loose shapes

TABLE

ledger: many

Platform	Spend	ROAS	Target	Share	Note	Owner	Status
Platform 1	₹1.2L	1.8x	2.0x	3%	Scale	Team	On track
Platform 2	₹2.2L	2.8x	2.0x	6%	Scale	Team	On track
Platform 3	₹3.2L	3.8x	2.0x	9%	Scale	Team	On track
Platform 4	₹4.2L	4.8x	2.0x	12%	Scale	Team	On track
Platform 5	₹5.2L	5.8x	2.0x	15%	Scale	Team	On track
Platform 6	₹6.2L	6.8x	2.0x	18%	Scale	Team	On track
Platform 7	₹7.2L	7.8x	2.0x	21%	Scale	Team	On track
Platform 8	₹8.2L	8.8x	2.0x	24%	Scale	Team	On track
Platform 9	₹9.2L	9.8x	2.0x	27%	Scale	Team	On track
Platform 10	₹10.2L	10.8x	2.0x	30%	Scale	Team	On track
Platform 11	₹11.2L	11.8x	2.0x	33%	Scale	Team	On track
Platform 12	₹12.2L	12.8x	2.0x	36%	Scale	Team	On track

ledger | narrow | BREAK | 42 loose shapes

TABLE

ledger: narrow

Commentary the LLM placed beside the block: what it means and what we do next.

Figure 10 · Older, still valid

Composition, atomic with the same look, and atomic with a new look. Atomic reaches the same looks at 30 to 51 times the output tokens, and swapped fields pass unseen. (atomic.png)

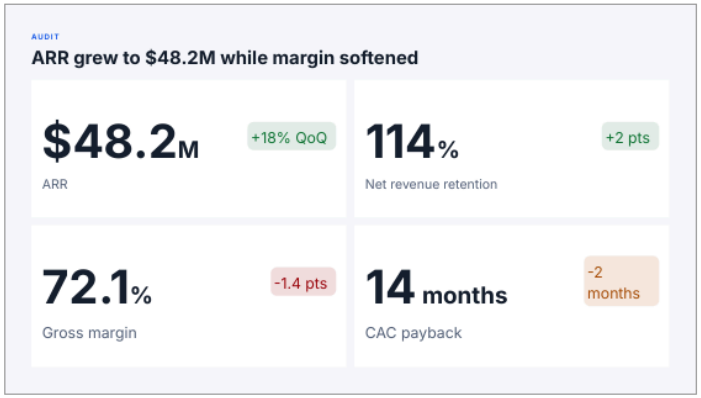
kpi_row composition



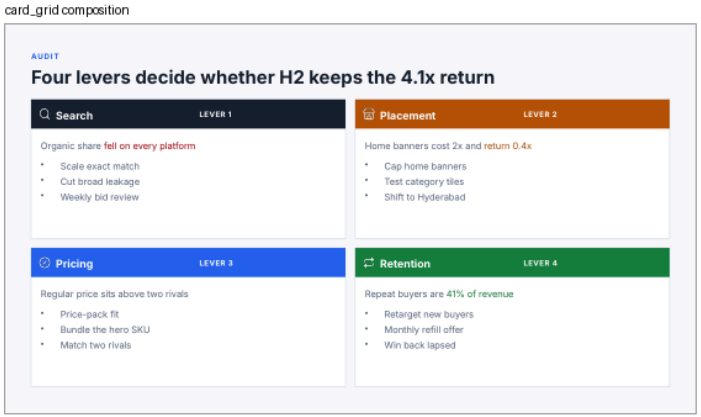
kpi_row atomic



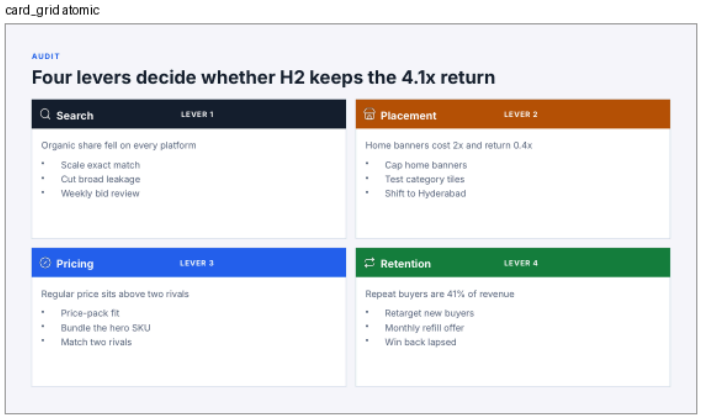
kpi_row atomic free



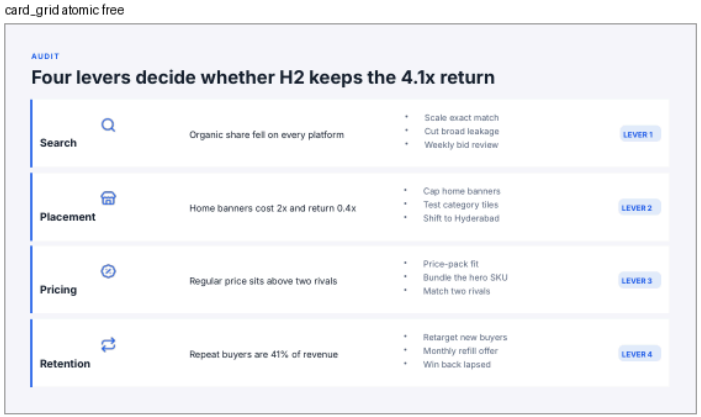
card_grid composition



card_grid atomic



card_grid atomic free



timeline composition



timeline atomic



timeline atomic free

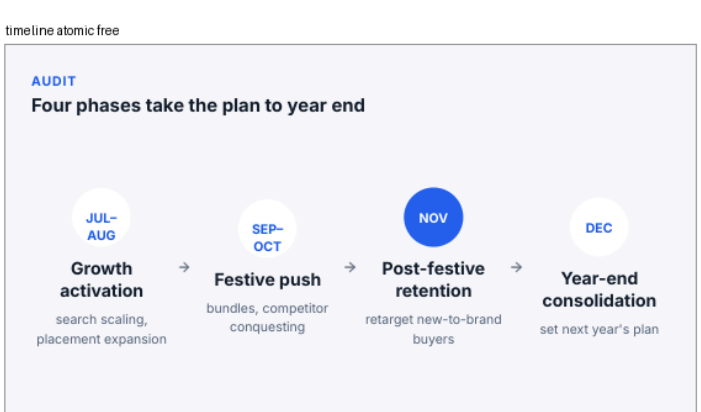


Figure 11 · Older, still valid

POM's flexbox behaviour, tested on small slides (audit section 6). (flex-probes.png)

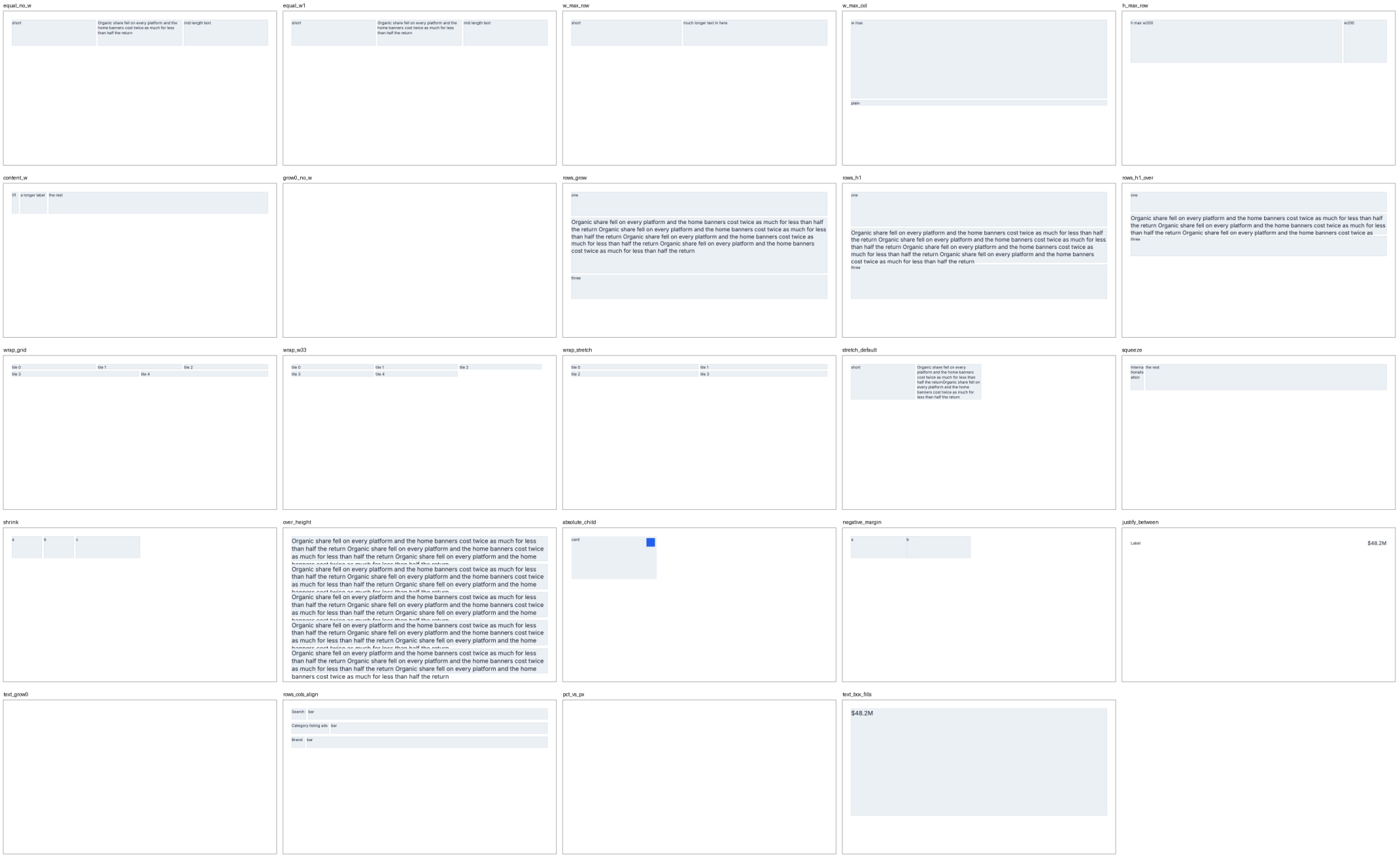


Figure 12 · Older, still valid

Flexbox changes tried: equal rows and others. Most were rejected on the render (audit section 6). (flex-before-after.png)

equal_rows_cards before

FLEX

Peers in a grid share a height

Search

Organic share fell on every platform

Placement

Home banners cost 2x and return 0.4x

- Cap home banners
- Review weekly
- Test category tiles
- Cut the worst slots
- Shift to Hyderabad

Pricing

Regular price sits above two rivals

Retention

Repeat buyers are 41% of revenue

equal_rows_cards after

FLEX

Peers in a grid share a height

Search

Organic share fell on every platform

Placement

Home banners cost 2x and return 0.4x

- Cap home banners
- Review weekly
- Test category tiles
- Cut the worst slots
- Shift to Hyderabad

Pricing

Regular price sits above two rivals

Retention

Repeat buyers are 41% of revenue

equal_rows_table before

FLEX

Peers in a table's rows share a height

Metric	H1	H2
Spend	₹31.2L	₹48L
A long row whose first cell runs on to fill two or three lines	2.8x	A longer second cell that wraps to several lines in its column as well
Share	41%	44%

equal_rows_table after

FLEX

Peers in a table's rows share a height

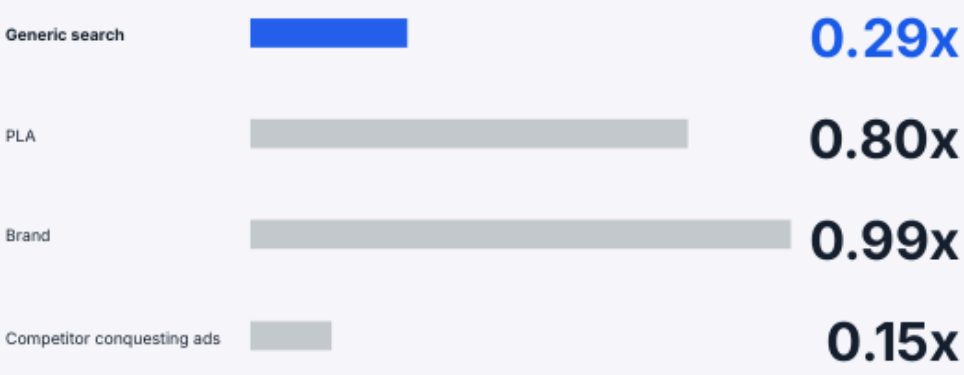
Metric	H1	H2
Spend	₹31.2L	₹48L
A long row whose first cell runs on to fill two or three lines	2.8x	A longer second cell that wraps to several lines in its column as well
Share	41%	44%

chart_rows_widths before

FLEX

Search takes the budget and returns the least

ROAS

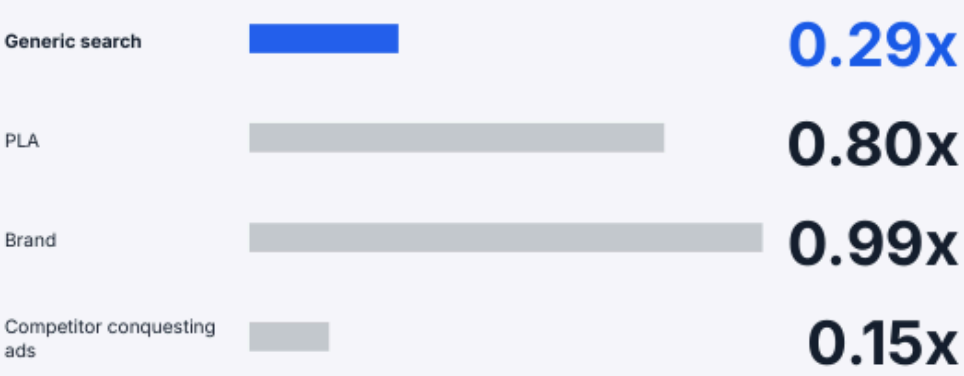


chart_rows_widths after

FLEX

Search takes the budget and returns the least

ROAS



drop_w1 before

FLEX

ARR grew to $48.2M while margin softened

ARR

$48.2M

+18% QoQ

Net revenue retention

114%

+2 pts

Gross margin

72.1%

-1.4 pts

CAC payback

14 months

-2 months

drop_w1 after

FLEX

ARR grew to $48.2M while margin softened

ARR

$48.2M

+18% QoQ

Net revenue retention

114%

+2 pts

Gross margin

72.1%

-1.4 pts

CAC payback

14 months

-2 months

centre_body before

A sparse slide centres its body by flexbox

One short line
A second line
A third line

centre_body after

A sparse slide centres its body by flexbox

One short line
A second line
A third line

delta_pill before

FLEX

ARR grew to $48.2M while margin softened

ARR

$48.2M

+18% QoQ

Net revenue retention

114%

+2 pts

Gross margin

72.1%

-1.4 pts

CAC payback

14 months

-2 months

delta_pill after

FLEX

ARR grew to $48.2M while margin softened

ARR

$48.2M

+18% QoQ

Net revenue retention

114%

+2 pts

Gross margin

72.1%

-1.4 pts

CAC payback

14 months

-2 months

band_height before

FLEX

ARR grew to $48.2M while margin softened

ARR	Net revenue retention	Gross margin	CAC payback
$48.2M	114%	72.1%	14 months
+18% QoQ	+2 pts	-1.4 pts	-2 months

band_height after

FLEX

ARR grew to $48.2M while margin softened

ARR	Net revenue retention	Gross margin	CAC payback
$48.2M	114%	72.1%	14 months
+18% QoQ	+2 pts	-1.4 pts	-2 months

Figure 13 · Before the 2026-11-04 changes

The bug behind the blank KPI slides: an attribute written by both the expander and the LLM. (duplicates.png)

KpiRow strip alignItems="center"



Table rows gap="20" beside a KpiRow



Figure 14 · Before the 2026-11-04 changes

Catalogue: kpi_row. These four columns are audit test arms, not options offered to the LLM. (catalogue-kpi_row.png)

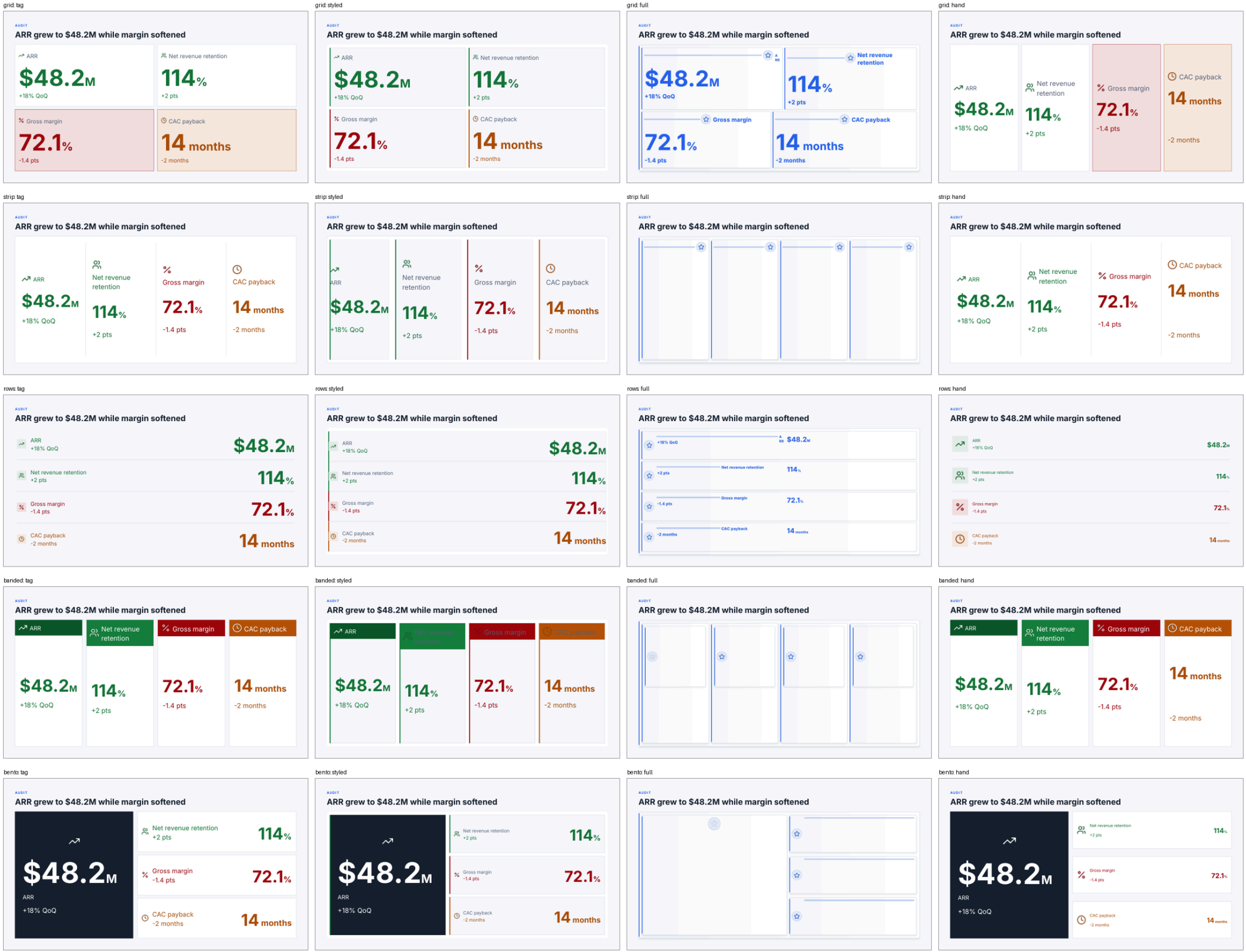
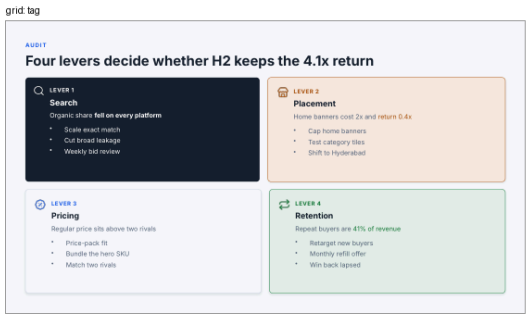


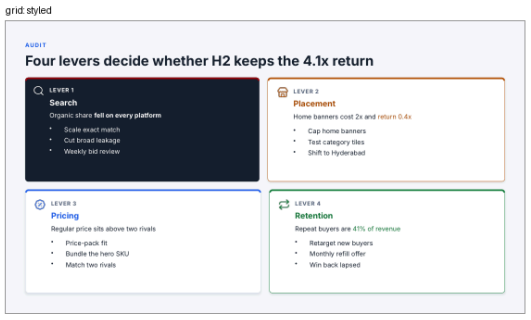
Figure 15 · Before the 2026-11-04 changes

Catalogue: card_grid. These four columns are audit test arms, not options offered to the LLM. (catalogue-card_grid.png)

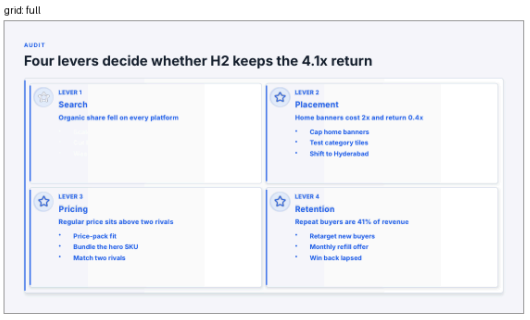
grid: tag



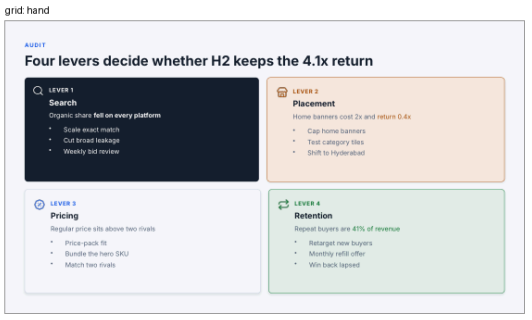
grid: styled



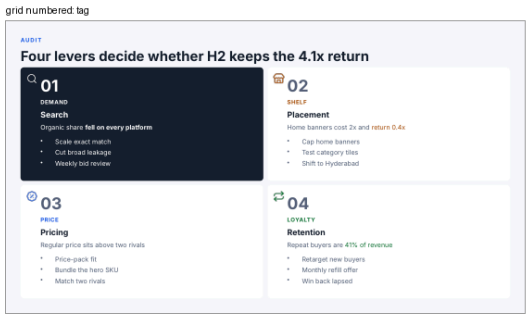
grid: full



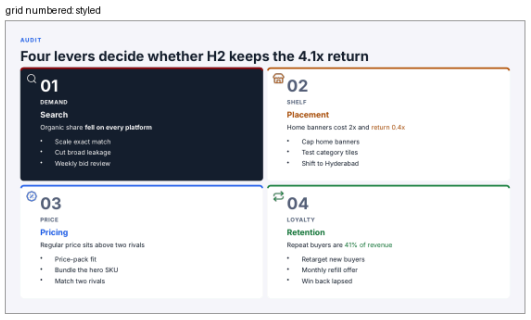
grid: hand



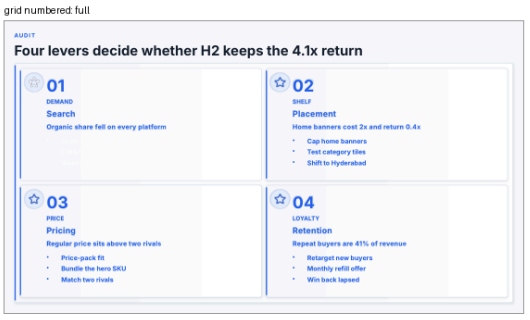
grid numbered: tag



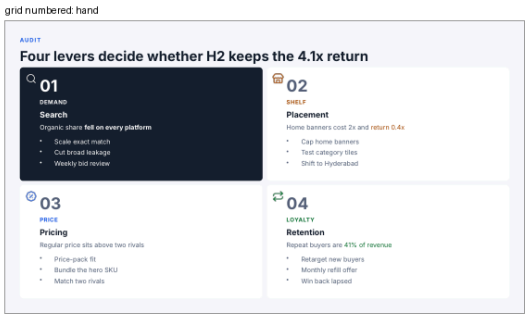
grid numbered: styled



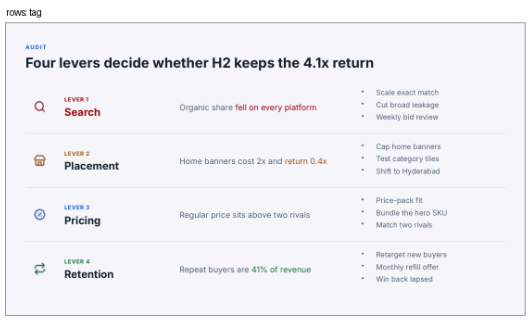
grid numbered: full



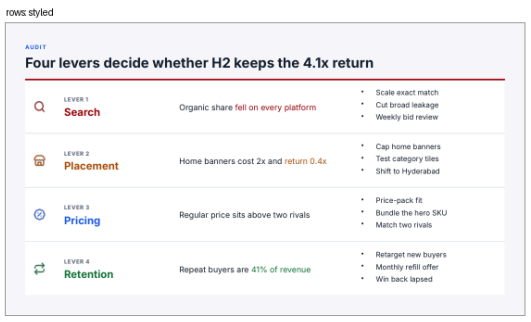
grid numbered: hand



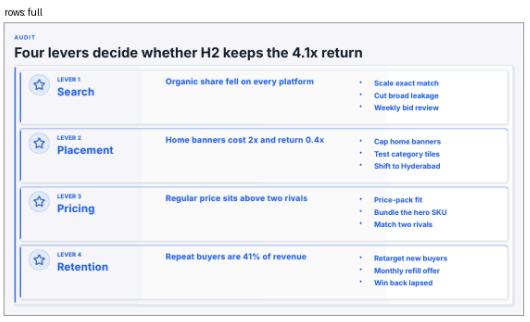
rows: tag



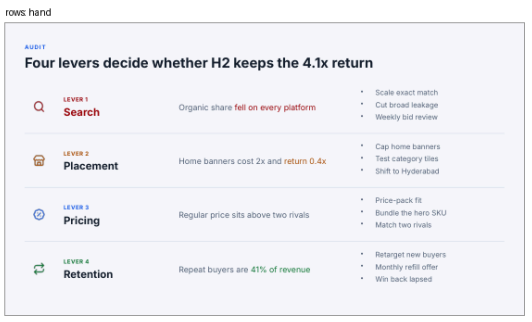
rows: styled



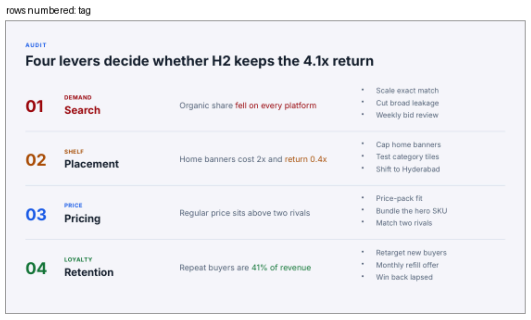
rows: full



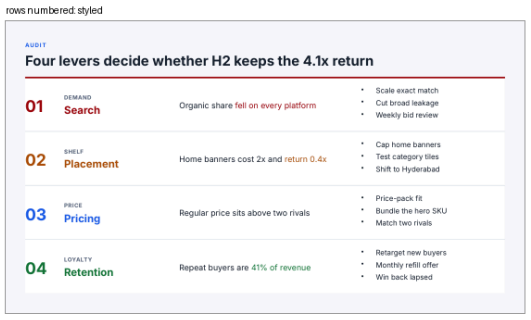
rows: hand



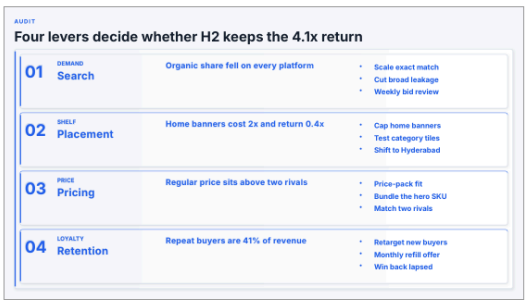
rows numbered: tag



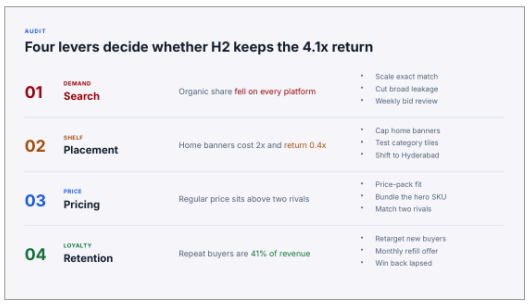
rows numbered: styled



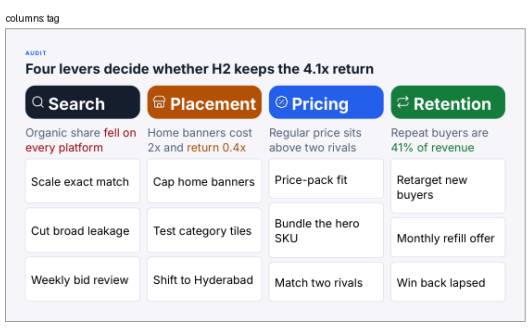
rows numbered: full



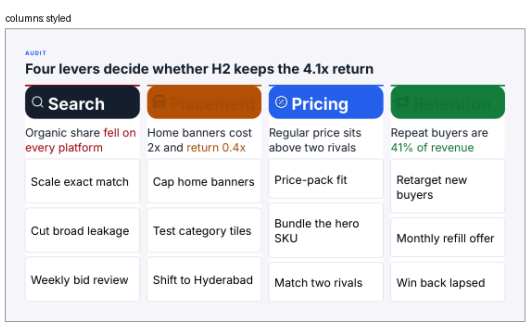
rows numbered: hand



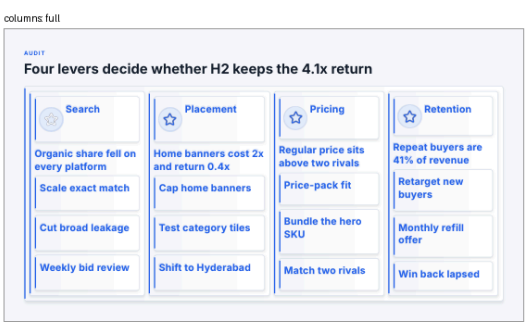
columns: tag



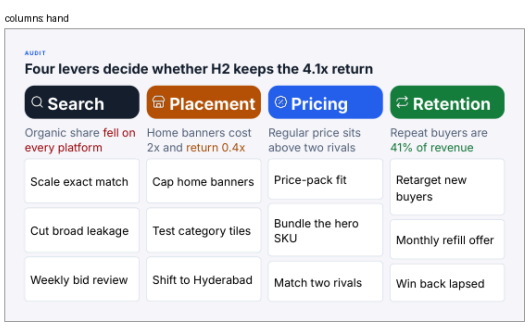
columns: styled



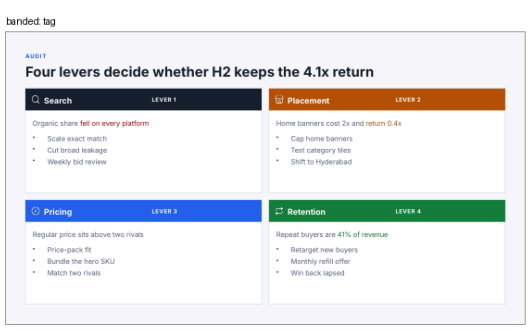
columns: full



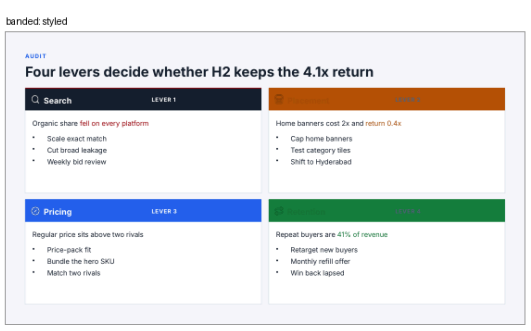
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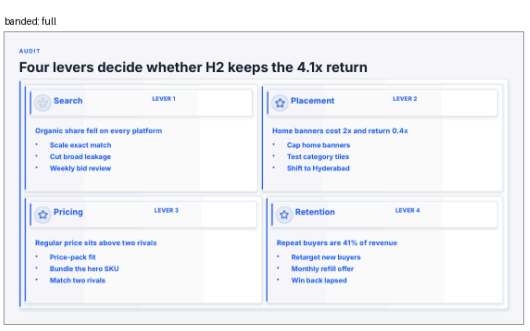
banded: tag



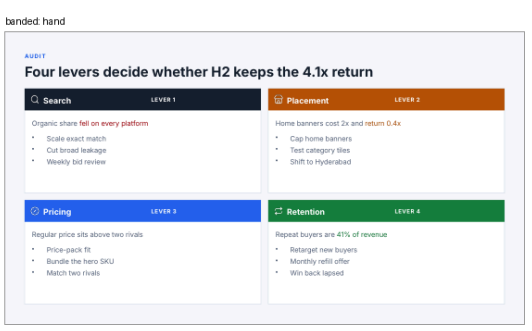
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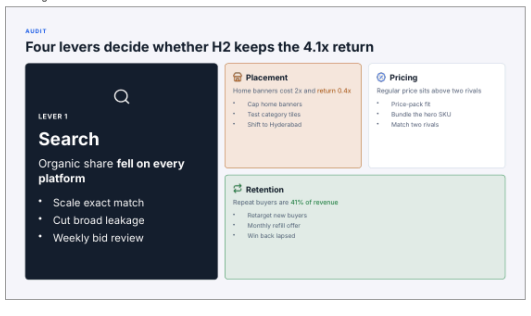
banded: full



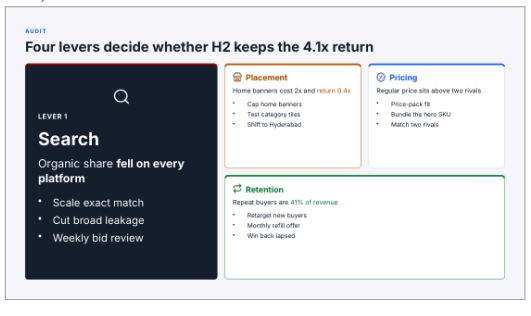
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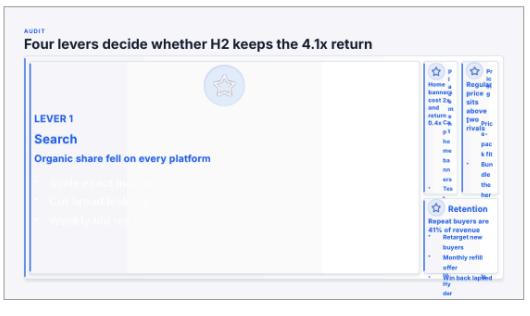
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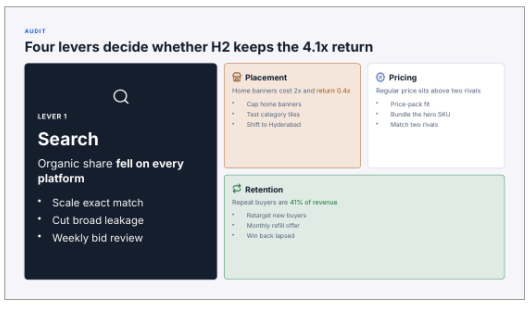
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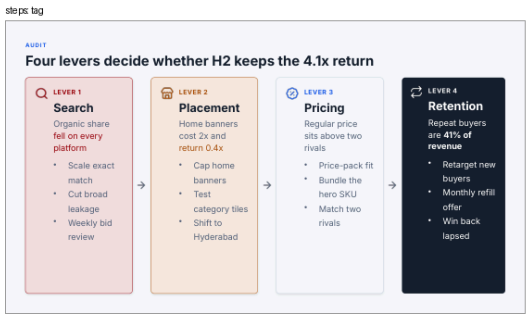
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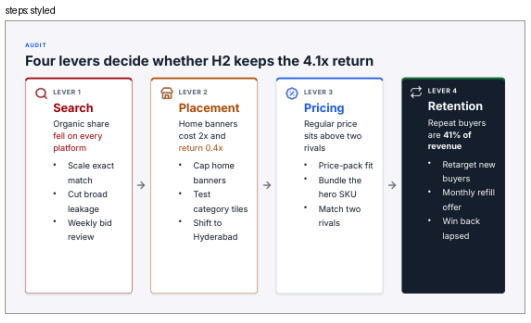
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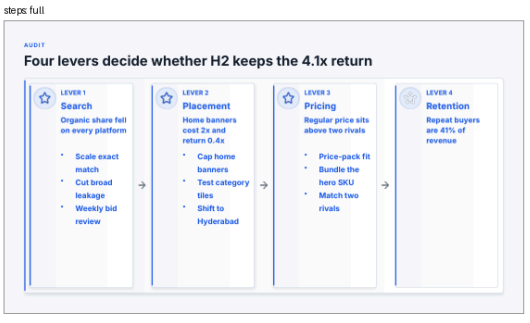
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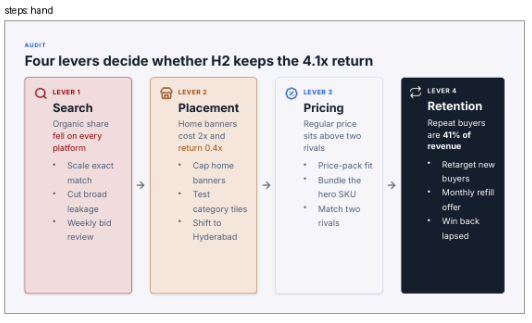
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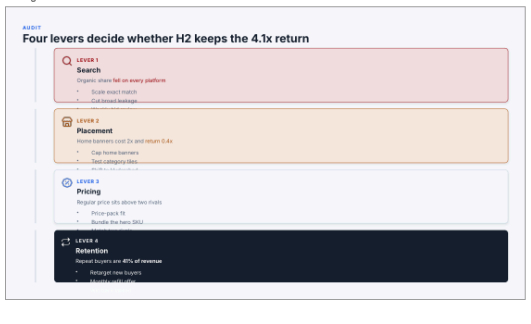
steps: full



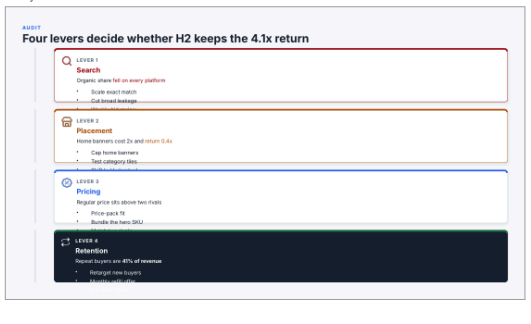
steps: hand



rail: tag



rail: styled



rail: full



rail: hand

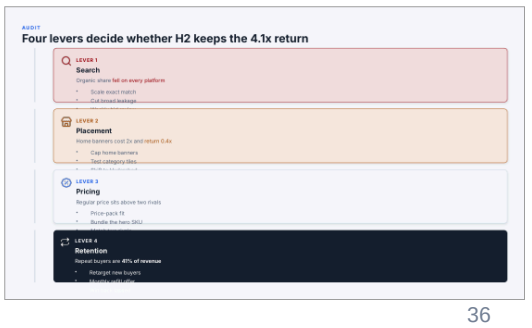
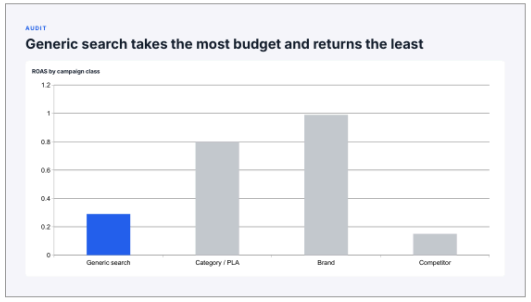


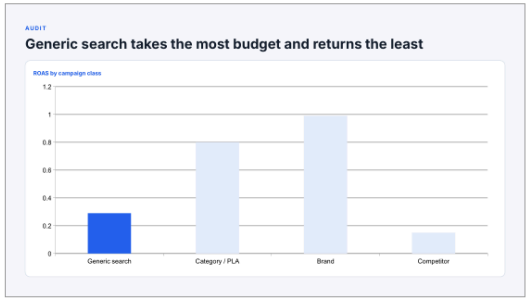
Figure 16 · Before the 2026-11-04 changes

Catalogue: chart. These four columns are audit test arms, not options offered to the LLM. (catalogue-chart.png)

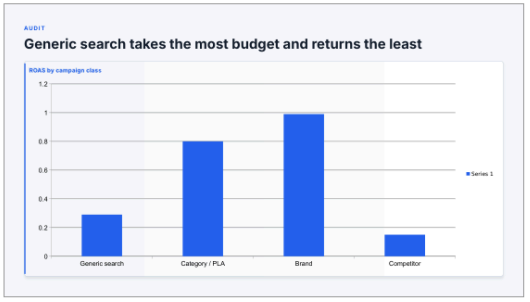
native: tag



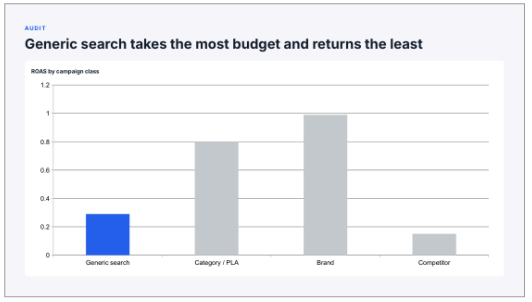
native: styled



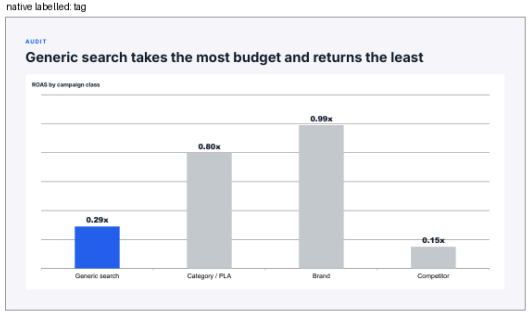
native: full



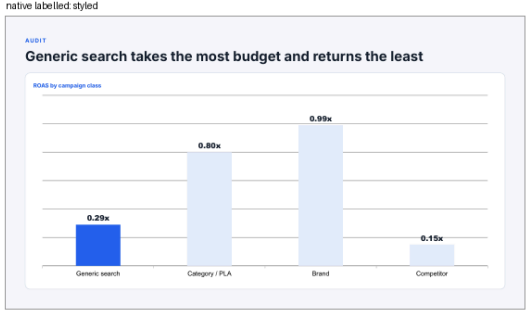
native: hand



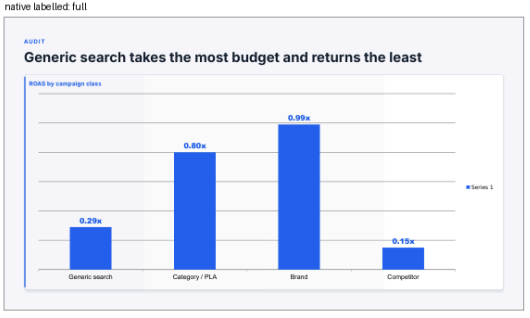
native labelled: tag



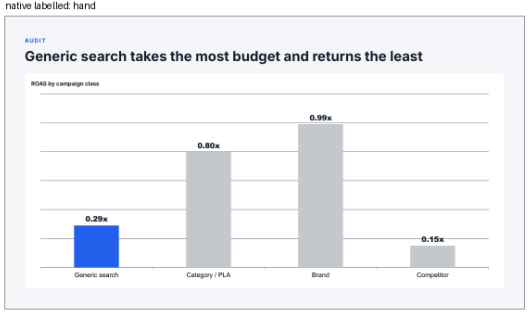
native labelled: styled



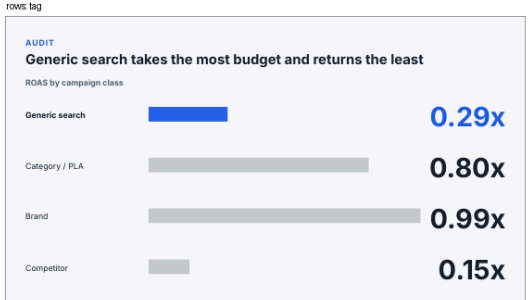
native labelled: full



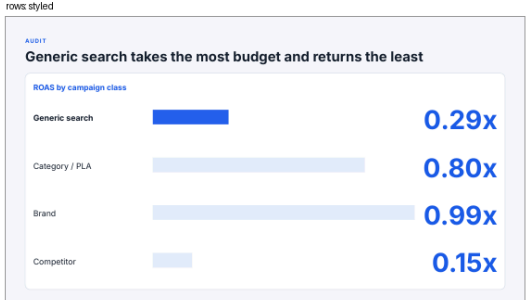
native labelled: hand



rows: tag



rows: styled



rows: full

rows: hand

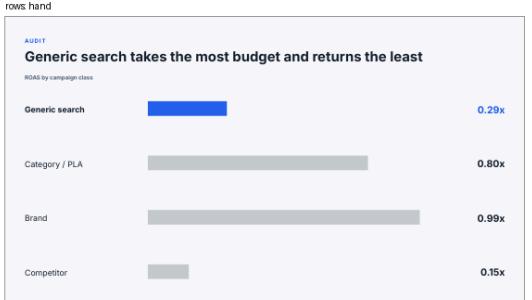


Figure 17 · Before the 2026-11-04 changes

Catalogue: timeline. These four columns are audit test arms, not options offered to the LLM. (catalogue-timeline.png)

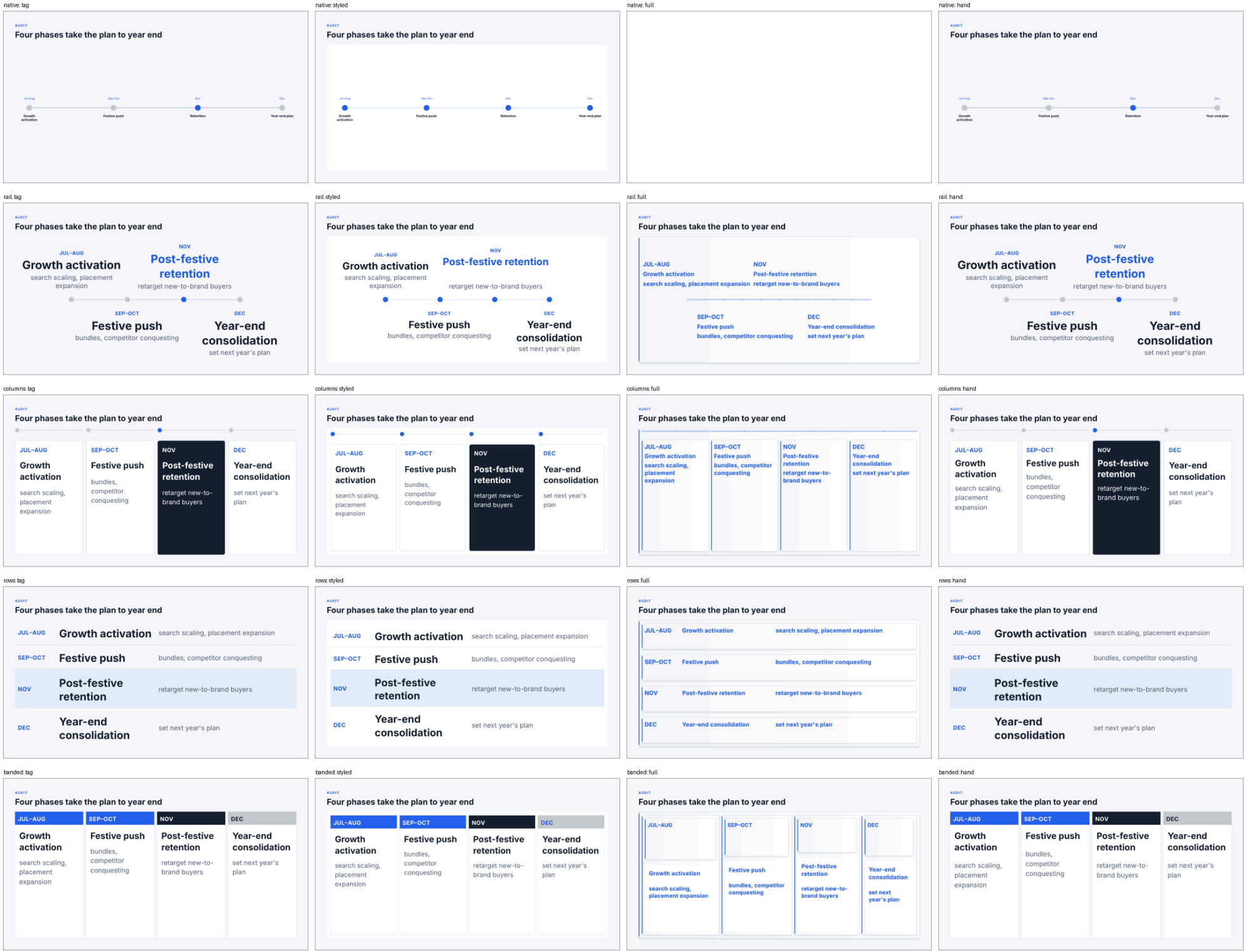


Figure 18 · Before the 2026-11-04 changes

Catalogue: table. These four columns are audit test arms, not options offered to the LLM. (catalogue-table.png)

native: tag

AUDIT

Four gaps cost us the return, and each has one fix

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

native: styled

AUDIT

Four gaps cost us the return, and each has one fix

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

native: full

native: hand

AUDIT

Four gaps cost us the return, and each has one fix

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

rows: tag

AUDIT

Four gaps cost us the return, and each has one fix

GAP	MARKETPLACE A	MARKETPLACE B	FIX
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

rows: styled

AUDIT

Four gaps cost us the return, and each has one fix

GAP	MARKETPLACE A	MARKETPLACE B	FIX
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

rows: full

rows: hand

AUDIT

Four gaps cost us the return, and each has one fix

GAP	MARKETPLACE A	MARKETPLACE B	FIX
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

rows numbered: tag

AUDIT

Four gaps cost us the return, and each has one fix

	GAP	MARKETPLACE A	MARKETPLACE B	FIX
01	CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
02	Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
03	Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
04	Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

rows numbered: styled

AUDIT

Four gaps cost us the return, and each has one fix

	GAP	MARKETPLACE A	MARKETPLACE B	FIX
01	CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
02	Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
03	Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
04	Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

rows numbered: full

rows numbered: hand

AUDIT

Four gaps cost us the return, and each has one fix

	GAP	MARKETPLACE A	MARKETPLACE B	FIX
01	CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
02	Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
03	Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
04	Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

columns: tag

AUDIT

Four gaps cost us the return, and each has one fix

CPC above allowable	Discovery leakage	Budget not following winners	Visibility spend mixed in
MARKETPLACE A CPC ₹31.1 vs ₹10.7 allowable	MARKETPLACE A Auto, phrase and broad weak	MARKETPLACE A Exact match under-scaled	MARKETPLACE A Display ads weak
MARKETPLACE B CPC ₹46.2 vs ₹14.8 allowable	MARKETPLACE B Phrase and generic weak	MARKETPLACE B Category ads under-funded	MARKETPLACE B Banner slots weak
FIX Break-even CPC bidding	FIX Weekly negative-keyword loop	FIX Budget tiers by ROAS and intent	FIX Separate visibility budget

columns: styled

AUDIT

Four gaps cost us the return, and each has one fix

CPC above allowable	Discovery leakage	Budget not following winners	Visibility spend mixed in
MARKETPLACE A CPC ₹31.1 vs ₹10.7 allowable	MARKETPLACE A Auto, phrase and broad weak	MARKETPLACE A Exact match under-scaled	MARKETPLACE A Display ads weak
MARKETPLACE B CPC ₹46.2 vs ₹14.8 allowable	MARKETPLACE B Phrase and generic weak	MARKETPLACE B Category ads under-funded	MARKETPLACE B Banner slots weak
FIX Break-even CPC bidding	FIX Weekly negative-keyword loop	FIX Budget tiers by ROAS and intent	FIX Separate visibility budget

columns: full

columns: hand

AUDIT

Four gaps cost us the return, and each has one fix

CPC above allowable	Discovery leakage	Budget not following winners	Visibility spend mixed in
MARKETPLACE A CPC ₹31.1 vs ₹10.7 allowable	MARKETPLACE A Auto, phrase and broad weak	MARKETPLACE A Exact match under-scaled	MARKETPLACE A Display ads weak
MARKETPLACE B CPC ₹46.2 vs ₹14.8 allowable	MARKETPLACE B Phrase and generic weak	MARKETPLACE B Category ads under-funded	MARKETPLACE B Banner slots weak
FIX Break-even CPC bidding	FIX Weekly negative-keyword loop	FIX Budget tiers by ROAS and intent	FIX Separate visibility budget

banded: tag

AUDIT

Four gaps cost us the return, and each has one fix

CPC above allowable		Discovery leakage	
Marketplace A	CPC ₹31.1 vs ₹10.7 allowable	Marketplace A	Auto, phrase and broad weak
Marketplace B	CPC ₹46.2 vs ₹14.8 allowable	Marketplace B	Phrase and generic weak
Fix	Break-even CPC bidding	Fix	Weekly negative-keyword loop
Budget not following winners		**Visibility spend mixed in**	
Marketplace A	Exact match under-scaled	Marketplace A	Display ads weak
Marketplace B	Category ads under-funded	Marketplace B	Banner slots weak
Fix	Budget tiers by ROAS and intent	Fix	Separate visibility budget

banded: styled

AUDIT

Four gaps cost us the return, and each has one fix

CPC above allowable		Discovery leakage	
Marketplace A	CPC ₹31.1 vs ₹10.7 allowable	Marketplace A	Auto, phrase and broad weak
Marketplace B	CPC ₹46.2 vs ₹14.8 allowable	Marketplace B	Phrase and generic weak
Fix	Break-even CPC bidding	Fix	Weekly negative-keyword loop
Budget not following winners		**Visibility spend mixed in**	
Marketplace A	Exact match under-scaled	Marketplace A	Display ads weak
Marketplace B	Category ads under-funded	Marketplace B	Banner slots weak
Fix	Budget tiers by ROAS and intent	Fix	Separate visibility budget

banded: full

banded: hand

AUDIT

Four gaps cost us the return, and each has one fix

CPC above allowable		Discovery leakage	
Marketplace A	CPC ₹31.1 vs ₹10.7 allowable	Marketplace A	Auto, phrase and broad weak
Marketplace B	CPC ₹46.2 vs ₹14.8 allowable	Marketplace B	Phrase and generic weak
Fix	Break-even CPC bidding	Fix	Weekly negative-keyword loop
Budget not following winners		**Visibility spend mixed in**	
Marketplace A	Exact match under-scaled	Marketplace A	Display ads weak
Marketplace B	Category ads under-funded	Marketplace B	Banner slots weak
Fix	Budget tiers by ROAS and intent	Fix	Separate visibility budget

grid: tag

AUDIT

Four gaps cost us the return, and each has one fix

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

grid: styled

AUDIT

Four gaps cost us the return, and each has one fix

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

grid: full

grid: hand

AUDIT

Four gaps cost us the return, and each has one fix

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

grid numbered: tag

AUDIT

Four gaps cost us the return, and each has one fix

	Gap	Marketplace A	Marketplace B	Fix
01	CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
02	Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
03	Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
04	Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

grid numbered: styled

AUDIT

Four gaps cost us the return, and each has one fix

	Gap	Marketplace A	Marketplace B	Fix
01	CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
02	Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
03	Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
04	Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

grid numbered: full

grid numbered: hand

AUDIT

Four gaps cost us the return, and each has one fix

	Gap	Marketplace A	Marketplace B	Fix
01	CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
02	Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
03	Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
04	Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

ledger: tag

AUDIT

Four gaps cost us the return, and each has one fix

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

ledger: styled

AUDIT

Four gaps cost us the return, and each has one fix

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

ledger: full

ledger: hand

AUDIT

Four gaps cost us the return, and each has one fix

Gap	Marketplace A	Marketplace B	Fix
CPC above allowable	CPC ₹31.1 vs ₹10.7 allowable	CPC ₹46.2 vs ₹14.8 allowable	Break-even CPC bidding
Discovery leakage	Auto, phrase and broad weak	Phrase and generic weak	Weekly negative-keyword loop
Budget not following winners	Exact match under-scaled	Category ads under-funded	Budget tiers by ROAS and intent
Visibility spend mixed in	Display ads weak	Banner slots weak	Separate visibility budget

Figure 19 · Before the 2026-11-04 changes

Catalogue: bullets. These four columns are audit test arms, not options offered to the LLM. (catalogue-bullets.png)

native (proof XML, no expander)

AUDIT

Recommendation ads and Product A carry the return

- Recommendation ads return 11.3x ROAS at a third of Search spend, but bring few new buyers
- **Product A is the star**: 8.0x ROAS at 12.5% ACOS, Product B close behind at 6.9x
- **Product C drags**: 4.3x ROAS and 23% ACOS, CPM ₹470 against ₹332
- **Generic keywords win new buyers** — 19.3% new-to-brand, keep them despite lower ROAS

rows (proof XML, no expander)

AUDIT

Recommendation ads and Product A carry the return

- **Recommendation ads return 11.3x ROAS at a third of Search spend, but bring few new buyers**
- **Product A is the star**
  8.0x ROAS at 12.5% ACOS, Product B close behind at 6.9x
- **Product C drags**
  4.3x ROAS and 23% ACOS, CPM ₹470 against ₹332
- **Generic keywords win new buyers**
  19.3% new-to-brand, keep them despite lower ROAS

rows numbered (proof XML, no expander)

AUDIT

Recommendation ads and Product A carry the return

01 **Recommendation ads return 11.3x ROAS at a third of Search spend, but bring few new buyers**

02 **Product A is the star**
8.0x ROAS at 12.5% ACOS, Product B close behind at 6.9x

03 **Product C drags**
4.3x ROAS and 23% ACOS, CPM ₹470 against ₹332

04 **Generic keywords win new buyers**
19.3% new-to-brand, keep them despite lower ROAS

grid (proof XML, no expander)

AUDIT

Recommendation ads and Product A carry the return

Recommendation ads return 11.3x ROAS at a third of Search spend, but bring few new buyers

Product A is the star
8.0x ROAS at 12.5% ACOS, Product B close behind at 6.9x

Product C drags
4.3x ROAS and 23% ACOS, CPM ₹470 against ₹332

Generic keywords win new buyers
19.3% new-to-brand, keep them despite lower ROAS

banded (proof XML, no expander)

AUDIT

Recommendation ads and Product A carry the return

Recommendation ads return 11.3x ROAS at a third of Search spend, but bring few new buyers

Product A is the star
8.0x ROAS at 12.5% ACOS, Product B close behind at 6.9x

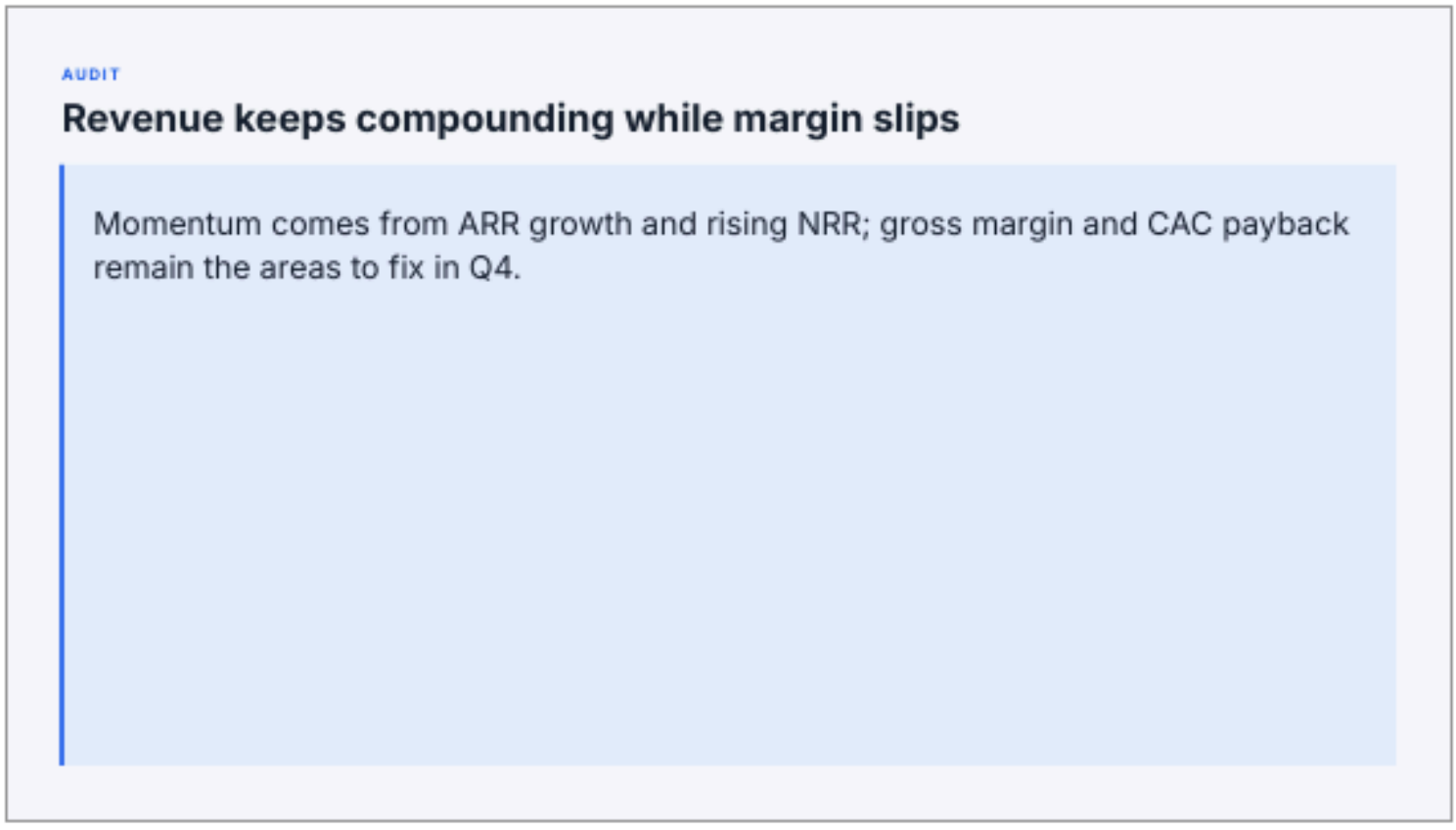
Product C drags
4.3x ROAS and 23% ACOS, CPM ₹470 against ₹332

Generic keywords win new buyers
19.3% new-to-brand, keep them despite lower ROAS

Figure 20 · Before the 2026-11-04 changes

Catalogue: callout. These four columns are audit test arms, not options offered to the LLM. (catalogue-callout.png)

native (proof XML, no expander)



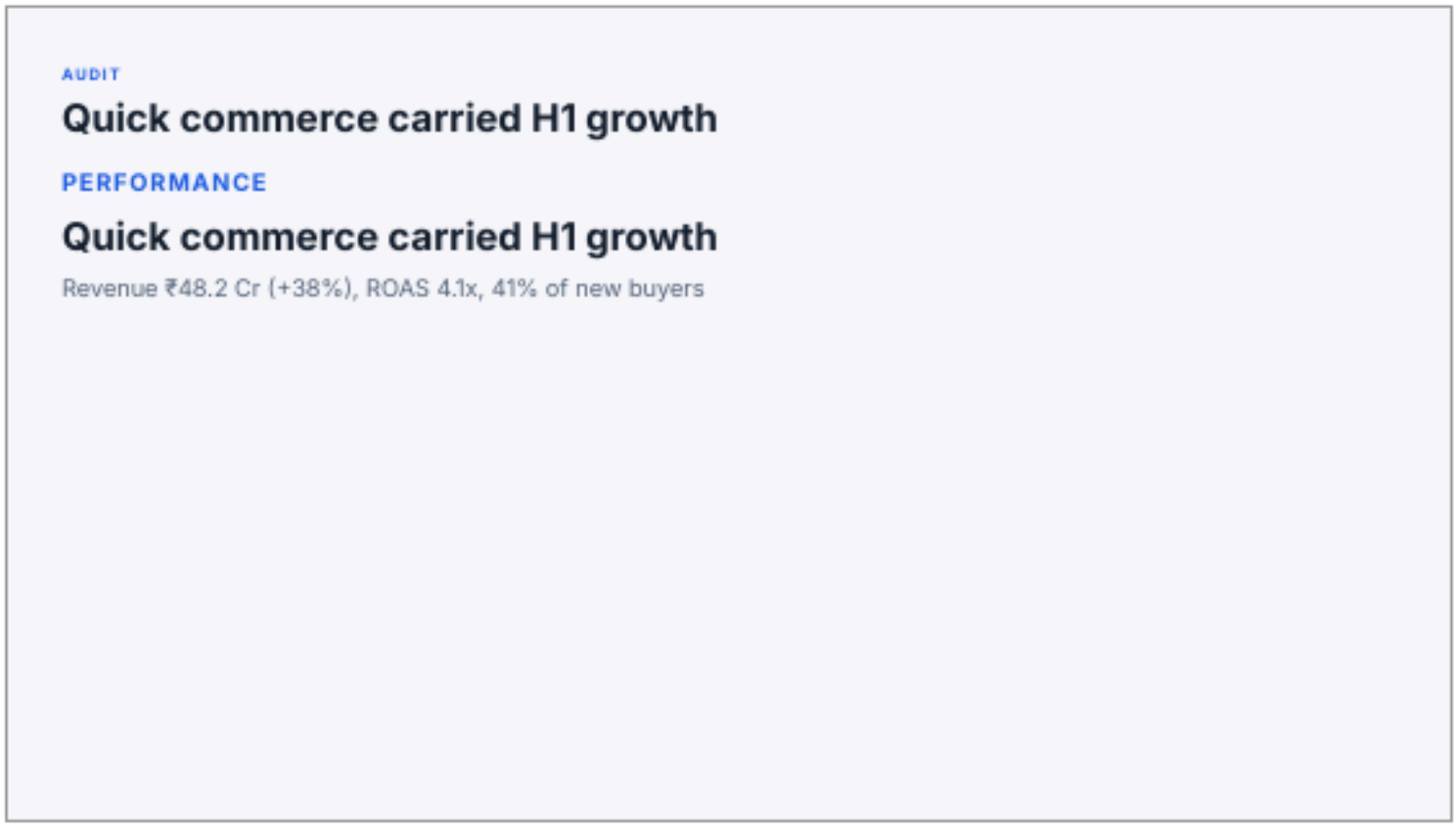
native (quote) (proof XML, no expander)



Figure 21 · Before the 2026-11-04 changes

Catalogue: header. These four columns are audit test arms, not options offered to the LLM. (catalogue-header.png)

standard (proof XML, no expander)



inverted (proof XML, no expander)

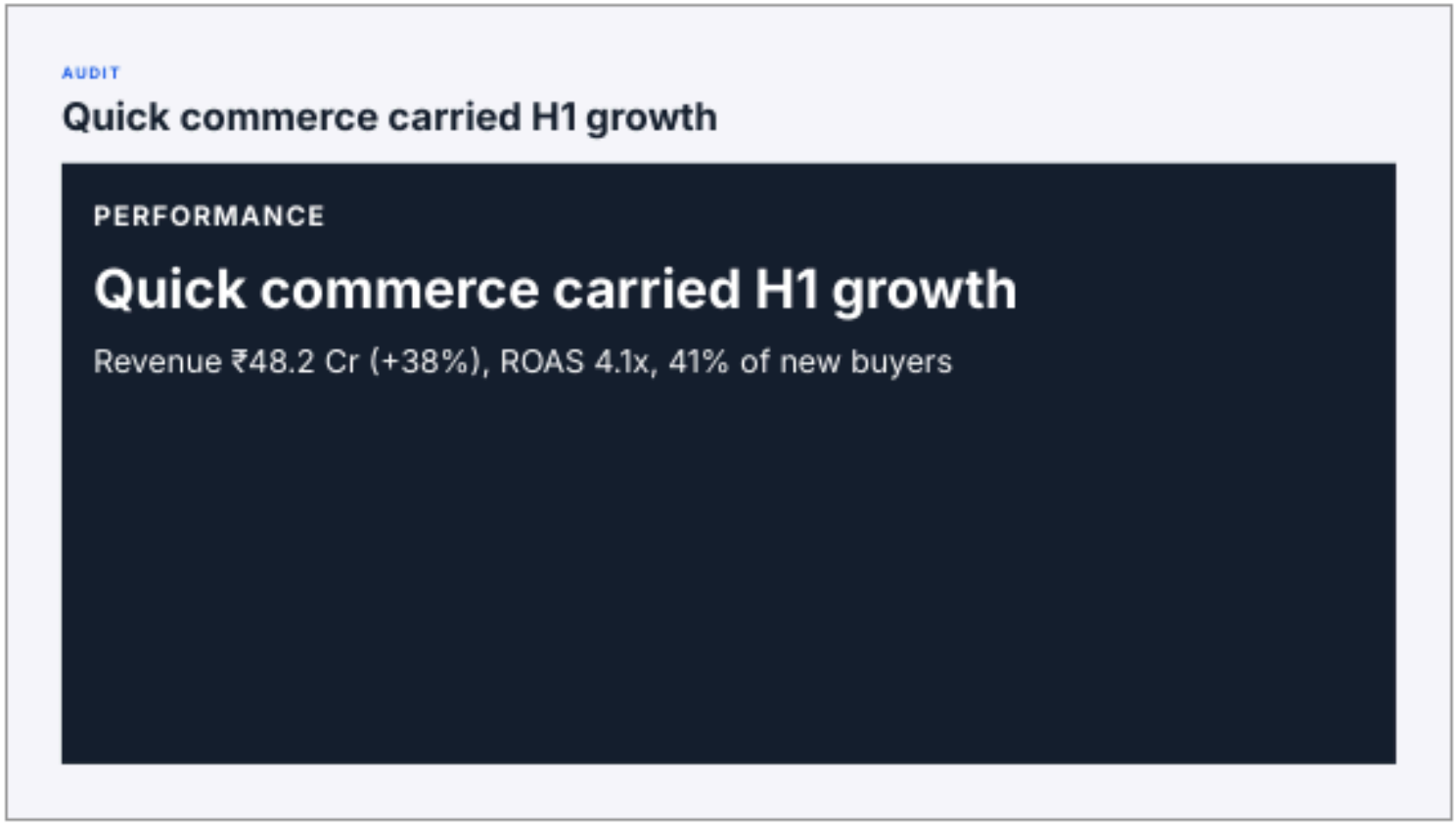


Figure 22 · Before the 2026-11-04 changes

Stress: kpi_row, layout cases. (stress-kpi_row.png)

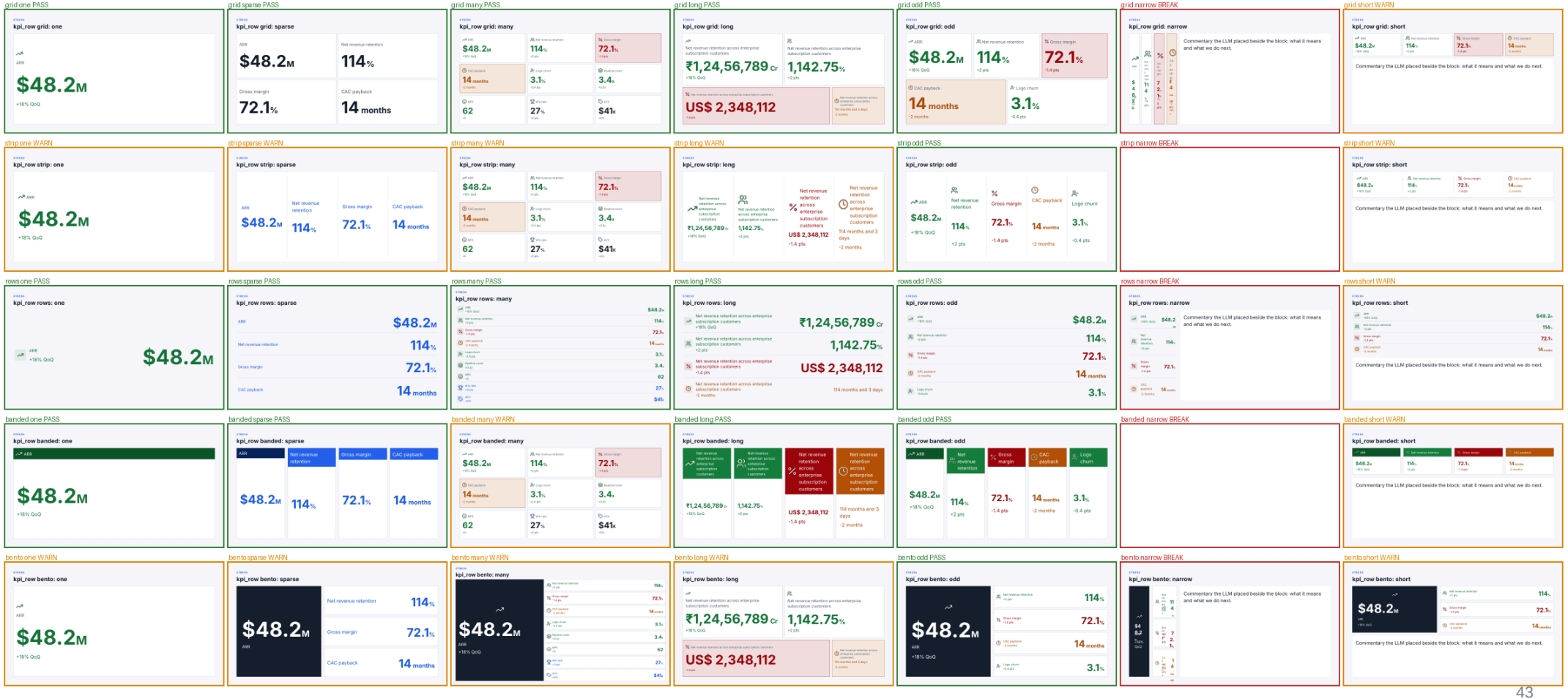


Figure 23 · Before the 2026-11-04 changes

Stress: kpi_row, node-level cases. (stress-kpi_row-node.png)

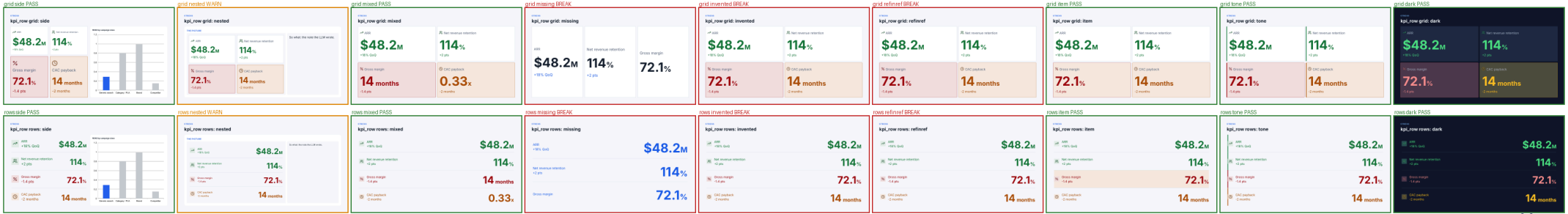


Figure 24 · Before the 2026-11-04 changes

Stress: card_grid, layout cases. (stress-card_grid.png)



Figure 25 · Before the 2026-11-04 changes

Stress: card_grid, node-level cases. (stress-card_grid-node.png)

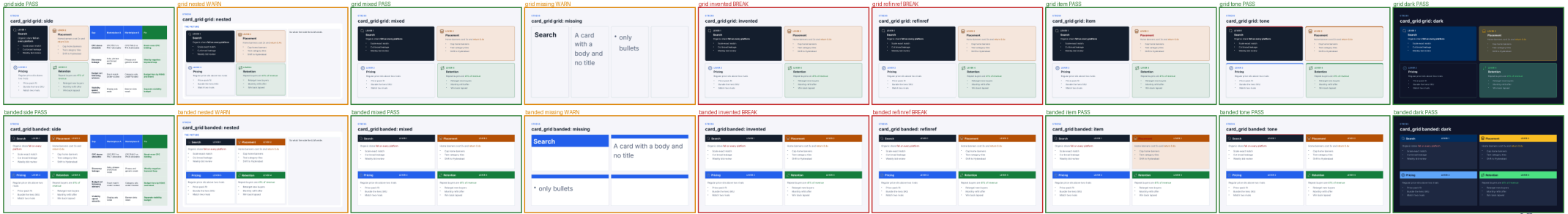


Figure 26 · Before the 2026-11-04 changes

Stress: chart, layout cases. (stress-chart.png)



Figure 27 · Before the 2026-11-04 changes

Stress: chart, node-level cases. (stress-chart-node.png)



Figure 28 · Before the 2026-11-04 changes

Stress: timeline, layout cases. (stress-timeline.png)

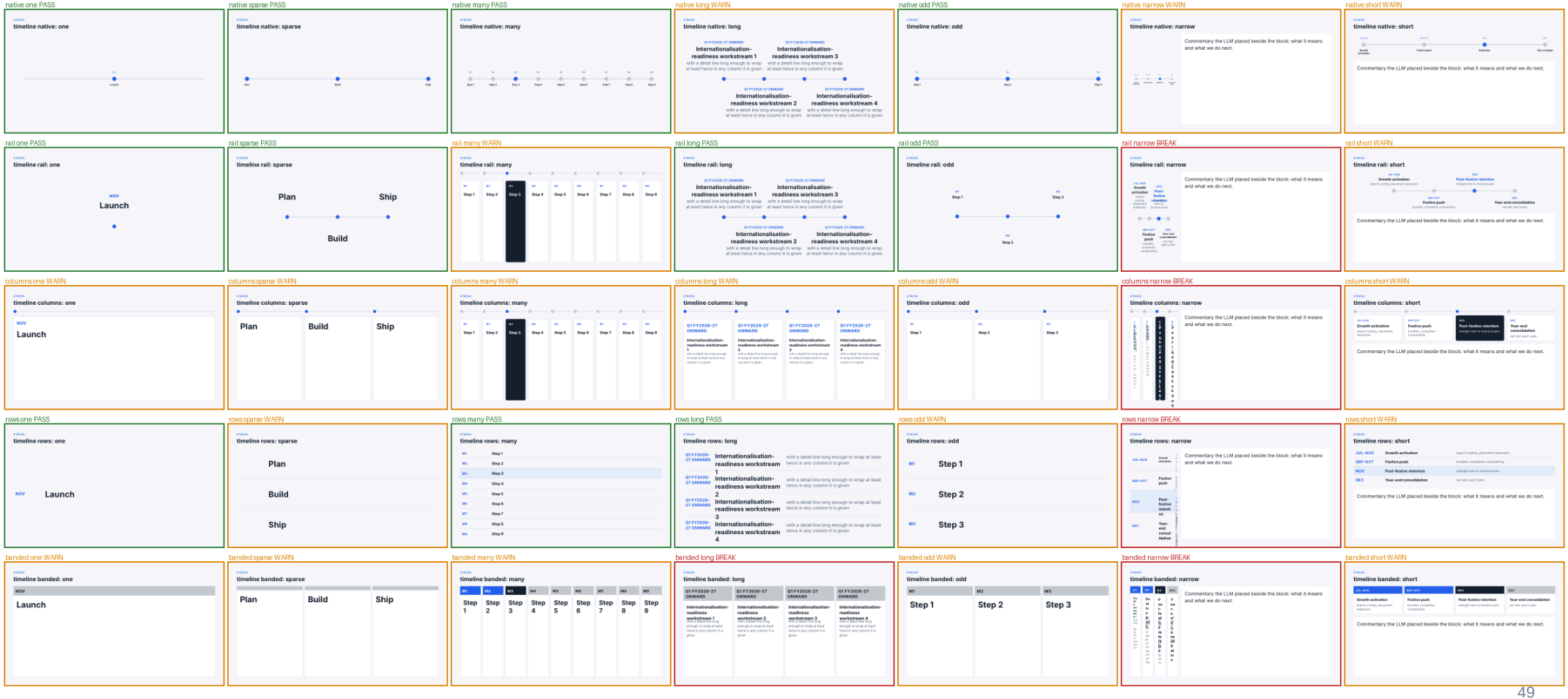


Figure 29 · Before the 2026-11-04 changes

Stress: timeline, node-level cases. (stress-timeline-node.png)

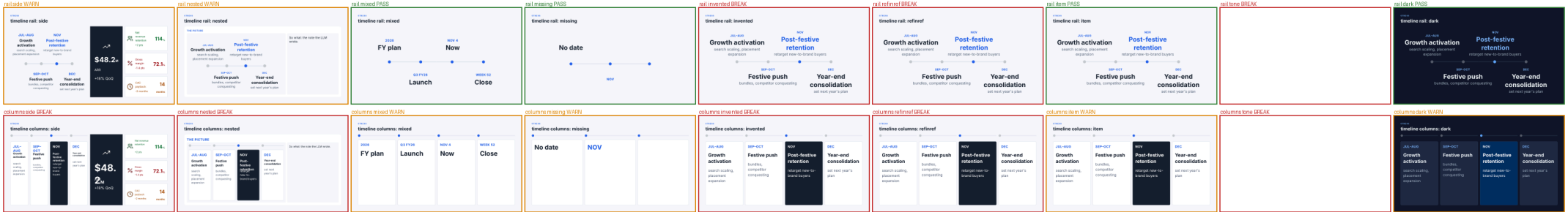


Figure 30 · Before the 2026-11-04 changes

Stress: table, layout cases. (stress-table.png)

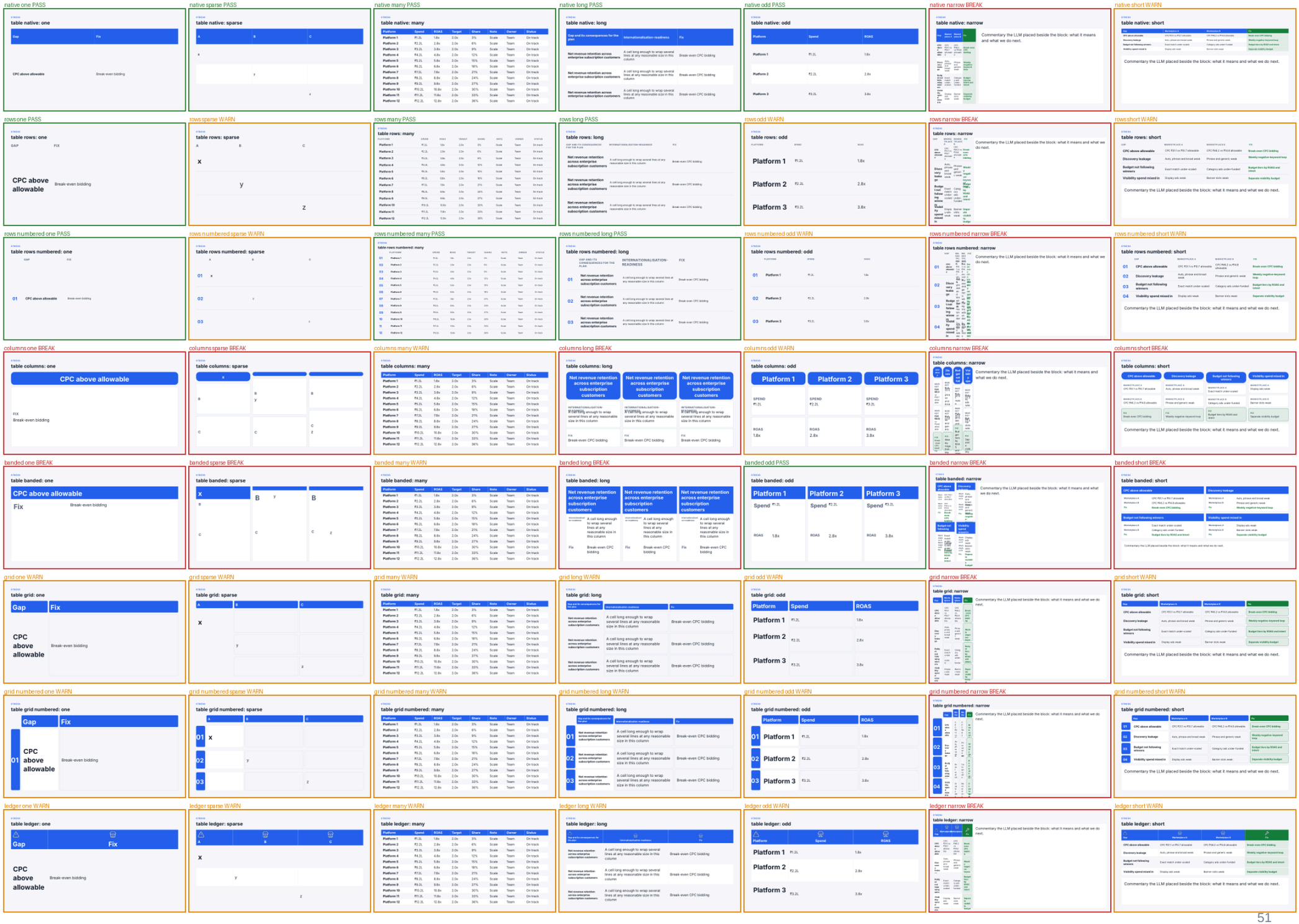


Figure 31 · Before the 2026-11-04 changes

Stress: table, node-level cases. (stress-table-node.png)



3. Syntax audit

> Dated 2026-11-04 and kept as written. Its catalogue, stress and reach figures predate the table and card deletions; §7 records those changes, and Part A is the current state.

Free work: no API calls, no LLM, `src/` untouched. Everything here re-runs with `python -m scripts.syntax_audit <command>` (`catalogue`, `consistency`, `stress`, `tables`, `duplicates`, `atomic`, `flex`, `bisect`, `census`). Sheets, numbers and the full catalogue are in eval/syntax-audit/.

The timeline is audited from `claude/left-blocks-build-0fe007` (53e74ca): the script loads its `sequences.py` and `nodes.timeline` from git in its own process and merges nothing. process_arrow and flow are treated as dropped.

Limits. "Blank", "breaks" and "survives" are measured on hand-written tags and plans built for the test. No real LLM wrote any tag here, so this audit cannot say how often the LLM would hit each case; the paid same-plan run is what measures that. Automatic checks caught most breaks, and I read every sheet by eye on top; where only my eye found something, the text says so.

Short answers to your questions

Question	Answer
Is the syntax sound?	**The tag is: 15–27 tokens against 108–1,739 expanded, and no way to get a word wrong. The attribute layer is not.** 31% of attribute × composition pairs silently do nothing, and one allowed attribute can blank a KPI row with no error
Where does it break?	**Boxes narrow for the content** (28 of 30 cases at 280 px: `requires` counts items and never looks at the box), card `rail` / `steps` with many or long items, and attributes the expander writes twice
Should anything be introduced?	**No new structure: no slots, spans or item overrides.** Fix the attribute gate first. One optional addition: a parts escape for looks no composition reaches
Build atomically instead?	**No, keep composition as the default.** Atomic reaches the same looks but costs 30–51 × the output tokens, and swapped fields pass unseen. Keep element refs as the escape hatch for looks a composition cannot draw
Are we really using flexbox?	**Yes.** POM is Yoga, and the expanders use it well for boxes. Flexbox cannot size type, and that is most of what fit-grow does. The real savings are small

Most serious finding: an allowed attribute blanks the KPI text

`<KpiRow ref="kpis" variant="strip" alignItems="center"/>` compiles, reports no error, and draws an empty strip. The same happens to a KPI row on the **same slide as** `<Table ref="gaps" variant="rows" gap="20"/>`: 12 of 12 KPI strings are gone, `node_ok` is true, and there are no codes (Figure 13, duplicates.json).

Cause (traced). Expanders write an attribute literally and then add the LLM's box attributes, e.g. `<HStack alignItems="stretch"{A(gbox)}>` (`tile_groups.py:274`), so the attribute appears twice. POM's parser accepts that and the last value wins (measured: `gap="8" gap="40"` lays out at 40). `tiles.fill` parses with lxml, gets a syntax error and returns the slide unfilled by design (`element_refs.py:155`), so every `<Text ref>` stays empty.

The attributes that do this are all on the allowed list. Full list per node in `consistency.json` → `duplicates`:

Node	Compositions that write it twice	Attributes
kpi_row	strip, banded, bento	`alignItems`; bento also `item.justifyContent`, `item.w`; banded and bento also `icon.name`
chart	native, native labelled	`alignItems`
timeline	rail, columns, rows, banded	`gap`, plus `alignItems` / `justifyContent` / `item.gap` / `item.grow` / `item.padding` per composition
table	all but native	`gap`, plus `item.padding`, `item.gap`, `item.grow`, `item.w`, `band.padding` per composition
card_grid	none	built through lxml, so one value wins

On the node's own slide the damage is a silently wrong value. **With a KPI row anywhere on the slide, its text disappears.**

This must be fixed before any paid run: a fixed slide would look like a generator failure.

1. Catalogue

catalogue.md has, for all 39 variants: the plan JSON, the minimal tag, a styled tag, the tag with every allowed attribute, the XML the real expander writes, and the atomic form. Each sheet has one row per variant and four columns: **tag · styled · every attribute · atomic form**.

kpi_row (Figure 14) · card_grid (Figure 15) · chart (Figure 16) · timeline (Figure 17) · table (Figure 18) · bullets (Figure 19) · callout (Figure 20) · header (Figure 21)

How the atomic form is made. It is the expansion with code's peer ids and the `alignItems="stretch"` default removed: every typed word, size and colour stays. So it is derived from the expansion, not typed by me (§5 has three really hand-written samples), and it is a floor on what an LLM must type for that look.

bullets, callout and header have no expander. Their tags are hypothetical and their XML is the 2026-10-27 proof's.

A worked example, KPI `rows`:

```
<!-- what the LLM writes: 17 tokens -->
<KpiRow ref="kpis" variant="rows" grow="1" />

<!-- what code writes: 748 tokens, 29 elements; the first of 4 rows -->
<VStack gap="0" padding.left="8" padding.right="8" grow="1">
  <HStack grow="1" gap="18" alignItems="center" padding.top="10" padding.bottom="10" id="card-kpis-0">
    <Icon name="trending-up" size="22" variant="square-filled" bgColor="$positiveSoft" color="$positiveText"/>
    <VStack gap="4" grow="1" id="card-kpisx-1"><Text color="$positiveText" fontSize="16" id="kpil-kpis-0">ARR</Text>
      <Text color="$positiveText" fontSize="16">+18% QoQ</Text></VStack>
    <Text color="$positiveText" textAlign="right" fontSize="28" bold="true" id="kpiv0-kpis-0">$48.2<Span fontSize="14">M</Span></Text>
  </HStack>...
```

Tokens (o200k_base), minimum to maximum across a node's variants. "Elements" is the number of XML elements, which is the line count when each sits on its own line.

Node	Variants	Plan JSON	Tag	Styled tag	Every attribute	Expanded	Atomic form	Elements
kpi_row	5	111	17–18	83	1,157	626–830	538–645	29–33
card_grid	9	289	16–17	62	1,427	826–1,312	751–1,170	43–60
chart	3	60	16–17	67	597	181–697	176–529	8–31
timeline	5	68	16–17	63	1,156	108–879	108–847	5–40
table	8	137	15–27	60	982	447–1,739	441–1,507	30–79
bullets (no expander)	5	113	9–14	–	–	154–605	154–573	8–24
callout (no expander)	2	22–30	8–9	–	–	65–101	65–101	2–4
header (no expander)	2	39	8–13	–	–	95–108	95–108	4

What the ids buy. Removing them changes KPI `grid` by 29% of pixels and table `banded` by 16%. KPI `grid` stays 4 across instead of regridding to 2 × 2, and table `grid` / `grid numbered` / `ledger` rows end up in different type sizes (table sheet (Figure 18), right column). Ids are small in tokens and carry fit-grow's peer rule, so any atomic route needs code to write them (§5).

What the every-attribute tag shows. It fails to compile on 10 of 30 variants, on attributes the validator allows:

Attribute on the tag	Where	Result
`gap`, `alignItems`, `justifyContent`	native `<Table>` and `<Timeline>`	`Unknown attribute`
`cellBorder.*` (`native_attributes`)	every table composition but native	`Unknown attribute "cellBorder"` on a VStack / HStack
`showLegend`	chart `rows`	`Unknown attribute "showLegend"` on a VStack
`stripe.color` and other text attributes	table `ledger`	`Unknown attribute "color"` and others (41 errors): `stripe.` is spread onto a box

Each of these goes to a repair call and is not reported before compile.

2. Consistency against the planner JSON

Tag and part names line up with plan keys, with these exceptions:

Plan key	Tag / part	Problem	Needed?
`kpi_deltas`	part `delta.`, but ref field `kpis.0.note`	one thing, two names on one node	no: use `delta` everywhere
`timeline_items[].label`	part `title.` (`label.` an alias)	breaks spec N3 (part = key), for POM's own word	decide: rename the plan key or the part; the alias costs little
`cards[].bullets`	text styled by `body.`; `bullets.` is a **box**, only in `columns`	`bullets.backgroundColor` does nothing on 8 of 9 compositions, and `bullets.color` is an error	no: `bullets.` should style the bullets
`cards[].key_phrase`	no part	its colour cannot be styled	fine: the phrase colour is code's
`table_rows[][0]`	`label.`	`label.` means the first column here, a KPI's label there, and chart labels elsewhere	acceptable: each is "the row's name"
`chart_type`, `chart_title`	code-owned; `title.`	fine	–
bullets / callout / header	`<Ul>` in `blocks`, `<BulletList>` in spec §8; `<Callout>` in `spec.py`, `<Narrative>` in spec §8; `<SlideHeader>` vs `<Title>`	three nodes have two tag names each	settle if built

Same idea, different name:

Idea	Names today	Suggestion
layout hint	`cols` (kpi, card), `direction` (chart, timeline), none (table)	fine: different questions
per-item icon override	`icon` (kpi, card; one value), `icons` (table; a comma list)	fine, but `icons` works only on `ledger` and is silently ignored elsewhere
restyle one item	item tags `<Kpi>` / `<Card>`, `<Highlight>` (table), nothing (chart, timeline)	two mechanisms for one idea; see §4
the emphasis	`focus_card` (card), hint words (kpi, table, timeline), headline / latest period / hint (chart)	fine as code's, but the rule differs per node and the prompt doesn't say which
fill	`backgroundColor` (boxes), `fill.color` (`bar.`, `marker.`, `connector.`)	unavoidable: POM's Shape vs box. The prompt must keep saying which part is which
tone tokens	`$tone*` resolve on kpi and card, on table as the accent only, and **stay literal on chart and timeline**	see below

Idea	Names today	Suggestion
composition words	`rail` = vertical numbered line (card) but horizontal dots with text above and below (timeline); `columns` = pill + boxes (card, table) but dot + box (timeline)	breaks spec §2.3 ("a word keeps its meaning")

`$tone*` fails on chart and timeline. `<Chart bar.fill.color="$tone"/>` and `<Timeline marker.fill.color="$tone"/>` end in a compile error: `Unknown theme token "$tone"` (stress case `tone`). The prompt for kpi and cards teaches these tokens, so an LLM can carry them across.

Silent no-ops. Each allowed attribute was tried alone on every composition and looked for in the expansion (reach.json): **1,601 of 5,091 pairs (31%) reach nothing, with no report.**

Node	Pairs	No effect	Examples
kpi_row	860	25%	`band.*` outside `banded`; `divider.*` on `grid`; `value.bold`; `item.borderRadius` on `grid` (code's square)
card_grid	1,899	36%	`band.*`, `bullets.*` on most; `number.*` outside numbered; `tag.*` on `columns`
chart	264	27%	`label.*`, `value.*`, `bar.*` on `native`
timeline	860	37%	`date.bold` and all text styling on `native`; `item.*`, `band.*` on `rail`
table	1,208	25%	`band.*`, `number.*`; `stripe.*` outside native and ledger

Part of this is by design (a part the composition doesn't draw), and `marker.*` needs `marker.shapeType` first, so the test overstates it slightly. But your rule is that styling freedom stays with the LLM, and today it cannot tell a style that worked from one that vanished.

One more silent loss: `Text` allows `highlight`, and the normalizer strips it (`_HIGHLIGHT_ATTR_RE`), so `delta.highlight` reaches the expansion and never the slide.

Aliases. In the 33 saved tile-tag slides: `CardGrid layout=` 13 times, `KpiGrid layout=` 6, `cols` 15, `variant="inverted"` twice, and `<KpiGrid>` 12 times. So only 2 of 37 group tags named a variant, and all of them wrote old names. Aliases are needed for replaying saved slides and for nothing else; the spec's rule (remove after one paid run shows zero) holds. **Whether the LLM picks a variant at all is still unmeasured**, and that is the largest open risk in the design.

Two sources for prompt lines. kpi and card keep theirs in `nodes.<kind>.prompts`, while chart, timeline and table read `blocks.<kind>.variants`. Move them to `nodes`.

3. Stress cases

300 rendered slides: 115 pass, 103 warn, 82 break by the automatic checks (words lost, overlap, broken words, spill, compile, node errors). Node-level cases ran on two variants per node. Table with causes: stress-table.md.

Sheets: kpi_row (Figure 22) (node (Figure 23)) · card_grid (Figure 24) (node (Figure 25)) · chart (Figure 26) (node (Figure 27)) · timeline (Figure 28) (node (Figure 29)) · table (Figure 30) (node (Figure 31))

✓ pass, ~ warn (mostly `GEOM_CARD_EMPTY`), ✗ break. Empty cells were not run.

Node	Variant	one	sparse	many	long	odd	narrow	short	side	nested	mixed	missing	invented	ref-in-ref	item	tone	dark
kpi_row	grid	✓	✓	✓	✓	✓	✗	~	✓	~	✓	✗	✗	✗	✓	✓	✓
	strip	~	~	~	~	✓	✗	~									
	rows	✓	✓	✓	✓	✓	✗	~	✓	~	✓	✗	✗	✗	✓	✓	✓
	banded	✓	✓	~	✓	✓	✗	~									
	bento	~	~	~	~	✓	✗	~									
card_grid	grid	✓	✓	✓	✓	✓	✗	✓	✓	~	✓	~	✗	✗	✓	✓	✓
	grid numbered	~	✓	~	✓	~	✗	~									
	rows	✓	~	~	✗	✓	✗	✓									
	rows numbered	~	~	~	✗	~	✗	~									
	columns	✗	~	~	✗	✗	✗	✗									
	banded	✓	~	~	✓	✓	✗	✓	✓	~	✓	~	✗	✗	✓	✓	✓
	bento	~	~	~	✓	✗	✗	✗									
	steps	✓	✓	✗	✓	✓	✗	~									
	rail	✓	✓	✗	✗	✗	✗	✗									
chart	native	✓	✓	✓	✓	✓	✗	~	✓	~	✗	✗	✗	✗	✓	✗	✓
	native labelled	✓	✓	~	✓	✓	~	~									
	rows	✓	✓	~	✓	✓	✗	~	✓	~	✓	✗	✗	✗	✓	✗	✓
timeline	native	✓	✓	✓	~	✓	~	~									
	rail	✓	✓	~	✓	✓	✗	~	~	~	✓	✓	✗	✗	✓	✗	✓
	columns	~	~	~	~	~	✗	~	✗	✗	~	~	✗	✗	~	✗	~
	rows	✓	~	✓	✓	~	✗	~									
	banded	~	~	~	✗	~	✗	~									
table	native	✓	✓	✓	✓	✓	✗	~	✓	✗	✓	✓	✗	✗	✓	✓	✓
	rows	✓	~	✓	✓	~	✗	~									
	rows numbered	✓	~	✓	✓	~	✗	~									
	columns	✗	✗	~	✗	~	✗	✗									
	banded	✗	✗	~	✗	✓	✗	✗									
	grid	~	~	~	~	~	✗	~									

Node	Variant	one	sparse	many	long	odd	narrow	short	side	nested	mixed	missing	invented	ref-in-ref	item	tone	dark
	grid numbered	~	~	~	~	~	✗	~									
	ledger	~	~	~	~	~	✗	~	~	~	✓	~	✗	✗	✓	✓	✓

Breaks by case: narrow 28, invented 10, ref-in-ref 10, long 7, short 5, missing 4, tone 4, one 3, odd 3, many 2, sparse 2, nested 2, side 1, mixed 1. The invented and ref-in-ref cases were planted, so their breaks are the expected error or a known gap.

Causes

Case	What happens	Cause	Severity
narrow (280 px beside a note)	28 of 30 break, with letter-stacked or overlapping text; **strip, KPI banded, card columns, bento and steps fail to compile** (`addTextBox: width must be a finite positive EMU value`)	`requires` checks counts only; no composition knows its box. Yoga has no min-content width, so columns shrink to nothing (flex test `squeeze`)	**high**
card rail (odd, long, many, short)	text overlaps and spills past the slide; even the sample of 4 full cards spills (`GEOM_SPILL`)	rail stacks 4+ full cards down the slide; its only `requires` is the layout	**high**: needs a cap, or a rule on bullets
card steps, many	9 cards in one row, letter-stacked	no count cap (`rows`, `columns`, `banded` have one)	medium
words dropped by design	card `columns` and small `bento` cards drop the `tag`; table `columns` / `banded` drop the first column's name	composition doesn't draw the part. **It breaks "every word, verbatim"** (14 of the breaks outside narrow)	your call
tone on chart, timeline	no compile, as §2	`$tone*` resolved only in `tile_groups` and `tables`	high
missing	KPI: a label with no value is dropped with **no report**. Chart: an error, then repair. Timeline and cards with empty items: an empty box, no report	checks differ per node	medium
mixed units on native chart	"₹31.2L", "2.8x", "41%" on one axis, every unit gone	`number_format` gives "General" and `requires` has no one-unit check. `rows` keeps each value verbatim, so it reads, but its bars still compare unlike things	medium
ref-in-ref	inner KPI: an error naming `<KpiGrid>`. Inner chart or table: **lost with only a report**	known (§13.3): `validate.check_skeleton` not wired	high
invented	error → repair, though the slide is right	known (§13.3)	medium
item on chart, timeline	`<ChartDataPoint>` / `<TimelineItem>` children dropped with a report	no item form there	low, see §4
dark	passes; the native chart stays a white block on a dark slide (by design, POM's black axis)	known	low

Found by eye (the checks passed)

- **Native timeline text is very small** on a full slide, in every case, and native is what a bare tag picks for short labels. Its size is fixed by POM, and fit-grow cannot change it.
- **Empty cards.** Timeline `columns` and `banded`, and card `banded` with sparse text, leave the lower half or more empty; `one` and `sparse` show it most. This is your "grow the type" rule failing. I did not trace why growth stops here.
- **Peers out of line.** KPI `banded` and `strip` put one-line and two-line labels in the same row, so the values do not line up. `strip` still decides the icon per tile (spec §6 finding 1, open as R5).
- **Uneven grid tiles.** KPI `grid` with long values or an odd count draws unequal widths in a row (content width plus a share), because no `w` is set.
- **A style that doesn't travel.** My styled tag `title.color="$toneText"` is fine on `grid` and **invisible on `columns` and `banded`**, where it sits on a `$tone` band. The same holds for `label.color` on table bands. `LOW_CONTRAST` reports it, and this follows your no-auto-coupling rule, so the choice is yours: the prompt could say which parts sit on bands.
- **Rail, rows and table `long`:** close line spacing where a long head wraps.

Where the syntax holds: `one`, `odd`, `sparse`, `dark`, `side` and `nested` on KPI `grid` / `rows`, card `grid` / `banded`, chart `rows` and table `native` / `rows`; `many` falls back correctly on KPI `strip` / `banded` and on most card and table compositions.

4. Should we introduce something?

Checked against the stress cases, composition-test (eval/composition-test/README.md) (table of data needs), genspark-variants (eval/genspark-variants/summary.md), and spec §7 / §11.

Fixes that come first (no new syntax):

#	Change	Evidence
F1	One merge helper: an LLM attribute replaces the expander's literal, never adds a second	the blank KPI row; 70 attribute × composition pairs write twice
F2	Filter per composition before the XML is written: an attribute this composition doesn't draw → `NODE_ATTR_NO_EFFECT` (report). Native tags pass only what POM accepts	1,601 silent no-ops; 10 of 30 variants fail compile on allowed attributes
F3	Parse check in `tiles.fill`: a syntax error is an error, not "the normalizer reports it" (it doesn't)	the blank KPI row reports nothing
F4	`$tone*` either resolve to `accent` on chart and timeline, or drop with a report	compile failure
F5	Remove `highlight` from `part_nodes.Text` (the normalizer strips it)	`delta.highlight` has no effect

Applied 2026-11-04 (free, no API calls). Render of the two formerly blank slides: Figure 6.

#	How	Measured after
F1	`validate.one_value`: one pass after the expanders and before any lxml parse; the last value wins, as in POM. The expanders' f-strings are untouched	both blank slides: 0 words lost
F2	`ignores` / `draws_only` per composition in `block-variants.yaml`, applied by `compose.undrawn` to the LLM's attributes only. An item placed apart keeps its own box attributes	every-attribute tag compiles on all 29 compositions (10 failed before); of 4,839 attribute × composition pairs, 0 are silent
F3	`element_refs.fill_refs`: `NODE_XML_INVALID` (error, goes to repair) when the slide has KPI content or a `<Text ref>`	unit tests
F4	`compose.no_item_tones`: `$tone*` on chart and timeline → accent, reported	all four tokens compile on both nodes
F5	`highlight` moved from the allowed Text attributes to `dropped.parts` (a report, not a repair)	unit tests

Bare and lightly styled tags are unchanged: 102 expansions identical to the commit before. Unit tests: 2,035 pass, the same 4 known failures (+29 in `tests/unit/test_attribute_layer.py`).

Corrections to this report's measure. The reach check first looked for the value in quotes, so it missed `bar.fill.color` in `chartColors` JSON; it also tested `marker.*` without a `shapeType`, and counted plans that fell back to another composition. Measured properly on this branch's nodes, 1,502 of 4,839 pairs (31%) do not reach the XML, the same share as the 1,601 of 5,091 above. The per-attribute lists in `reach.json` predate this and use the other branch's timeline.

Open from this fix.

- `ignores` for `timeline` were re-measured after `claude/left-blocks-build-0fe007` merged (2026-11-04): 0 of 3,904 attribute x composition pairs silent on chart, timeline, KPI, cards and table; `bullet_list` also clean on its own probe. Sheet of every variant: Figure 1.
- "Reaches the XML" is not "changes the picture": an attribute can reach the XML and still look the same.
- `POM_BLOCKS=tiles` without `POM_NODES` (the older tile route in `tiles.py` and `cards.py`) is not filtered. It gets F1, F3 and F5, not F2.

Candidates for new syntax:

Candidate	Evidence	Verdict
Slots (`<Card ref><slot .../></Card>`)	no case needs it: an LLM card around a ref holds (probe d1, `nested`)	**not needed**
Spans (true bento)	no plan field; `<Card ref w="34%"/>` apart already gives uneven widths	**not needed now**
Item overrides for chart / timeline	`item` case: no way to mark one point. But the emphasis already comes from the plan, and table has `<Highlight>`	**wait**: if the paid run shows the need, reuse `<Highlight>` rather than add tags
Space-aware fallback (not syntax)	28 of 30 narrow breaks; 5 compile failures	**yes, top need.** Fit-grow can measure the box and pick the fallback, as `regridTiles` already re-lays out

Candidate	Evidence	Verdict
Parts escape (`<Text ref="levers.0.title"/>` on every node)	Looks no composition reaches, rendered in Figure 10, right column: the KPI delta as a pill beside the value (Genspark KPI-E, spec §11.1, still "find"), a card as title · body · bullets · tag pill, the date inside a large marker	**optional.** Exists for KPI, about 70 lines of prototype for cards and timeline. Only if you want those looks
Per-row / per-column icons on table	composition test rows 2–3; `icons` works only on `ledger`	**small, optional**: extend `icons` to `rows` and `grid`, or say in the prompt that it's ledger only

The pill. Spec §11.1 asked whether POM can draw a pill from Text attributes alone. **No**: `highlight` is stripped and a Text has no box, so it needs a box part or the escape.

Nothing is needed for odd counts, per-item KPI and card styling, nesting, or dark themes: those already work.

5. Atomic vs composition

Three nodes built as small parts the LLM assembles, each part drawn from the plan by ref. KPI uses the real `fill_refs`. Card and timeline refs (`<Text ref="levers.0.title"/>`, `<Ul ref="levers.0.bullets"/>`, `<Icon ref="levers.0.icon"/>`, `<Text ref="roadmap.0.date"/>`) **don't exist**: a prototype of about 70 lines in the script (`atomic_fill`) stands in, writing the same ids the expanders write.

The limit of this comparison: I wrote the atomic XML by hand, correctly, knowing the target. An LLM would make mistakes I didn't, so atomic's risk and its look quality are both best cases here.

Figure 10: composition · atomic, same look · atomic, a new look.

	Composition	Atomic
Output tokens, KPI rows	**17**	513 (30 ×)
Output tokens, card banded	**17**	783 (46 ×)
Output tokens, timeline rail	**16**	814 (51 ×), and the LLM must get the half-unit grid right
Same look reachable?	yes	**yes**: 1–2% of pixels differ, 0% on the rail. The card lost the key-phrase colour (the LLM would have to write spans)
Looks a composition can't reach	no	**yes**: 3 rendered, all compile and pass
Fit-grow and the fixes	all	**survive if code writes the ids**: same growth (76 px, 2.73 ×), peers equal, regrid and guards work. Without ids, peers split (§1)
Type sizes across peers	code's	code's where refs are used. Boxes, padding and part order are the LLM's, per item
Prompt, fixed part	1,826 tokens for 5 nodes (compositions, parts, recipes)	286 for the grammar, **but it teaches no looks**: an equal-quality prompt needs a look library, not measured

	Composition	Atomic
Prompt, per component	51–209 tokens	same

What the validator sees, with mistakes planted (atomic.json):

Mistake	Caught?	Possible under composition?
item left out; index past the plan; same ref twice; word typed instead of a ref	**yes** (`NODE_ITEM_MISSING`, `NODE_REF_*`)	no
value ↔ delta swapped on an item	**no**	no
values of two items swapped (the $48.2M under the wrong label)	**no**	no
title ↔ tag swapped; card order changed	**no**	no
body or bullets forgotten on one card	**no** (the prototype checks items only; KPI `note` has a check that a real fill could copy)	no
one peer styled apart	**no**	only if the LLM does it on an item tag

Swaps are the serious ones: a number under the wrong label is a wrong slide that passes every check. The validator could check "every field placed" and part order in peers, but no check can tell that a value under a label is the right pair when the refs inside one box are all legal, except comparing indices within a box, which would forbid some free layouts.

Evidence from the KPI element-ref arm (same-plan (eval/tile-nodes/same-plan/README.md), paid 2026-10-14, 12 plans): generator output 683–1,619 tokens with refs vs 192–1,124 with tile tags; cost −26% for tags; 0 values off the plan both ways; larger numbers with tags in 7 of 12, smaller in 3. Blind marks: tags 0, refs 3, same 9 (0.375). Two of the three losses were layout bugs fixed afterwards; the third was "the LLM decorating its own tiles more". So with refs the LLM did sometimes draw livelier tiles, at more output. Later free gates passed on compositions (kpi_row, card_grid 0.949, chart 0.868, timeline 0.739, table 0.97), but those drew bare tags with no LLM.

Recommendation: a hybrid, with composition as the rule.

Where	Use	Why
table, chart, timeline	**composition only**	atomic would be 800–1,700 tokens and has real geometry (rail units, bar lengths, the highlight rules)
KPI, cards, default	**composition**	17 tokens, no swaps, peers alike
KPI, cards, a look no composition draws	**parts escape, opt-in**	KPI element refs already exist. Extend to cards only if you want those looks, with the "every field placed" check. Allow only when the plan's hint asks for something the variants can't do, and count how often the LLM takes it in the paid run: if it takes it everywhere, the savings are gone
bullets, callout, header	see decision 8	gains are small on two of them

6. Flexbox audit

POM's engine is Yoga (Facebook's flexbox) for every box (`calcYogaLayout.js`, `registry/definitions/stack.js`). Tested with POM's own layout on small slides (Figure 11; numbers in flex.json):

Feature	POM	Measured	Used by code / LLM
`gap`, `padding`, `grow`, `justifyContent`, `alignItems`	yes	as CSS. `alignItems` **defaults to stretch**, so writing it does nothing	heavily
Equal columns	**automatic**: HStack children with no `w` get basis 0, grow 1	`equal_no_w` = `w="1" grow="1"` to the pixel (393 / 394 / 393)	code writes `w="1"` 166 times across 30 expansions; redundant in an HStack, tested there
`w="max"`	`flexGrow: 1` **on the parent's main axis**	in a VStack it grows the **height** (367 of 400 px); `h="max"` in an HStack grows the **width**	the LLM wrote `w="max"` 309 times; fit-grow's `respectHeights` exists to undo the ballooning
Content width, no growth	`w="max" grow="0"` works on a stack	30, 123 px. `grow="0"` **on a Text is rejected**; with no `w` a stack collapses to 0	unused
Equal rows	`h="1" grow="1"`	160 / 160 / 160 vs `grow="1"` alone 111 / 258 / 111; **but text overflows a full row**	unused
`flexWrap`	yes	wraps at `minW`, last line widens	LLM 0 uses; code only `wrapChipRows`
`alignContent`	**not exposed**	**wrapped lines stay at the top: 23 px tall in a 500 px box**	verified limit
`minW`, `maxW`, `minH`, `maxH`	yes	`minW` holds while siblings shrink	LLM minW 1, maxW / maxH 0; code minW 4
Min-content width	**no** (Yoga default)	a 60 px column breaks "Internationalisation"	verified limit: why the word guard exists
Shrink	all stack children but icons	over-tall slides crush boxes below their text	verified: why `fixSpills` exists
`position="absolute"` in a stack	yes	anchored to its parent (`top` / `right`)	0 uses; a ref can't take children, so it doesn't help layers
Negative margin	yes	40 px overlap	unused
Percent width	yes	of the parent's content box	code 67, LLM 136
Column alignment across sibling rows	**impossible**	content leads 67 / 174 / 61 px: no grid in flexbox	verified limit: why rows use `w="24%"`
Sizing type to a box	**impossible**	a 500 px box, its type unchanged	verified limit: fit-grow's main job
`Table`, `Chart`, `Timeline`	not flex inside	fixed rows, fixed node type	`RIGID` in fit-grow

Census of stacks, by attribute (first LLM output of paid runs; code = the 30 catalogue expansions):

	Free LLM XML	KPI refs	Tile / card tags	Code
Slides / stacks	104 / 1,428	30 / 389	33 / 172	30 / 513

	Free LLM XML	KPI refs	Tile / card tags	Code
`gap`	1,308	389	159	321
`grow`	165 (12%)	38 (10%)	42 (24%)	**365 (71%)**
`alignItems` (of them stretch)	711 (279)	159 (89)	101 (76)	162 (51)
`justifyContent`	489	156	13	119
`w` px / % / max / share `1` / none	146 / 136 / 309 / 0 / 837	31 / 80 / 85 / 0 / 193	33 / 0 / 6 / 0 / 133	30 / 67 / 2 / 166 / 248
`h` px	147	30	33	**0**
`minW` + `minH`	34	3	12	6
`alignSelf`	68	100	0	3
`flexWrap`, `maxW`, `maxH`, absolute	0	0	0	0

The px rows include each slide's root (`w="1280" h="720"`). Saved slides are from older prompts, so the free column is the better read of LLM habit. Code never hardcodes a height, and that part is good flexbox.

Literal sizes in expander source, per file (`census.json`): `tiles` 1 px / 1 %, `cards` 3 / 1, `tile_groups` 3 / 5, `charts` 2 / 2, `tables` 7 / 3, `sequences` 4 / 1. Most are one of:

- lead columns for numbers and dates (`w="72"`, `"64"`, `"150"`);
- first columns (`22%`, `24%`, `34%`) and value columns (`16%`), which must be shared across rows.

fit-grow, 2,874 lines, 23 phase calls:

Phases	Could flexbox do it?
`growText`, `growStats`, `growStatSides`, `evenStats`, `evenTableParts`, `oneLineValues`, `reserveWrap`, `sizeDiagramWords`, `fitTableText`	**no**: type. Verified limit
`guardWords`, `fixSpills`	**no**: no min-content, and shrink crushes. Verified limit
`fitTables`, `sizeTables`, `growDiagrams`, `splitLists`	**no**: rigid nodes
`regridTiles`, `regridCards`	**no**: wrap can't stretch lines (`wrap_stretch`), and the decision is "do numbers grow 15% more", which needs measured type
`shareHeight`, `fillLoneRows`	**partly**: the `grow` they write is flexbox, but when to write it needs a measure. An expander could write `grow` on its row where it knows
`centreBody`	**yes**: a band with `grow="1" justifyContent="center"` is **pixel-identical** to the margin it computes

Phases	Could flexbox do it?
`respectHeights`, `wrapChipRows`	they repair LLM flex: `w="max"` ballooning, tags in a row. They write flex attributes, so a prompt line on the `w="max"` trap might cut them

Before and after, through the full pipeline (Figure 12; edits on real expansions in the script, `src/` untouched):

Change	Result	Verdict
Drop `w="1"` and `alignItems="stretch"` from a KPI grid	**0.0% of pixels differ**, −2% tokens, regrid intact	Safe on this slide; **replay before adopting**. Tidiness only: low priority
`centreBody` → flex	**0.0%**	Works; small gain, simple case only
Equal rows: `h="1" grow="1"` on card rows	rows equal, but **peers split: sizes 20–48 px on the slide (before: 25–34), row 1 small and row 2 large**	**Rejected**: worse under your rule, today's stays
The same on table grid rows	long row still taller after the spill fix; type smaller	**Rejected**
Chart rows: value column `w="16%"` → content-sized	values 66 → 75 px, labels 20 → 23 px here, but **bar tracks differ when value widths differ**, and a bar's length is a share of its track, so **the chart would misstate the data** (`rows_cols_align`)	**Rejected as written. Better:** code computes that shared share from the longest value string (not rendered)
`band.minH` to line up KPI bands	no fixed value works: type grows afterwards	**Rejected**: flexbox can't align across siblings. A fit-grow step that sets peer bands to the tallest would do it (not built)

So: we do use flexbox, for boxes, and well: 71% of code's stacks use `grow`, and no fixed heights. fit-grow isn't doing flexbox's job: nearly all of it is type, which flexbox cannot do. Two real improvements come out of this: teach the `w="max"` axis trap in the prompt, and let code size shared columns from their content. Removing redundant attributes is cosmetic.

7. Decisions for you, ranked

1. **Fix the attribute layer before any paid run (F1–F5).** *Done 2026-11-04, see §4.*
2. **Narrow boxes.** *Part done 2026-11-04:* fit-grow phase `reflowNarrow` gives a row of code-drawn items fewer columns when a word cannot fit even at the guard's floor (`POM_FIT_REFLOW=0` turns it off). The 30 narrow cases: compile 25 → 30; verdicts BREAK 28 → 24, WARN 2 → 6; four KPI compositions go BREAK → WARN (Figure 7). `long`: no verdict changed. `many`: card `steps` BREAK → PASS, nothing worse. The break map's short `steps` point moved from 7 to 12 cards (re-measured, 12 is the top of the sweep). *Still breaking:* every table composition, card and timeline `rows`, timeline `columns`, and `rail`. Cards now keep whole words but overlap or spill: four full cards do not fit the height of that box in any layout. A stacked `steps` card keeps its right arrow. *Still open: should a composition switch to another one when its box is narrow?* That needs a Python change, since fit-grow runs after the tag is expanded. How often real slides get a box this narrow is not measured.
3. **Keep composition as the default, element refs as an opt-in escape.** *Recommend: yes. Not atomic: 30–51 × the tokens and swapped values pass unseen.*

4. **Caps and defaults.** Card `rail`: cap it (or ban it with bullets on 4+ cards). Card `steps`: at most 6. Timeline bare tag: start at `rail`, not `native`, whose type is fixed and tiny. *Recommend: all three.*
5. **Words dropped by design.** Card `columns` and small `bento` drop `tag`; table `columns` and `banded` drop the first column's name. Does "every word" hold here? *Recommend: draw them, or say in the prompt the part is omitted; your call.*
6. **Empty cards in timeline `columns` / `banded` and sparse cards.** I did not find why growth stops. *Recommend: trace it as its own small task, since it's your rule.*
7. **Names.** `note` → `delta`; `bullets.` should style bullets; timeline's `rail` and `columns` mean something else than on cards; prompts all in `nodes`. *Recommend: fix `note` and `bullets.` now; rename the timeline words before the prompt reaches the paid run.*
8. **Undecided nodes.** *Recommend: callout and header stay LLM-drawn: 65–108 tokens is little to save, one item leaves no composition, and the header is the LLM's voice. Bullets: run the free replay first (66 of 142 saved lists split, 154–605 tokens); `rows` and `grid` are the cases.*
9. **The look-parts question.** Do you want the pill, the side-by-side card and date-in-marker looks? If yes, that decides the escape in 3 for cards and timeline. *Recommend: decide from the right column of Figure 10.*
10. **Flexbox tidying** (`w="1"`, stretch, `centreBody`). *Recommend: later, after a replay. A prompt line on the `w="max"` axis trap is the one piece worth doing soon.*

Table variants (user, 2026-11-04). `rows numbered`, `columns`, `banded`, `grid` and `grid numbered` are removed, so a person can edit tables by hand: only `native` is a real PowerPoint table (Figure 9, Figure 5). `native`, `rows` and `ledger` remain; the removed names are aliases. The catalogue, stress and reach files above predate this.

Card variants (user, 2026-11-04). Card `rail` is removed (the name draws `steps`); the timeline's `rail` is another drawing and stays. Card `bento` stays: its breaks were small cards dropping the `tag`, now drawn, and it holds from 500 px wide (400 px overlaps, `grid` only warns there; 280 px breaks every variant). Atomic cards stay rejected (§5). Flexbox placement needs no new syntax: `<HStack><Card ref w="40%"/><CardGrid ref cols="2"/></HStack>` compiles and loses no words, but its focus card gets no larger type and sits half empty, and 3 of 4 layouts spill on long cards, so it is the LLM's freedom, not bento's replacement. Card `columns` now draws the `tag`. Card `rows` and `rows numbered` fall back to `grid` when their padding and bullet lines would not fit (`cards.rows_fit`): bullets set the height, since rows with a body alone held to 367 characters at 6 rows. Measured over 8 variants x 8 cases: 33 pass, 23 warn, 8 break (27 / 19 / 18 at the start of this review); all 8 breaks are the 280 px box. Left as is: `grid` spills on 5 or 6 cards with three bullets each, but of 230 distinct saved card plans, none of the 63 `grid` plans with 5+ cards has bullets (longest card 177 characters), so no fix until a real plan needs one. Open: a long tag on one `columns` card drops its pill below its peers; `rows_fit` assumes a full slide body, so a smaller box can still break. Sheets: Figure 2, Figure 3, Figure 8, Figure 4.

After these, the paid same-plan run (`scripts/composition_paid.py`) answers what this audit can't: does the LLM pick variants at all? In saved tile-tag slides only 2 of 37 group tags named one.

4. Design record (Part B, 2026-10-27 to 2026-11-03)

Kept as written, with section numbers unchanged, apart from notes marked **Now:** where the text would mislead. Part A is the current state.

Superseded since this was written: `variant` on an item is read as an alias, not dropped (§2.2). A nested or missing component is now an error through `composed.unplaced`, although `validate.check_skeleton` stays unwired (§12.6, §13.3). Of §13.5, the merge and the unit tests are done; the `NODE_ATTR_UNKNOWN` severity and the anchor decision are not, and the paid run went ahead without them. The undated evidence in §6 and the old variant names in §5, §7 and §11 predate the deletions.

One composition convention for every derived node (spec draft, 2026-10-27)

Status: **approved by the user, 2026-10-27**: D1–D5, plus the §8 replacement and edits C1–C14 from `docs/eval/composition-left/README.md` (folded in below). Build per §10; the 8 remaining blocks go one block per step, behind `POM_NODES` (chart first). Free proof renders: `scripts/composition_convention.py` (KPI / cards) and `scripts/composition_gallery_left.py` (the other blocks), sheets in `docs/eval/composition-convention/` and `docs/eval/composition-left/`. Builds on:

- `docs/derived-nodes-spec-left-blocks.md` §2.1 (naming rules N1–N4: group tag = plan `kind`, item tag = list key, part = key without its prefix).
- `docs/eval/composition-test/README.md`: 10 of 12 style variants render identically from attributes, and only `divided`, `numerals`, `arrange` and `layout` are compositions.
- The user's definition (2026-10-27):
  - **Composition** = the same plan JSON drawn a different way: which parts appear and where. Code draws it.
  - **Style** = attributes. The generator LLM has complete power over them.

0. Summary (the decisions this spec asks you to approve)

#	Decision	Proposal
D1	The composition attribute	**`variant`, on the group tag only, meaning composition only.** It absorbs today's `layout`, `arrange`, `divided` and `numerals`. Nothing else on a tag chooses structure, apart from the measured knobs `cols` / `direction`, which code may overrule.
D2	Value grammar	**One structure word from a shared vocabulary of 8, plus at most one part modifier (`numbered` or `labelled`).** Examples: `variant="rows"`, `variant="rows numbered"`. A word means the same structure on every block (§2).
D3	How style is written	Dot-name attributes on the group, item or part, plus two small additions. **`item.` on the group** styles every item's box. **`$tone`, `$toneSoft`, `$toneText`** are colour tokens that code resolves per item from the plan's tone. Group box attributes stay on the group's own box (user, 2026-10-26).
D4	Today's style variant names	**They leave the tag.** `plain filled toned bar dot bare inverted` (KPI) and `lifted column filled outline toned accent_top inverted` (cards) become **style recipes**: 1–4 attributes each with `item.` / `$tone`, down from 6–34. They stay in `block-variants.yaml` for the prompt's examples and for the hint mapping. Emphasis and tone colours become part of each composition's default look (§3).

#	Decision	Proposal
D5	Where the theme defaults live	**One place: `block-variants.yaml` (renamed `nodes`), one `look` per composition, written only in palette tokens and `$tone*`.** `palettes.yaml` keeps supplying the token values, and `house-style.yaml` keeps the type ramp. Resolution order: composition look → group attributes → item attributes (§4).
D6	KPI compositions	`grid` (default) · `strip` · `rows` · `banded` · `bento` (rendered, §6)
D7	Card compositions	`grid` (default) · `grid numbered` · `rows` / `rows numbered` · `columns` · `banded` · `bento` · `steps` · `rail` (rendered: `rows numbered`, `columns`, `banded`, `bento`, §6). **Now:** `rail` removed (A.5)

1. Why one convention

- **One pattern to learn.** Today a generator meets `variant` (a look on KPI and cards), `layout` (group arrangement), `arrange` (part order inside an item) and `cols`. On top of that come the left-blocks draft's per-block names (`labelled`, `zebra`, `cards`, `step_cards`, `tiles`, `tinted` …). Every block names a different thing with the same attribute. In the combined check the LLM wrote 0 variants on 13 `<KpiGrid>` and 1 on 32 `<CardGrid>`. A choice the generator has to relearn on every block goes unused, or it gets invented.
- **The renders agree with the user.** In the composition test, two of today's style variants of one block looked like the same slide, while two compositions looked like two slides (Q2 verdict). The proof sheets here show the same for KPI tiles and cards (§6).
- **Style is cheap once it has two helpers.** The test's cost of 6–34 attributes per group came from two gaps. No attribute reached every item (group box ≠ items), and no colour token followed the plan's tone. `item.` and `$tone` close both gaps (D3). The LLM then gets complete power over style at a few attributes per group.

2. The convention

2.1 Tags and parts (unchanged from the left-blocks spec)

N1–N4 of `derived-nodes-spec-left-blocks.md` §2.1 hold: `<KpiRow ref>` / `<Kpi ref>`, `<CardGrid ref>` / `<Card ref>`, `<Timeline ref>` / `<TimelineItem ref>` … Parts carry the plan key's name (`label.`, `value.`, `delta.`, `title.`, `tag.`, `body.`, `bullets.`, `key_phrase.`, `icon.`). Look parts that hold no plan content are named for what they draw. This spec fixes their list, so that one name means one thing on every block:

Look part	Node	Drawn by	Appears in
`number.`	Text	code's 01 02 03	the `numbered` modifier; `steps` (undated)
`marker.`	Shape	a dot / square before a label	any composition, when `marker.shapeType` is set (style)
`band.`	HStack box	the solid head of an item	`banded`, `columns` (the pill)
`divider.`	the hairline between items	—	`strip`, `rows`
`connector.`	Shape / line	arrows, rail line	`steps`, `rail`
`item.`	each item's own box	—	every composition (D3)

2.2 `variant` = composition only (D1)

- **Only on the group tag.** An item tag (`<Card ref="levers.3" w="34%"/>` placed apart) is drawn in its group's item form. A `variant` on an item is dropped with `NODE_ATTR_DROPPED`. Today's item variants were looks (`inverted`, `toned`): they become attributes or the emphasis default (§3).
- **No `variant`** → code picks a default from the plan: `card_layout` (`steps` → `steps`), the count (a KPI row of 6+ → `grid` in two tiers, as today), and the hint (`look_from_hint`, extended to name compositions: "hero" / "spotlight" → `bento`, "ruled list" → `rows`, "strip" → `strip`), else `grid`. It is logged with `NODE_LOOK_FROM_HINT`, as today.
- **A composition the content cannot hold** (`requires` fails) → the block's `fallback`, with `NODE_VARIANT_UNMET`. The test is the same as today's, per composition (§5).
- **`cols` / `direction`** stay as measured knobs. They do not choose structure, and code overrules them on the break-map limits (`NODE_DIRECTION_UNMET`, `regridTiles`).
- **Part order inside an item is the composition's, never the LLM's.** Today's `arrange` values map to compositions (§7). fit-grow's `stackTileIcons` already moves icons above sparse labels on its own.

2.3 The value grammar (D2)

```
variant = WORD [ MODIFIER ]
WORD     = grid | strip | rows | columns | banded | bento | steps | rail | native
MODIFIER = numbered | labelled
```

Shared structure words. Each word has one meaning on every block:

Word	Structure (which parts appear, and where)	Evidence
`grid`	equal separate boxes in rows × columns (code picks the columns); each item's parts stacked	today's KPI / card default
`strip`	one shared box, the items side by side in **one** row, a divider between them; never regridded	KPI `divided` (2026-10-22 review of 25 example decks)
`rows`	one item per full-width row, its parts left to right (lead · title / label · body · value or detail at the right), a divider between rows	KPI `inline` / `side`, example 7 "ruled list"; Genspark "3 items → stacked rows", CARD-F; Perceptis `numbered_rows`; the LLM's 62 hand-built icon rows
`columns`	one column per item: a `band.` head on top, then each sub-item (bullet / sub-row) in its own box	Perceptis `columns` (composition test)
`banded`	equal boxes; each box has a solid `band.` head (icon + title / label) over the rest	Perceptis `title_bar`, `icon_header`
`bento`	the plan's emphasis item as one large block (value or title first), the others grouped beside it	Genspark KPI-E / KPI-F, CARD-C featured card; tile examples 3 and 7; `bento` in the composition test

Word	Structure (which parts appear, and where)	Evidence
`steps`	items in one row joined by `connector.` arrows; the destination item is the emphasis	cards `steps`; Genspark PROCESS-B / -D
`rail`	items along a `connector.` line with a dot per item (down the slide by default)	cards `rail`; Genspark STEPS-A, TIMELINE-B
`native`	the block's own POM node, filled by code (`Table`, `Chart`, `Ul`, `Timeline`, `ProcessArrow`, `Flow`)	left-blocks spec: the D10 node contract

Modifiers (parts code adds that the plan does not name; each also means the same everywhere):

Modifier	Adds	Rule
`numbered`	`number.`: 01 02 03 leading each item (above it in `grid`, at the left in `rows`, over it in `steps`)	not when the items carry dates (today's dated-steps rule) **or their own ordinal** ("Phase 1: …", "01. …"; C10)
`labelled`	the item's value written on its mark (chart bars, rail dots)	only where an item has one value (left-blocks spec, chart `labelled`)

Block-specific words. None are needed for KPI tiles, cards or the 8 other blocks (C5: a callout's quotation is a content rule of `native`, not a word). A later block that needs a structure no word covers adds one word to this table, with its meaning. It never reuses a word with a new meaning.

- **`ledger`** (table, user 2026-11-01): one solid header band, an icon over each column name (the tag's `icons`), over joined rows tinted by turns. The table also took the shared `grid` / `grid numbered`. **Now:** both removed, along with `labels` (A.5).

Head / detail split (C10). Where a composition shows a string item as a head and a detail, code splits it at the first " — " and only then at ": ", so "Phase 1: Foundation — detail" gives the head "Phase 1: Foundation". Every word is kept (`NODE_TEXT_SPLIT`, info).

2.4 How style is written (D3)

Where	Reaches	Example
group box attribute	**the group's own box only** (user, 2026-10-26)	`<KpiRow backgroundColor="$surfaceAlt" padding="22">`
`item.<box attribute>` on the group	**new:** every item's box	`item.borderLeft.color="$tone" item.borderLeft.width="6"`
`<part>.<attribute>` on the group	that part in every item (as today's `shared: parts`)	`value.color="$toneText"`, `band.backgroundColor="$panelDark"`
box attribute on an item tag	that item's box	`<Card ref="levers.2" backgroundColor="$warningSoft"/>`
`<part>.<attribute>` on an item tag	that part of that item	`<Kpi ref="kpis.0" value.glow.size="6"/>`

- **`$tone`, `$toneSoft`, `$toneText`** (new, code-resolved): the item's plan tone as a role token. good → `$positive*`, bad → `$negative*`, watch → `$warning*`, none → `$accent*`. An item's own `color="$negative"` sets its tone. They are accepted wherever a `$token` is accepted, and code swaps them per item before `normalize_xml`. So `LITERAL_COLOR` and the palette rules are unchanged. This is what the test could not write (`${role}`). With it, `dot` drops from 34 attributes to 3, and card `toned` from 32 to 3.

- **`item.` on `native` (C1).** `item.` reaches every item of a code-drawn composition. On a `native` node no item tag takes a box attribute (R2), so `item.` is limited to the colours each item tag takes: table `item.backgroundColor`, which code writes onto every `Td`, and diagram items' `item.color` / `item.textColor`. Any other `item.` attribute on `native` → `NODE_ATTR_DROPPED`. So do attributes that POM accepts and silently ignores (`Tr`, `ChartSeries`, `ChartDataPoint`, `TreeItem` apart from its colours): they are dropped and logged, never passed on (F12). The exception is a row's colour, which code writes onto that row's cells.
- **Dropped** (`NODE_ATTR_DROPPED`), unchanged: font sizes, font family, positions, `id`, typed text. Corner radius stays code's for KPI tiles (square) and cards (8 px).
- **Code still owns:** every word (verbatim, in plan order), type sizes (one per part per group, fit-grow), word fit, numbers / arrows / rails, readable ink on dark fills (`ON_FILL`), and **which item is the emphasis** (the plan's). The LLM may restyle the emphasis item but cannot move the emphasis. At most one dark item per block stays a check (`DARK_PANEL_SOFTENED`). It is not an attribute ban.
- **Table highlights (user, 2026-11-02).** A table colours by meaning with `<Highlight>` children inside `<Table ref>`, written by the generator (it reads the cells in its slide line, never types them); the planner only says in its `design_hint` what matters. Targets by name: `row` (its first cell), `column`, `cell="<row> / <column>"`, `phrase` (words inside a cell). Tones good / bad / watch. `fill` = the cell's background (`$<role>Soft`), `color` = its text (`$<role>Text`): separate, so a tinted row can carry a differently coloured phrase. Phrases are `<Span color>`, never POM's `<Mark>`. Value rules colour the text, code compares the numbers: `above` / `below` a number or another column (`else` for the rest), `max` / `min` (highest / lowest). A name or phrase not in the table: `NODE_HIGHLIGHT_UNKNOWN` (error, the LLM fixes it); more than 6: `NODE_HIGHLIGHT_LIMIT` (a warning, all drawn: no cap, user 2026-11-02); cells without numbers in one unit: `NODE_HIGHLIGHT_RULE` (left uncoloured). With highlights there is no emphasis from the hint; a bare tag keeps it.

3. Today's style variants: where they go (D4)

The test shows three costs of losing the names. Each one gets a home:

What the name carried	New home
the look (fill, borders, part colours)	a **style recipe**: a few attributes, listed in `block-variants.yaml` `styles:` (as `treatments` are today) and shown in the prompt as examples. The LLM writes the attributes, not the name
tone-driven colours (`with_tone`, `with_color`)	`$tone*` in the composition's **default look** (§4). With no attributes, a bad tile is tinted and a good number is green, as today. A recipe or an LLM attribute that names `$tone` follows the plan item by item
emphasis rules (`inverted` only on the emphasis item, card focus dark, `size="hero"`)	the composition's **`emphasis` look**: code applies it to the plan's emphasis item (dark block in `bento`, dark band in `banded` / `columns`, the dark focus card in `grid`)
the default look (`plain`, `lifted`)	each composition's `look`. Because the default is a layer the LLM writes over, not a variant to undo, the resets of the test (`border.width="0"` ...) are no longer needed for the common case

Recipes after the change (written against the `grid` default; `{t}` = `$tone`):

Today	Recipe	Attributes (test → now)
KPI `filled`	`item.backgroundColor="$toneSoft" value.color="$toneText"`	8 → 2
KPI `toned`	`item.backgroundColor="$toneSoft" item.border.color="$tone" value.color="$toneText"`	9 → 3

Today	Recipe	Attributes (test → now)
KPI `bar`	`item.borderLeft.color="$tone" item.borderLeft.width="6"`	18 → 2
KPI `dot`	`marker.shapeType="ellipse" marker.fill.color="$tone" label.color="$toneText"`	34 → 3
KPI `bare`	`item.backgroundColor="$surface" item.border.width="0"` (fit-grow: see §9 risk 2)	17 → 2
KPI `inverted`	the emphasis look (no attribute)	6 → 0
KPI `divided`	**composition** `strip`	—
card `column`	`item.backgroundColor="$surface" item.shadow.opacity="0" item.borderLeft.color="$tone" item.borderLeft.width="3"`	28 → 4
card `filled`	`item.backgroundColor="$surfaceAlt" item.shadow.opacity="0"`	20 → 2
card `outline`	`item.shadow.opacity="0"`	17 → 1
card `toned`	`item.backgroundColor="$toneSoft" item.borderLeft.color="$tone" title.color="$toneText"`	32 → 3
card `accent_top`	`item.borderTop.color="$tone" item.borderTop.width="4"`	24 → 2
card `numerals`	**composition** `grid numbered`	—
card `inverted`	the emphasis look	—

4. Complementing the theme: one approach (D5)

Every composition has one default look in `block-variants.yaml`, written only in palette tokens and `$tone*`. Palettes give the tokens their values. The LLM's attributes layer on top.

```
nodes:                          # today's tile_nodes, renamed (left-blocks spec §3)
  kpi:
    compositions:               # first = default
      grid:
        look:     {item.backgroundColor: $surfaceAlt, item.border.color: $border, item.border.width: 1,
                   item.padding: 16, label.color: $textMuted, value.color: $textMain, delta.color: $toneText}
        tone:     {bad: {item.backgroundColor: $toneSoft, item.border.color: $tone, value.color: $toneText},
                   watch: same as bad, good: {value.color: $toneText}}
        emphasis: {item.backgroundColor: $panelDark, label.color: $onPanelDark, value.color: $onPanelDark, size: hero}
      strip:   {requires: [count_le_5], fallback: grid, look: {...}, emphasis: {...}}
      ...
    styles:                     # recipes (§3): prompt examples and the hint mapping, never a tag value
      bar: 'item.borderLeft.color="$tone" item.borderLeft.width="6"'
```

- **Resolution order (later wins):** composition `look` → `tone` (per the item's plan tone) → `emphasis` (the plan's emphasis item) → group attributes → item attributes. This is today's order with "house look + style variant" folded into one composition layer, so the expander loses a layer instead of gaining one.
- **Why not `palettes.yaml` / the Theme:** a Theme is hex only, with no fonts or alpha (`pom-chartcolors-tokens` note). It holds colours, not structure. And the same composition must look right on every one of the ~20 palettes. Tokens give that: the proof draws **one XML** per composition on corporate-slate and graphite-dark, and the theme alone restyles it (§6).
- **Why not `house-style.yaml`:** it is prompt prose for the LLM's own drawing, and no expander reads it as data. The type ramp and spacing scale stay there, and code keeps reading sizes from its own constants as today.
- **Per-template house looks** (gj-h1 and others) later override a composition's `look` by palette name in the same file (`look_by_palette:`). Not proposed now. It is noted so the place is fixed.

5. KPI and card compositions from the same plan JSON

Fields: `kpi_row` = `kpi_labels / kpi_values / kpi_deltas / kpi_tones / kpi_icons`, emphasis from `design_hint`. `card_grid` = `cards[].title / tag / body / bullets / icon / tone / key_phrase`, `focus_card`, `card_layout`.

5.1 KPI (`<KpiRow ref variant>`)

Composition	Parts and where	requires → fallback	Default look (theme)	Emphasis item
`grid` (default)	separate tiles; icon + label, value, delta stacked	—	today's `plain` with tones	dark tile, `size=hero`
`strip`	one shared box, tiles side by side, dividers, one row	≤ 5 items → `grid`	`$surfaceAlt` box, `$border` dividers, value in `$toneText`	value hero size only (no dark item inside a strip)
`rows`	icon badge (`$toneSoft`) · label over delta · value right-aligned; dividers	—	no fill, `$border` dividers	its row's value hero size
`banded`	a `band.` head (icon + label) in `$tone` over value + delta	≤ 5 items → `grid`	band `$tone` / `$onAccent`, body `$surfaceAlt`	band `$panelDark`
`bento`	the emphasis tile large and value first, the rest as `rows` beside it	an emphasis item and ≥ 3 items → `grid`	big block `$panelDark`; small tiles as `grid`	is the big block

5.2 Cards (`<CardGrid ref variant>`)

Composition	Parts and where	requires → fallback	Default look	Emphasis (focus card)
`grid` (default for `card_layout: grid`)	today's card: icon, tag, title, body (+ key phrase span), bullets	—	today's `lifted` with tones	dark card
`grid numbered`	+ `number.` above the title	no dated tags → `grid`	`$surfaceAlt`, number `$textMuted`	dark card

Composition	Parts and where	requires → fallback	Default look	Emphasis (focus card)
`rows` / `rows numbered`	(number) · tag over title · body · bullets, one card per row, dividers	≤ 6 cards → `grid`	no fill, `$border` dividers, number / tag `$toneText`	its title in `$toneText`, bold number
`columns`	a `band.` pill (icon + title), body, each bullet in its own box	bullets on every card, ≤ 5 cards → `grid`	pill `$tone` / `$onAccent`, boxes `$surfaceAlt`	pill `$panelDark`
`banded`	a `band.` (icon, title, tag at right) over body + bullets	≤ 6 cards → `grid`	band `$tone`, body `$surfaceAlt`	band `$panelDark`
`bento`	the focus card as a large block, the others 2 + 1 beside it	`focus_card` set, 3–5 cards → `grid`	big `$panelDark`; small cards toned	is the big block
`steps` (default for `card_layout: steps`)	today's steps (arrows; numbered unless dated)	as today	`lifted`	the destination
`rail`	**Now:** removed, the name draws `steps` (A.5)			

6. Proof (free): the sheets

`python -m scripts.composition_convention` renders one sample plan per block, `kpis` (4 KPIs, tones good / good / bad / watch, icons, hint "highlight ARR as the hero") and `levers` (4 cards with tag, body, bullets, icon, tones, key phrases, focus card 0). The plans are validated against today's typed schema. Each block gets:

- **Rows 1–2:** today's real `<KpiGrid>` / `<CardGrid>` code (`POM_BLOCKS=tiles`) in three style variants, on corporate-slate and graphite-dark.
- **The rows below:** one proposed composition per row, written as the XML its node would expand to, with only its default look (tokens + `$tone` roles resolved by hand, no LLM attributes). The **same XML** is drawn on both palettes.

KPI parts are element refs (`<Text ref="kpis.0.value"/>`), so code writes the text and the `kpiv-` peer ids. Card boxes carry the cards' `card-<cid>-<i>` ids, so fit-grow grows every card of a group to one size.

- kpi.png: today `plain` / `filled` / `bar` | `strip`, `rows`, `banded`, `bento`
- card.png: today `lifted` / `filled` / `accent_top` | `rows numbered`, `columns`, `banded`, `bento`

What the sheets show

- **Today's three styles are one slide.** The KPI `plain`, `filled` and `bar` are the same 2 × 2 of tiles with a different fill or edge, and the card styles are the same 2 × 2 of cards. Each proposed composition is a different slide at a glance: a ruled strip, a ruled list with right-hand numbers, banded columns, a hero block beside a list.
- **The theme carries the colours.** Each composition drawn on graphite-dark reads as the same design in the dark palette, with no attribute changed. This supports D5: default looks in tokens are enough to complement any palette.
- **Peers share one size.** In `rows` all four values grew to 76 px and all labels to 23 px (`fitted.xml`). The composition test's uneven `numbered_rows` came from missing peer ids, not from the composition.

Render findings to carry into the build (none blocks the convention): Each was traced in fit-grow's own layout (`layout` / `natural` / `fill`, measured on the slides' `in.xml`):

1. `strip` **icons sit unevenly: a per-tile decision.** `stackTileIcons` (`src/node/fit-grow.js`) moves a tile's icon above its label when *that tile* is below `LOW_FILL` (0.70) after growth. It decides tile by tile, while the values next to it are grown as one group (`refGroups`). ARR (short label, one line) stayed just above 0.70, and the three tiles whose labels wrapped fell below it. **Fix (small, fit-grow):** decide once per ref group. If any tile of the group qualifies, move the icon on every tile of the group, so that peers match. This holds for every composition, not just `strip`. Under the convention, a composition node could also fix the icon position itself (`banded` / `rows` / `bento` already put it in the band, badge or lead).
2. **Tall bands stay empty: the width binds the numbers, and nothing can use the height.** In `strip` the values grew 28 → 65 px and stopped at the tile width (4 across, ~270 px each). The box's fill went only 20% → 30% (`banded`: 15% → 26%). A one-row composition cannot use the height by definition, and neither can a `regridTiles` split, because `strip` never regrids. **Proposal (a fit rule in code, not an LLM rule):** one-row compositions (`strip`, `banded` with ≤ 5 items) take content height and do not `grow`, so that the slide's other bands take the spare height. When the block is the slide's only growing band and its fill after growth stays under 50%, code draws the fallback instead (`grid`, which can regrid to 2 × 2) and logs `NODE_VARIANT_UNMET`. The same check is useful for any composition whose `requires` depends on space.
3. **"Card text does not grow" was a misreading: those cards are full by height.** fit-grow does run (`compile-result.json` `fitGrow: []`, nothing changed). By its measure today's `lifted` 2 × 2 cards are **87–92% full**. Card 3 has an inner height of 208 px: tag 18 + title 26 + body 22 + three bullets 97 + gaps 24 = 188 px. The proposed `banded` cards measure 88%, and `rows numbered` 96–97%. The cards look small on the sheet because each line uses ~40% of a 494 px wide card: the stacked parts fill the **height**, and the **width** stays spare. fit-grow grows type only into spare height (`LOW_FILL` on height), so it is right not to grow them. A bigger size would overflow the card.
   - **What does grow the type is a composition that spends the width.** `columns` (each bullet in its own box) grew to 28 / 39 px. The `bento` big block grew 30 → 44 px. The small `bento` cards sit at 0.71, just above the 0.70 threshold, so they stayed.
   - **Consequence for the convention:** for cards with a body *and* bullets, `grid` is the composition that keeps type smallest. A default chosen from the content (body + ≥ 3 bullets per card, ≤ 4 cards → `columns`, else `grid`) would follow the user's rule "grow the type, never shrink the card" better than a fixed `grid` default. This is a code choice from the plan's shape, checked on the saved card slides (free replay) before it becomes a default.
   - **My proof bug, now fixed:** in `rows numbered` the "01" Text had no width, so its column took 248 px of the row. The number now sits in a 72 px column (`scripts/composition_convention.py`), and the card sheet is re-rendered.
4. On graphite-dark the `banded` band's `$onAccent` ink sits on light tone fills. It reads, but the contrast check should run on `band.` like any dark fill (`ON_FILL` table: add `$positive` / `$negative` / `$warning` → `$onAccent`, after a contrast check per palette).

7. Migration

Today	After	How
`<KpiGrid>`	`<KpiRow>`	alias for one release, `NODE_TAG_ALIAS` (left-blocks §3)
`note.`	`delta.`	alias

Today	After	How
KPI `variant="plain|filled|toned|bar|dot|bare"`	no variant (`grid`) + the recipe's attributes	the expander reads an old name as its recipe, `NODE_ATTR_ALIAS` (info), for one release
KPI `variant="inverted"` on an item	dropped: the emphasis look already draws it	info
KPI `variant="divided"`	`variant="strip"` (or `rows` with `arrange="inline"`)	alias
KPI `arrange="label-first|icon-top"`	`grid` (the icon placement is fit-grow's)	alias
KPI `arrange="value-first"`	`bento` for the emphasis item; otherwise `grid`	alias
KPI `arrange="side|inline"`	`rows`	alias
KPI `layout="row|grid|column"`	`grid` (+ `cols`); `column` → `rows`	alias
card `variant="lifted|column|filled|outline|toned|accent_top"`	`grid` + recipe	alias as above
card `variant="numerals"`	`grid numbered`	alias
card `layout="grid|steps|rail"`	`variant="grid|steps|rail"`	alias
card `arrange="side"`	`rows`	alias
`look_from_hint` cues → variant names	cues → a composition (hero / ruled / strip) **or** a recipe's attributes	rename in `plan_checks`

Galleries:

- `scripts/tile_gallery.py` and `card_gallery.py` loop over compositions (rows) × one style row of recipes, instead of `variants × arranges`.
- `scripts/composition_convention.py` (this proof) becomes the composition gallery once the nodes exist: its hand-written bodies turn into bare `<KpiRow ref variant>` / `<CardGrid ref variant>` tags, and the sheets must match.
- `kpi_gallery.py`: no change. The blind-mark sheets of the passed card gate stay valid, because `grid` + `lifted` is today's default pixel for pixel.

Saved slides and replays keep working through the aliases. Remove them after one paid run shows 0 `NODE_*_ALIAS` from the generator (left-blocks spec, same rule).

Plan fields each composition needs

Composition	Needs	In today's plan?
KPI `grid`, `strip`, `rows`, `banded`	labels, values, (deltas, tones, icons optional)	yes
KPI `bento`	an emphasis item	partly: from `design_hint` only. **New field recommended:** `focus_kpi` (index, mirroring `focus_card`; `settle_card_fields`-style check). Until then the hint, and `bento` falls back to `grid` when the hint names none

Composition	Needs	In today's plan?
card `grid`, `grid numbered`, `rows`, `banded`, `steps`, `rail`	title (+ optional tag, body, bullets, icon, tone, key_phrase)	yes
card `columns`	bullets on **every** card	yes (field exists; `requires` checks it)
card `bento`	`focus_card`	yes
card true bento (blocks spanning 2 cells, Perceptis)	a span per card	**no, new field; not proposed** (composition test Q2 row 7)
table `banded` with icons per column	an icon per column	**no, new field** (`column_icons`), or the LLM's `icon` per `<TableColumn>` as for bullets today

8. The other blocks on the same convention (approved 2026-10-27)

Drawn and rendered in `docs/eval/composition-left/` (R1; the hand-written XML each node expands to is in `scripts/composition_gallery_left.py`). Today's style names become recipes or the default / emphasis look.

Block (tag)	Compositions (`variant`, default first)	requires → fallback	Becomes a style recipe
table (`<Table>`)	**Now:** `native` · `rows` · `ledger` (A.5). Approved then: `native` · `rows` / `rows numbered` · `columns` · `banded`	`ledger`: ≤ 6 columns → `native`	`plain` (default look), `zebra` (`stripe.`), `dark_header` (`header.` / `label.`); `highlight_row` = emphasis look; `minimal` dropped (C2)
chart (`<Chart>`)	`native` · `native labelled` · `rows`	`labelled` / `rows`: one series, ≤ 8 points (bar for `labelled`) → `native`	`focus_label` = emphasis look of `native labelled`; `horizontal` → `direction`
bullet_list (`<Ul>`, user 2026-11-04: `grid` / `banded` deleted as the card grid's looks, `docs/eval/bullet-node/`)	`native` · `native numbered` · `rows` (optional `icons`, ≤ 6 points) / `rows numbered`	numbered: no own numbers → `native` / `rows`; anything that cannot be composed → `native`, POM's real list	`lead` = `native` default; `icon` = `rows` + `icons`; `columns` = fit-grow's automatic split (4+ short points)
timeline (`<Timeline>`)	`native` · `rail` · `columns` · `rows` · `banded`	`native`: every label ≤ 24 characters with no detail → else `rail` (dots on a line, date / head / detail alternating above and below); `rail`: ≤ 7 points → else `columns` (a box per point under the line). Default chosen from content (2026-11-02)	`vertical` → `direction`; `cards` → `columns`; `alternating` → `rail`
process_arrow (`<ProcessArrow>`)	**no node (user, 2026-11-02)**	—	the LLM draws it as before (`chevrons` / `numbered` / `alternating` / `step_cards`)
flow (`<Flow>`)	**no node (user, 2026-11-02)**	—	the LLM draws it as before (`boxes` / `numbered` / `step_cards`)

Block (tag)	Compositions (`variant` , default first)	requires → fallback	Becomes a style recipe
callout (`<Narrative>`)	`native` (a quotation draws its `mark.` and attribution by content rule, C5)	—	`tinted` (default look), `rule` , `dark` , `plain`
header (`<Title>`)	`native`	—	`standard` (default), `inverted` ; `ruled` needs a `rule.` look part (C13), not built
matrix (`<Matrix>`)	`native` · `grid`	`grid` : `quadrants` present → `native`	—

Pyramid and tree stay `native` (C9).

Notes carried from the check:

- **Table look parts (C2, F11).** Table `native` gets the look parts `header.` (the header row), `label.` (the first column) and `stripe.` (alternate rows), which code writes onto the `Td` s. `cellBorder` is uniform and `Td` takes no border, so `minimal` cannot be written.
- **Chart `native labelled` (C6)** needs two code fixes: the label ink from `$chartInk` , and a number format derived from the plan's unit when the values share one ("0.29x" → `0.00"x"`). Chart text takes no attribute, so style power on charts lives in `rows` , where every part is a Text / Shape.
- **Timeline (C4):** `banded` uses the date as the band, coloured done / now / ahead from the "now" item (hint only). **2026-11-02 (user):** process_arrow's alternating `rail` moved to the timeline as `rail` , the default for labels with detail (the timeline's problem was long text: POM's 120 px titles); the old timeline `rail` (a line with a box per point) is now `columns` , used past 7 points. A vertical timeline that is not native draws `rows` . **Fields follow POM's `<TimelineItem date title description>` (user, 2026-11-02):** the parts are `date.` `title.` `description.` (`label.` read as an alias of `title.`). With `POM_NODES=timeline` the planner gets the `description` field back (label = the milestone's name, ≤ 3 words; description ≤ 20 words, capacity.yaml) and the label-room check leaves timelines alone; with the node off its prompt is unchanged (short labels only, the user's overflow fix for POM's own timeline). Splitting a label at " — " / ": " / an unspaced dash between letters stays as the fallback for plans without a description.
- **Process arrow and flow dropped (user, 2026-11-02).** POM's chevrons (150 × 80) and flow nodes (120 × 60) hold about two words at a fixed size and only shrink (`process-arrow.yaml` / `flow.yaml` facts), and the code-drawn `steps` were card grids under another name (`docs/eval/sequence-nodes/native-vs-derived-*.png`). Their nodes are gone; the planner still plans them and the LLM writes POM's `<ProcessArrow>` / `<Flow>` as before.
- **Inner boxes (C11).** A composition node gives its inner boxes one peer group, `card-<cid>x-<n>` , as well as the item ids `card-<cid>-<i>` , so that fit-grow grows peers to one size when an item's children stretch.
- **Value refs (C12).** A code-drawn `rows` writes its value / label as element refs (role ids), so that fit-grow grows them as values, not as body text held to its caps.
- **Knowledge (C14).** List per item tag what it takes (from `r2-attributes.json`) in the knowledge files.

9. Risks and open questions

#	Question / risk	Draft answer
Q1	Approve D1–D5 as one convention (one `variant` = composition, the shared words, `item.` + `$tone`, recipes, defaults in `block-variants.yaml`)?	yes
Q2	`bare` as attributes leaves fit-grow treating tiles as boxes (test note 1)	a recipe value of `item.border.width="0"` with no fill counts as "no box" in `hasBoxStyle`. A small fit-grow change, checked on the KPI gallery
Q3	Should the LLM still be able to make a non-emphasis item dark?	yes (complete power over style). `DARK_PANEL_SOFTENED` stays a report and does not undo it
Q4	New planner field `focus_kpi`?	build only if the free replay shows `bento` falling back on hints that do name a KPI (paid planner check, ask first)
Q5	The KPI gate failed on today's tiles. Do compositions change that?	unknown until marked. Next free step: a blind sheet of saved KPI slides, LLM vs `grid` vs the composition the hint asks for (no cost)
Q6	Prompt size	the shared vocabulary is ~120 cached tokens once (§2.3 table as one line per word), and per component one line names the compositions its content allows (~20 tokens). Recipes replace today's per-variant prompts (net ≈ 0)

10. Build order (after approval; free until the last step)

1. `block-variants.yaml`: `compositions:` + `styles:` for `kpi` and `card`, the `$tone*` resolver, `item.` in `_read`, and aliases (§7). Unit tests: every alias, every `requires` / fallback, `$tone` per tone.
2. Expanders: KPI `strip` reuses `build_group` (divided), `rows` reuses `inline`, `bento` reuses the item placed apart + `size=hero`. New: the `band.` part (banded / columns) and card `rows` / `columns` / `bento`. Inner boxes carry the peer ids `card-<cid>x-<n>` (C11), and code-drawn `rows` write value refs (C12).

The 8 other blocks (§8) follow the same steps, one block per step behind `POM_NODES=<kind>` (left-blocks spec §2.5, §13): chart, then timeline (+ process_arrow + flow, dropped 2026-11-02), then table, then bullet_list. Callout comes later, and header is not built now. Each block stops at its gate for the user's marks.

3. Composition gallery = this proof with bare tags. It must match these sheets.
4. Free replay on the saved KPI / card slides: the bare tag with the hint's composition, scorecard + blind sheet (your marks ≥ 0.5).
5. Paid check only with your OK: same-plan generator run on the saved plans (≈ $0.5, ceiling $1).

11. Left to find or define (2026-10-27)

Everything this spec has not settled, in one list. **find** = needs evidence (a render, a census or a replay). **define** = needs a decision or a written rule. **build** = understood, but waits for approval.

11.1 Status of today's names (from the user's status board, 2026-10-27)

Block	Built (style → recipe)	Built (composition → word)	Left to build	Dropped
KPI tiles	`plain` `filled` `toned` `bar` `dot` `inverted` `bare`; `hero` → `size="hero"`	`divided` → `strip`; `row` / `grid` / `column` → `grid` (+ `cols`) / `rows`; `label-first` / `icon-top` → `grid`; `value-first` → `bento`; `side` / `inline` → `rows`	**hero note pill** (the delta as a pill under a hero value, Genspark KPI-E). Decide: a part of `bento`'s emphasis look (style), or a part `delta.` drawn as a pill (a recipe: `delta.backgroundColor` needs a box around the Text, which a Text part cannot carry, so this is **find**: can POM draw a pill from Text attributes alone, or does it need a `pill.` look part?)	—
Cards	`lifted` (default) `column` `outline` `filled` `toned` `accent_top` `inverted`	`grid` `steps` `rail`; `numerals` → `grid numbered`; `stack` → `grid`; `side` → `rows`	**featured left column** = `bento` (§5.2; rendered, not built). **matrix** (deferred): a card_grid with `columns` / `rows` axes. **find**: is a card matrix `grid` with axis labels (a new word `matrix`), or slide layout?	`columns` and `title-band` as old card layouts. Note: the **word** `columns` and the composition `banded` (= title band) come back here as compositions with new meanings (§2.3). Confirm this is wanted, since the user once dropped them
The other 8 blocks	—	§8 (approved 2026-10-27): every name has a home except table `minimal` (dropped)	each block's node, in the §10 order	table `minimal`

11.2 Define (decisions still open)

#	What	Why it is open
F1	D1–D5 approval (§9 Q1)	the whole convention waits on it
F2	Whether `$tone*` tokens and `item.` are accepted by the normalizer as written, or swapped by the expander before `normalize_xml`	the spec assumes the expander swaps them. `LITERAL_COLOR` / the token check must not reject `$tone` first
F3	The fit rule for one-row compositions (§6 finding 2): no `grow`, and the fallback when they are the only band, under 50% full	a threshold to agree on; checked on the KPI gallery
F4	The default composition chosen from the plan's shape (§6 finding 3: body + ≥ 3 bullets per card → `columns`)	needs the free replay on saved card slides before it becomes a rule (no one-example rules)
F5	`look_from_hint` cues → composition words (hero / spotlight → `bento`, ruled → `rows`, strip → `strip`)	the cue list today maps to style names only (`plan_checks.look_from_hint`)
F6	Per-template looks (`look_by_palette:`) for gj-h1 and others	place fixed (§4), content not written
F7	`focus_kpi` planner field (§9 Q4)	only if the replay shows `bento` falling back
F8	Whether the LLM may still put `variant` on an item placed apart (today `<Kpi ref variant="inverted">`)	the spec drops it. Saved slides use it (tile example 3)

#	What	Why it is open
F9	~~The `quote` word for callouts~~	**resolved (C5):** no `quote` word; `native` draws a quotation's `mark.` and attribution from the content
F10	`ON_FILL` for `band.` on `$positive` / `$negative` / `$warning` fills (§6 finding 4)	needs a contrast check on every palette
F11	The look parts the native table needs (`header.` `label.` `stripe.`), written onto `Td`s by code	needed for `zebra` / `dark_header` (C2); built with the table step
F12	Attributes POM accepts and silently ignores (`Tr`, `ChartSeries`, `ChartDataPoint`, `TreeItem`)	drop with `NODE_ATTR_DROPPED`, except a row's colour, which code moves onto its cells (§2.4)
F13	Defaults chosen from the content for timeline / process (with F4)	one rule per block with a measured limit (break-map label room); checked by free replay before it becomes the default

11.3 Find (evidence still missing)

#	What	How (free unless marked)
R1	**Done (2026-10-27, `docs/eval/composition-left/`).** Compositions for the 8 remaining blocks: which of the shared words each block's plan JSON can fill, drawn and rendered from today's `slide_component_planner` output, beside today's style names	a composition gallery like `scripts/composition_convention.py`, one sample plan per kind from saved runs (`output/**/slides.json`). **Started as its own session, 2026-10-27**
R2	**Done (`docs/eval/composition-left/r2-attributes.md`).** Which style attributes POM accepts on each native node, because style can only be "complete power" where the syntax allows it. Known limits: `<Td>` holds no `Icon` / HStack; borders are rejected on `Td` / `Li` / `FlowNode` / `TimelineItem` / `ProcessArrowStep` / `PyramidLevel` / `MatrixItem` (`ITEM_BORDER_REMOVED`, POM 10.3.0); `Chart` text is black on dark palettes (`chartSurface`); no data labels in POM charts	compile a probe slide per native node with every box / part attribute (`attributes.yaml`, `llm.md`); record accept / reject / ignored. It decides where a composition must be drawn by code (boxes) instead of the native node
R3	**Done:** the kinds with no item list are `title`, `narrative`, `caption` and `tree`, and `layer` is free-form (`pyramid` and `matrix` have lists; C9). Was: which plan kinds have no item list to compose, and whether a composition word applies to them at all	from `planner_schema_typed.py` + R1
R4	**Done (README R4).** The plan fields each composition of the 8 blocks needs, marked "new planner field" where missing (as §7 does for KPI / cards): known gaps are per-column / per-row icons (table), a table title, a span per card, a narrative icon, timeline `description` (in the schema, filled 0 of 28 times)	R1 + the left-blocks census
R5	The per-group icon fix (§6 finding 1) on the KPI gallery	**build** after approval (fit-grow, a few lines)
R6	A blind sheet of saved KPI slides: LLM vs `grid` vs the composition the hint asks for (§9 Q5)	free; your marks
R7	The prompt cost of the vocabulary and the per-component line, measured rather than estimated (§9 Q6)	count tokens on 10 saved slides
R8	A unit test that every tag / variant / style name a prompt or knowledge file uses exists in `block-variants.yaml` (promised in its header, missing; left-blocks §3)	**build** with step 1

#	What	How (free unless marked)
R9	Code fixes found by R1–R4 (C6, C10, C11, C12): chart label ink from `$chartInk` and a number format from the plan's unit; the split " — " before ": " and the `numbered` ordinal rule; inner-box peer ids; value refs for code-drawn `rows`	**build**, each with its block (chart: the label ink and unit format)

12. Who owns what (2026-11-03)

One rule for every block from here on. The free probe that tested it (§12.6) is `scripts/ownership_probe.py`, with its sheet and results in `docs/eval/ownership-probe/`.

12.1 Three layers

Layer	What	Who writes it	Changed by us?
1 native POM nodes	`Text`, `Shape`, `VStack` / `HStack`, `Layer`, `Line`, `Arrow`, `Table`, `Chart`, `Ul`, `Timeline`, `ProcessArrow`, `Flow`, `Pyramid`, `Tree`, `Matrix`…	the LLM (free XML), or code inside a composition node	**never.** Fixed syntax (POM 10.3.0); we only compose around it
2 composition nodes	`<Tag ref variant …/>`: `KpiRow`, `CardGrid`, `Chart`, `Table`, `Timeline`, and **now** `Ul` (`src/compiler/nodes/`: `tile_groups`, `cards`, `charts`, `sequences`, `tables`, `bullets`)	the LLM places the tag; code expands it from the plan into layer-1 nodes	yes, one kind at a time behind `POM_NODES`
3 free LLM XML	slide layout, header (`title`), decor and arrow layers (`layer`), and any block with no node	the LLM	prompt and knowledge only

12.2 The rule

- **Code owns everything inside a block's box:** every word and number (verbatim, in plan order), the parts and their order, type sizes, word fit, the emphasis item, and the ids.
- **The LLM owns everything outside it and every style choice on it:** layout, placement, size (`w h grow min* max* alignSelf`), the `variant` pick, and every styling attribute (box, `item.`, `<part>.`, `$tone*`, gradients, borders, per-item looks).
- **Colour freedom stays.** Fill, text colour and visuals are separate explicit choices with no auto-coupling (2026-11-02). Colours are `$tokens` only (`LITERAL_COLOR`), so a theme switch restyles the slide. A group's own box attributes stay on its own box (no cascade, 2026-10-26).
- **Content attributes the LLM writes are dropped, never followed** (`NODE_ATTR_DROPPED`): text, `fontSize`, `fontFamily`, `id`, `chartType`, `title`, positions.

12.3 One nesting level

The LLM places refs inside its own layout, and code builds a ref out of native nodes. **A ref is never placed inside another ref**, and an LLM-placed ref never holds LLM children, except the documented ones (`<Card>` / `<Kpi>` item tags, `<Highlight>` in `<Table>`). A ref inside an LLM card is fine (probe d1).

12.4 Admission and fallback

- **A kind becomes a node only if its free replay and the user's blind marks beat the LLM.** process_arrow and flow were dropped on 2026-11-02 (`claude/left-blocks-build-0fe007`, f49fab6): POM's fixed-size chevrons and flow boxes hold about 2 words, and the code-drawn steps were card grids under another name. They stay layer-1 native nodes that the LLM draws. **Now:** merged.
- **Every kind keeps a native fallback** (`variant="native"`, or the `requires` → `fallback` chain).
- **One source of truth:** `block-variants.yaml` (`nodes.<kind>`: tag, parts, compositions, styles, aliases). It is read by `spec.py` and guarded by the R8 test (`tests/unit/test_block_variant_names.py`).

12.5 Layers are the LLM's: the open risk

`NO_REF_KINDS` = `title`, `layer` stay with the LLM. The risk is absolute-positioned arrows and overlays over blocks whose final size code or fit-grow decides. The LLM writes `<KpiRow ref/>` and never sees where the tiles end up.

12.6 Free probe (results: `docs/eval/ownership-probe/README.md`)

Case	Result
a refs in LLM stacks with styling	holds
b Table ref + LLM native ProcessArrow	holds (the native chevrons stay small: an LLM / native-node problem)
c Line layer over ref blocks, fit-grow on	lands within ≤ 14 px, **given the pixels**. fit-grow moved boxes ≤ 6 px. The generator never has those pixels, and `regridTiles` / `fillLoneRows` (not triggered here) do move boxes
c POM `<Arrow from to>` on code ids	works with the real ids (`kpiv0-kpis-0` → `cardt-levers-0`). With the guessed ids it fails silently (`ARROW_REF_NOT_FOUND` is only a compile warning). Box ids (VStack) cannot be connected
d ref in LLM card / ref-in-ref	in a card: holds. Ref-in-ref: the inner KPI row is an error (`NODE_ITEM_MISSING`, which names the old tag `<KpiGrid>`). **The inner chart is lost with only a report.** `validate.check_skeleton` (`NODE_MISSING`) is not wired
e content attributes	all dropped. Invented names (`x`, `y`, `kpi_values`, `table_rows`) are `NODE_ATTR_UNKNOWN` **errors**: a repair call for a slide code already drew correctly

13. Implementation plan from here (2026-11-03; code changes wait for the user's OK)

13.1 Which kind is where

Status	Kinds
node (layer 2)	kpi_row (KpiRow), card_grid (CardGrid), chart, table, timeline, bullet_list (all merged, **Now:** 2026-11-04)

Status	Kinds
native, drawn by the LLM (layer 1 / 3)	process_arrow, flow (dropped 2026-11-02), pyramid, tree, matrix (C9; matrix `grid` only rendered), title / header, layer
undecided	**Now:** bullet_list is a node (A.5). **callout** (narrative: one item, so the gain is only the quote rule C5; candidate to stay LLM-drawn), **header** (`title` is in `NO_REF_KINDS`; keep it LLM-drawn unless a blind sheet says otherwise)

13.2 Fixing the layer risk (proposal, in order)

1. **Anchor layers to block ids, not pixels.** Each node writes one stable, documented anchor id per item: `<cid>-<i>` on a connectable `Shape` (a `rect` with no fill, the item's size) or on the item's main Text. The prompt line names them, and the LLM draws `<Layer><Arrow from="kpis-0" to="levers-2"/></Layer>`. POM's connector then follows the boxes after fit-grow (probe c4).
2. **Turn a missed anchor into a node error** (`NODE_ANCHOR_UNKNOWN`, before compile), not a compile warning that leaves no arrow.
3. **Pixel layers over a ref block**: a `<Layer position="absolute">` whose children overlap a ref's box → `NODE_LAYER_OVER_REF` (warning, report the miss). fit-grow keeps `regridTiles` / `fillLoneRows` off for a block that a layer points at.

13.3 Validator gaps found

- Wire `validate.check_skeleton`'s plan-level check (`NODE_MISSING`, `NODE_REF_DUPLICATE`, `NODE_CHILDREN_DROPPED` → `NODE_REF_NESTED`) into `validator.py` before `composed.expand`, so that a ref-in-ref or a dropped component is one error with one message (d2).
- `NODE_ITEM_MISSING` names `<KpiGrid>`: under `POM_NODES=kpi_row` it should name `<KpiRow>`.
- Decide: should invented attribute names (`NODE_ATTR_UNKNOWN`) be an error (repair) or a dropped report? Code already draws the block right without them (e). A proposed split: unknown *part* attributes stay errors (the LLM meant a style), and unknown bare names on the tag are dropped with a report.

13.4 Per-node checklist (every new or changed kind)

1. `block-variants.yaml` `nodes.<kind>`: tag, parts, compositions (`requires` → `fallback`, ending in `native`), look, emphasis, styles, aliases. R8 passes.
2. Expander in `src/compiler/nodes/` from `compose.py` helpers: reads only the plan, writes the anchor ids (§13.2), peer ids (C11) and value refs (C12).
3. Prompt line (`prompt_line`, `node_lines`): the compositions the content allows, the style parts, and the anchor ids.
4. Validator codes: dropped / unknown / variant / nesting / missing, each with a unit test.
5. Gallery: a row per composition, two palettes, plus an ownership-probe row (refs in LLM stacks, a layer anchored to the node).
6. Free replay on saved slides (LLM layout kept, bare tag): scorecard + blind sheet.
7. The user's blind marks ≥ 0.5 (admission, §12.4); else the kind stays native.
8. Paid same-plan check, only with the user's OK, a case list and a cost estimate.

13.5 Before the paid same-plan runs (`scripts/composition_paid.py`, ≈ $0.87)

- The timeline (`0fe007`) and table (`3cf2e1`) branches are merged into one tree with f49fab6's process_arrow / flow drop, and unit tests pass apart from the 4 known failures.
- §13.3 is wired (no silent component loss), and the `NODE_ATTR_UNKNOWN` severity is decided, so that a paid run does not pay for repairs of slides that are already right.
- The anchor-id decision (§13.2) is made, at least for the prompt line: either the ids go into the prompt, or the prompt says "no layer over a ref block".
- The ownership probe and every gallery re-render clean on the merged tree.